

ABCD ADHD menarche analysis: Output for longitudinal supplementary/sensitivity analyses

Model 2 longitudinal ADHD present output

```
      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8266999 0.004585641 0.8172962 0.8353203
```

```
Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.844	0.015	0.815	0.873	1.000	1907	3757

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.109	0.033	0.052	0.181	1.004	994	1511

Regression Coefficients:

	Estimate	Est.Error	1-95% CI
Intercept	1.219	0.037	1.145
menarche_status_p_wave_2Y	0.170	0.033	0.106
adhd_status_presY	0.208	0.055	0.102
age_wave_2_sc	-0.017	0.030	-0.078
ethnraceblack_nh	-0.404	0.055	-0.512
ethnracehispanic	-0.103	0.048	-0.198

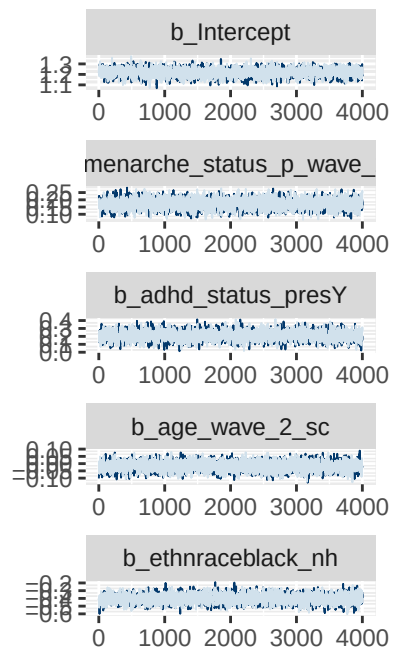
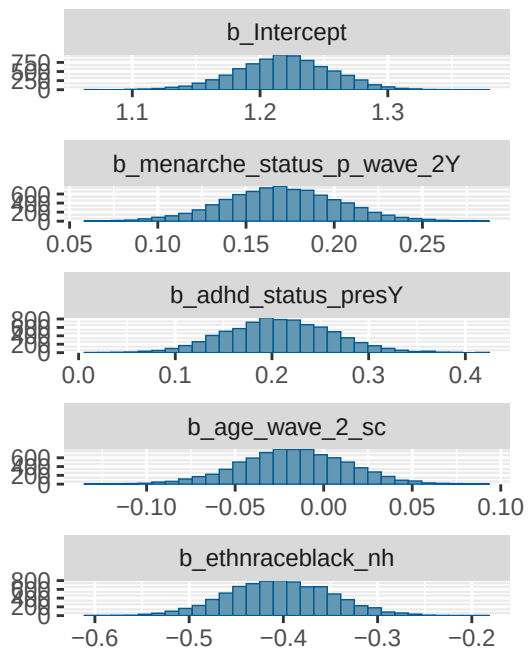
ethnraceother	0.001	0.051	-0.098	
inr_wave_0_sc	0.079	0.035	0.010	
internalising_wave_0_sc	0.850	0.027	0.797	
menarche_status_p_wave_2Y:adhd_status_presY	0.058	0.087	-0.113	
	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.290	1.000	2055	3567
menarche_status_p_wave_2Y	0.236	1.001	1993	3577
adhd_status_presY	0.318	1.000	1975	3949
age_wave_2_sc	0.041	1.000	2401	4049
ethnraceblack_nh	-0.296	1.000	2056	3413
ethnracehispanic	-0.009	1.000	1693	3494
ethnraceother	0.102	1.001	1711	2988
inr_wave_0_sc	0.148	1.000	1899	3569
internalising_wave_0_sc	0.902	1.001	1499	2913
menarche_status_p_wave_2Y:adhd_status_presY	0.227	1.001	1589	3268

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 2 longitudinal ADHD present diagnostics

prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_status_presY
normal(0, 1)	b	age_wave_2_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	inr_wave_0_sc
normal(0, 1)	b	internalising_wave_0_sc
normal(0, 1)	b	menarche_status_p_wave_2Y
normal(0, 1)	b	menarche_status_p_wave_2Y:adhd_status_presY
student_t(3, 1.4, 2.5)	Intercept	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
student_t(3, 0, 2.5)	sd	
student_t(3, 0, 2.5)	sd	Intercept
group resp dpar nlpar lb ub tag	source	
	user	
	(vectorized)	

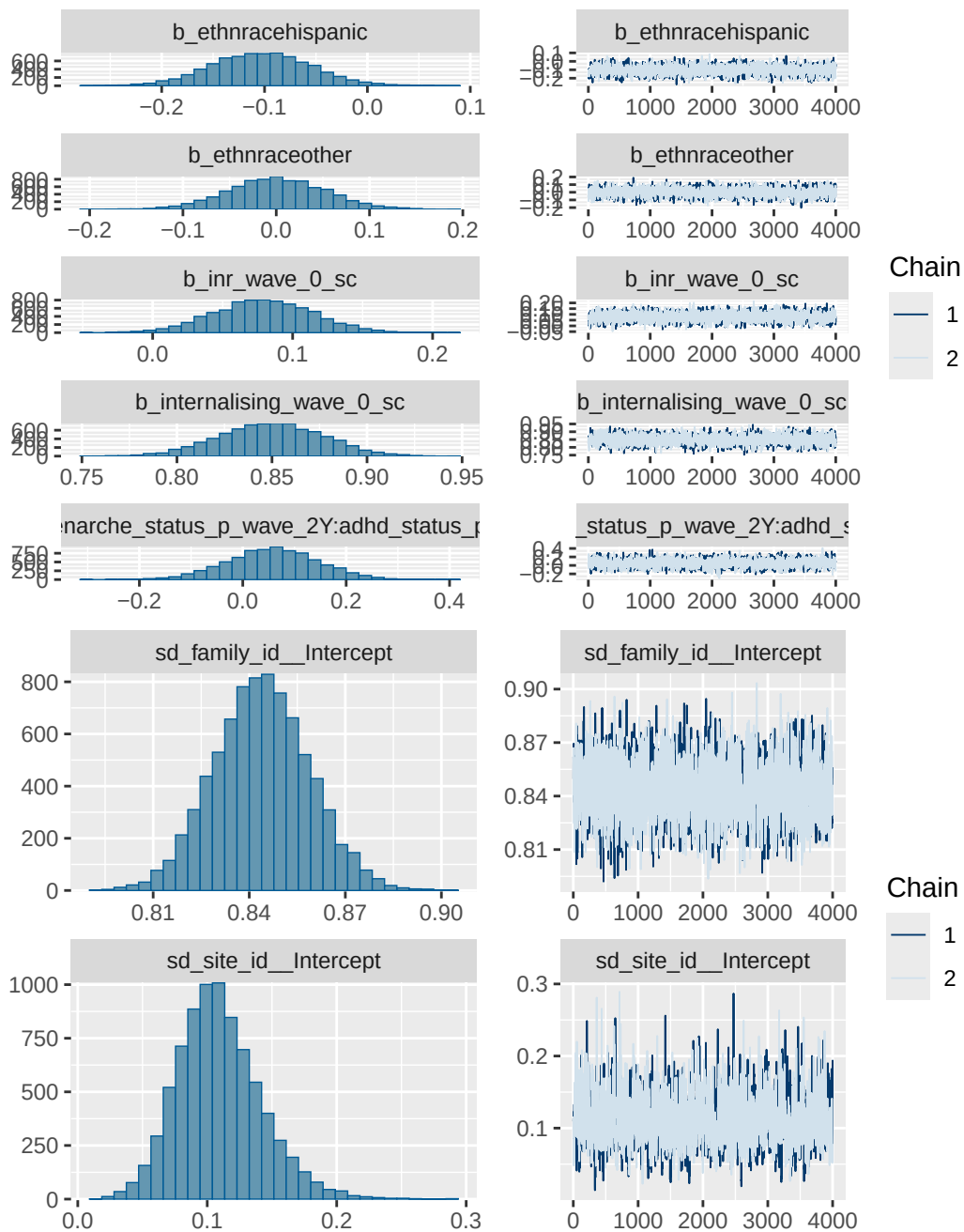
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	(vectorized)
	default
	default
family_id	0
family_id	0
site_id	0
site_id	0

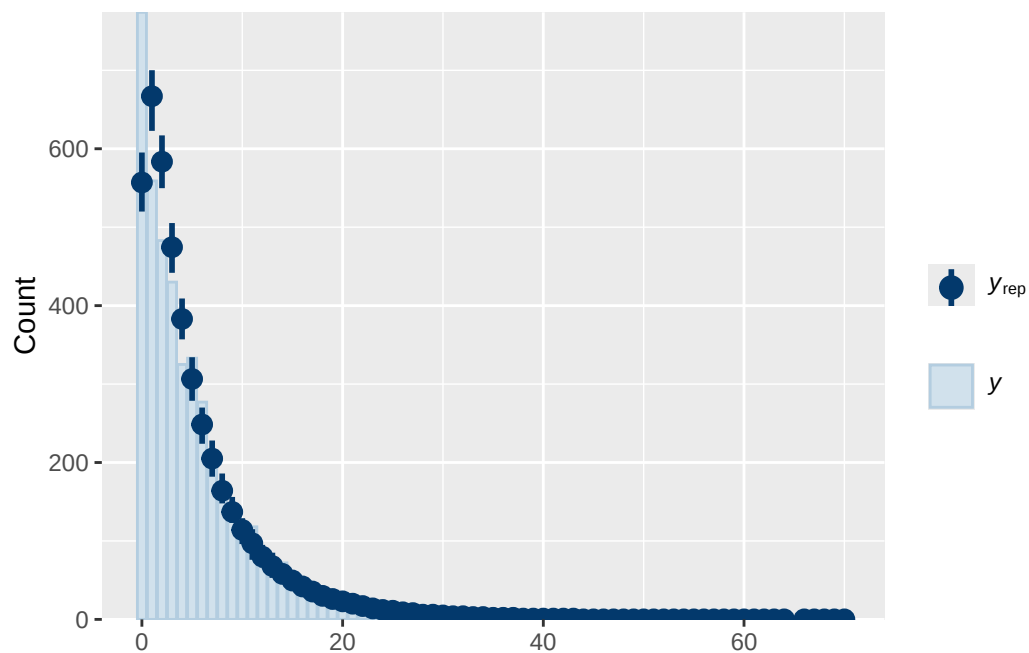


Chain

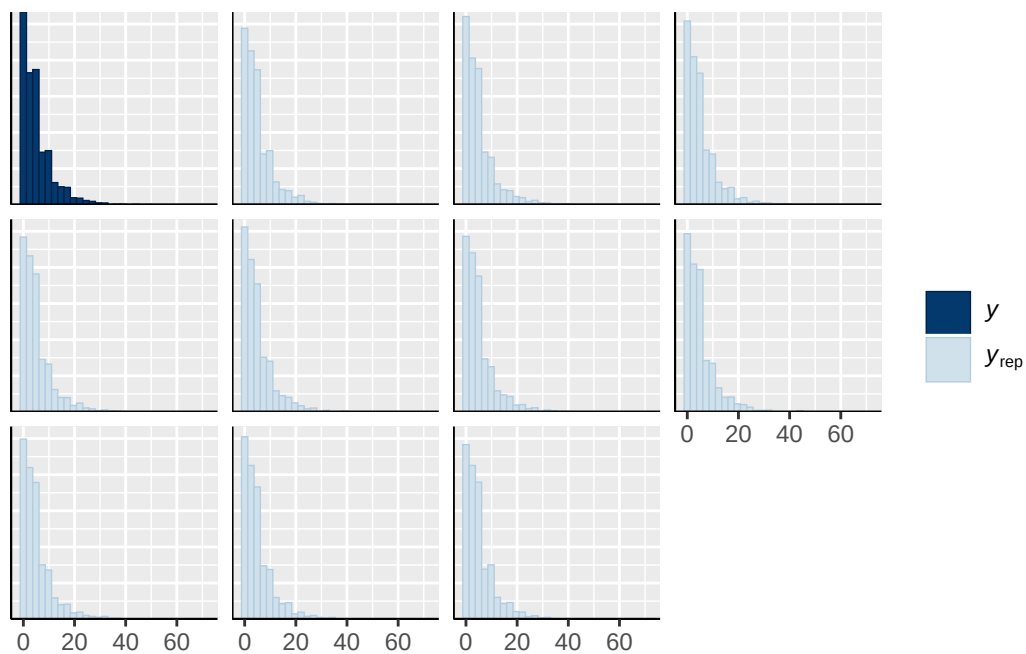
1

2

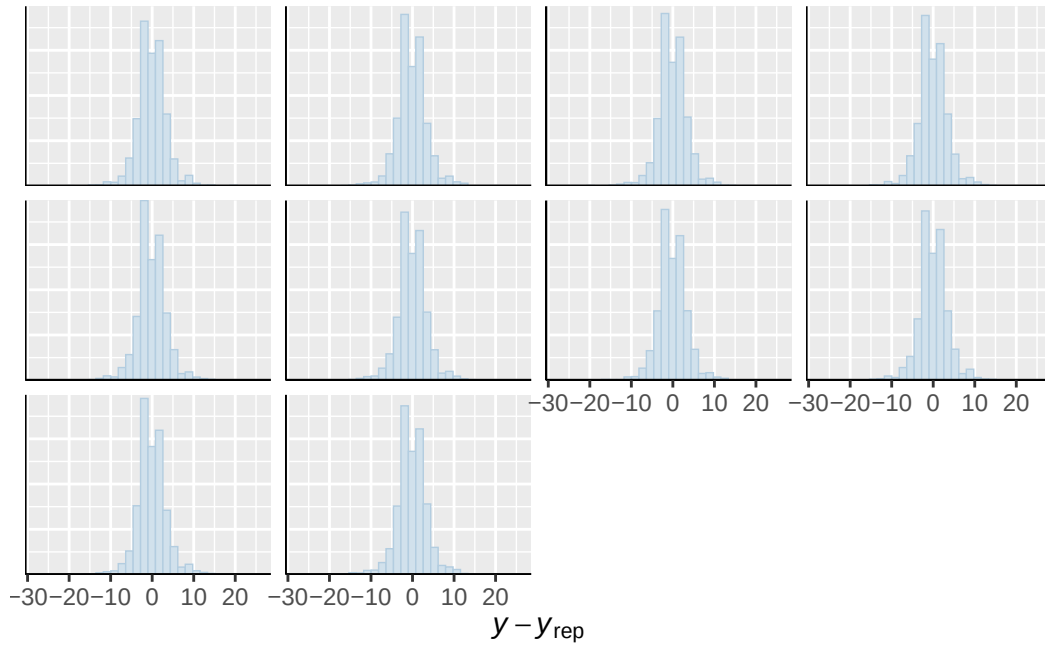




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

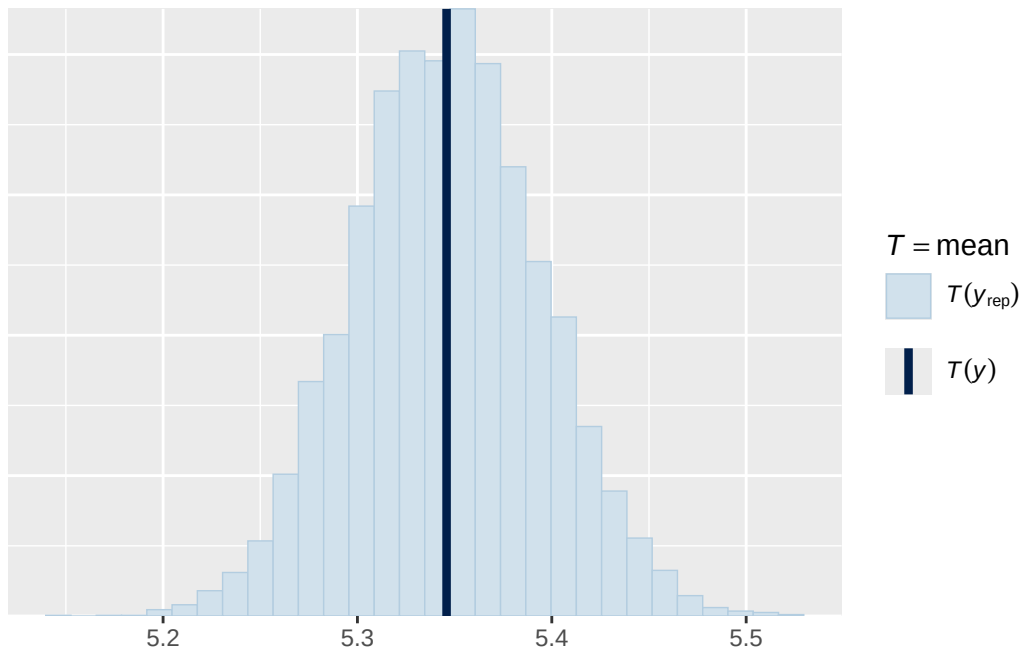


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

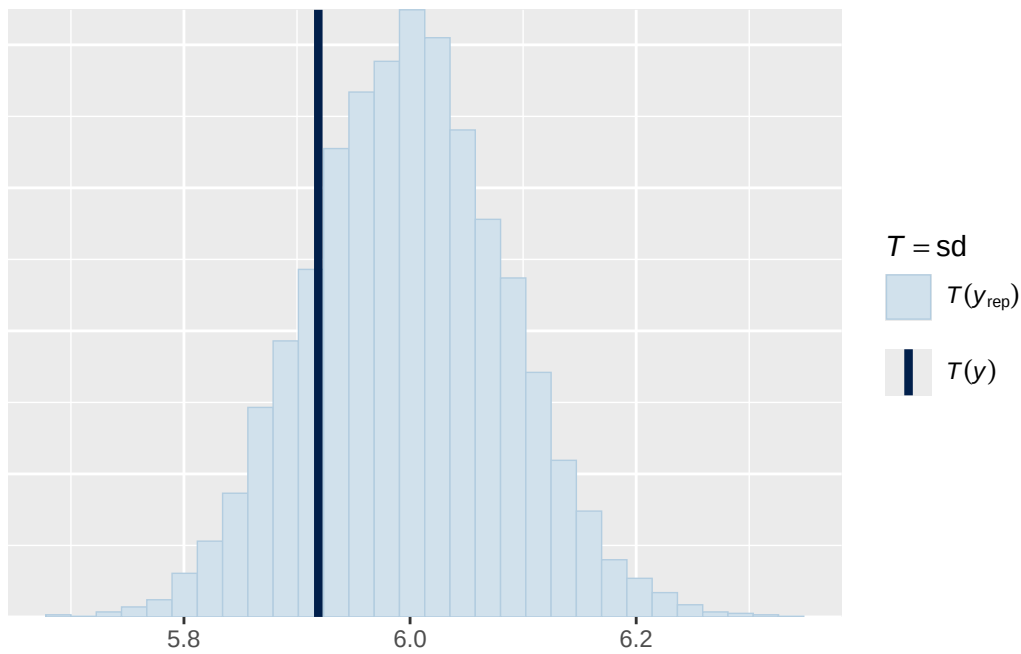


Using all posterior draws for ppc type 'stat' by default.

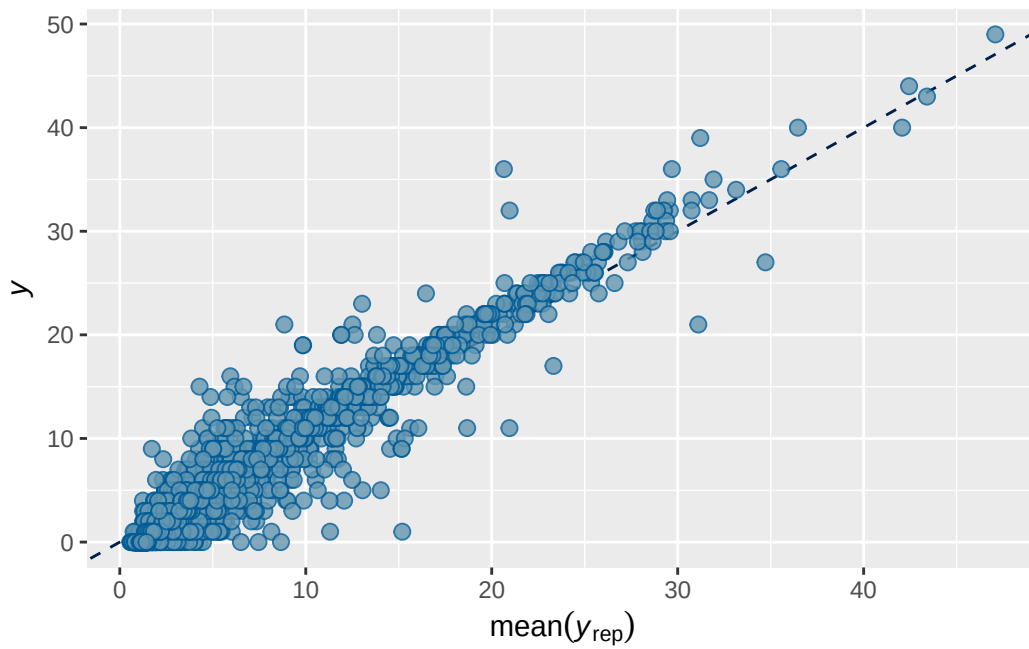
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 2B longitudinal ADHD present output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8274823 0.004586237 0.8183964 0.8362702

```

```

Family: poisson
Links: mu = log
Formula: internalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.849    0.015   0.821   0.878 1.001    1081    2057

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.114    0.033   0.057   0.186 1.003     431     364

```

Regression Coefficients:

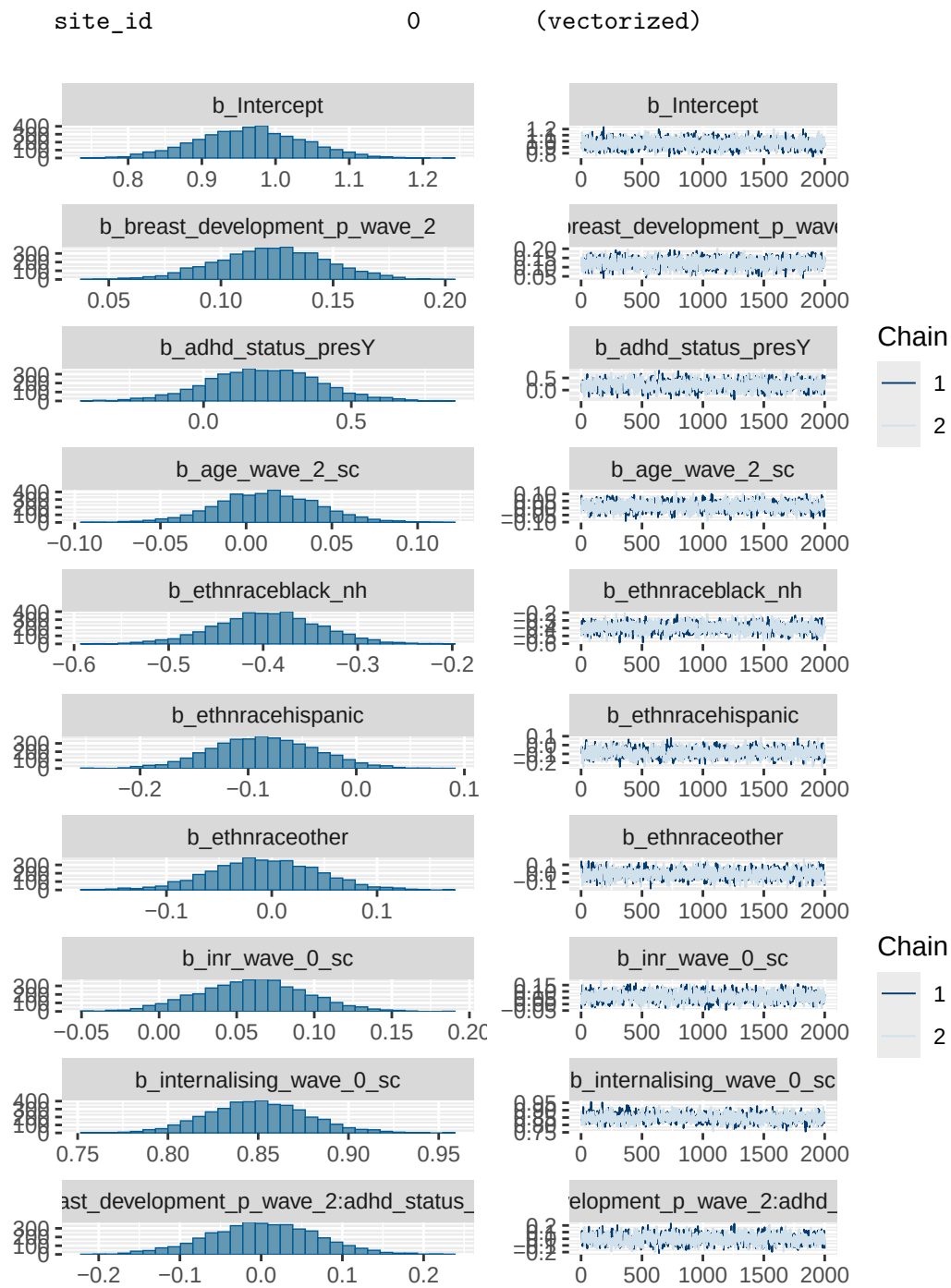
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.965	0.072	0.825				
breast_development_p_wave_2	0.123	0.024	0.074				
adhd_status_presY	0.204	0.183	-0.156				
age_wave_2_sc	0.013	0.029	-0.046				
ethnraceblack_nh	-0.395	0.055	-0.505				
ethnracehispanic	-0.085	0.048	-0.179				
ethnraceother	-0.007	0.051	-0.106				
inr_wave_0_sc	0.060	0.035	-0.007				
internalising_wave_0_sc	0.850	0.027	0.797				
breast_development_p_wave_2:adhd_status_presY	0.002	0.067	-0.131				
	u-95% CI	Rhat	Bulk_ESS	Tail_ESS			
Intercept	1.106	1.001	1261	1930			
breast_development_p_wave_2	0.169	1.000	1223	2060			
adhd_status_presY	0.571	1.002	1089	1992			
age_wave_2_sc	0.071	1.001	1292	2223			
ethnraceblack_nh	-0.288	1.001	1174	2007			
ethnracehispanic	0.010	1.002	1073	1838			
ethnraceother	0.092	1.001	1146	1819			
inr_wave_0_sc	0.129	1.001	1134	2084			

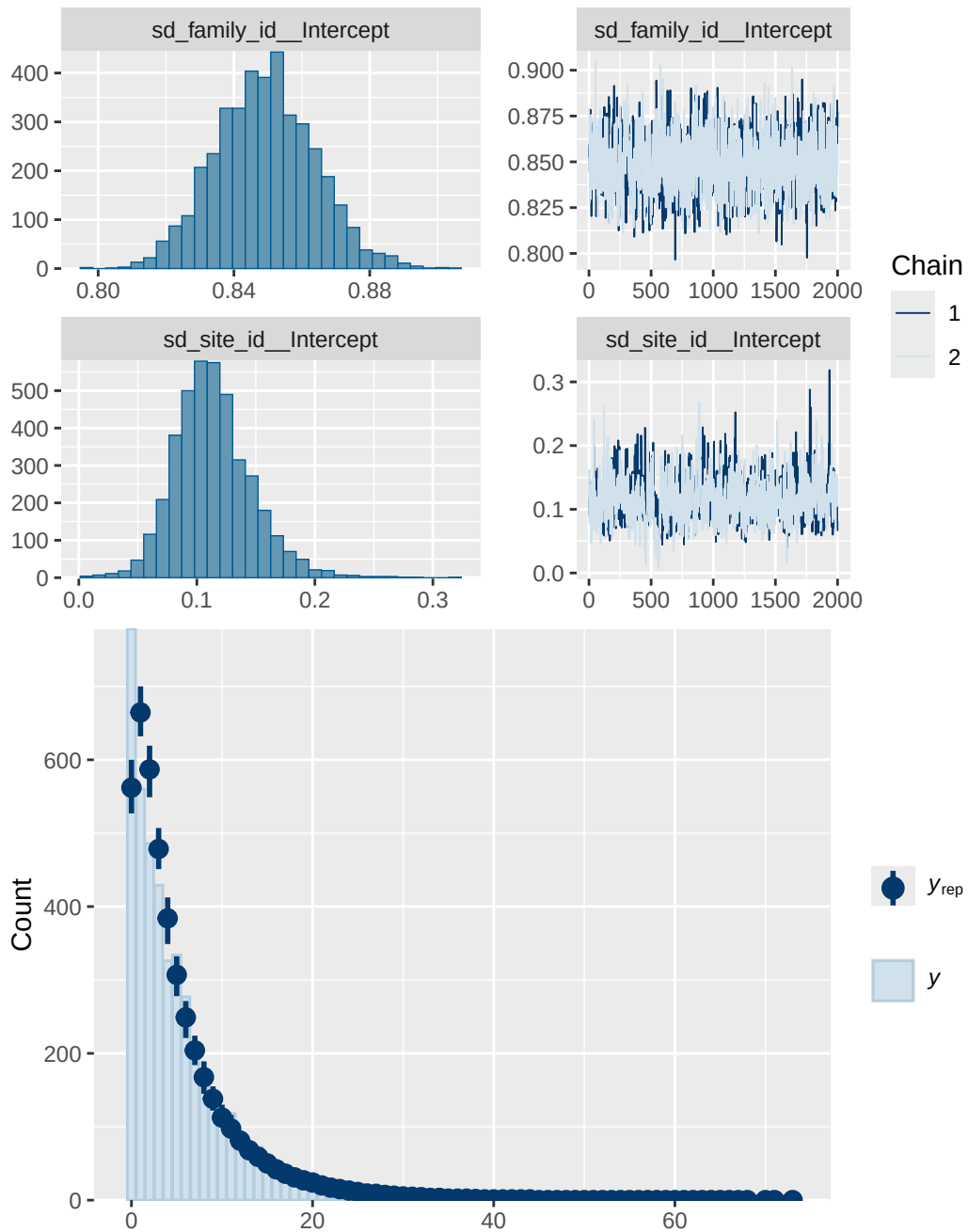
internalising_wave_0_sc	0.905	1.015	880	1394
breast_development_p_wave_2:adhd_status_presY	0.133	1.001	1084	1958

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

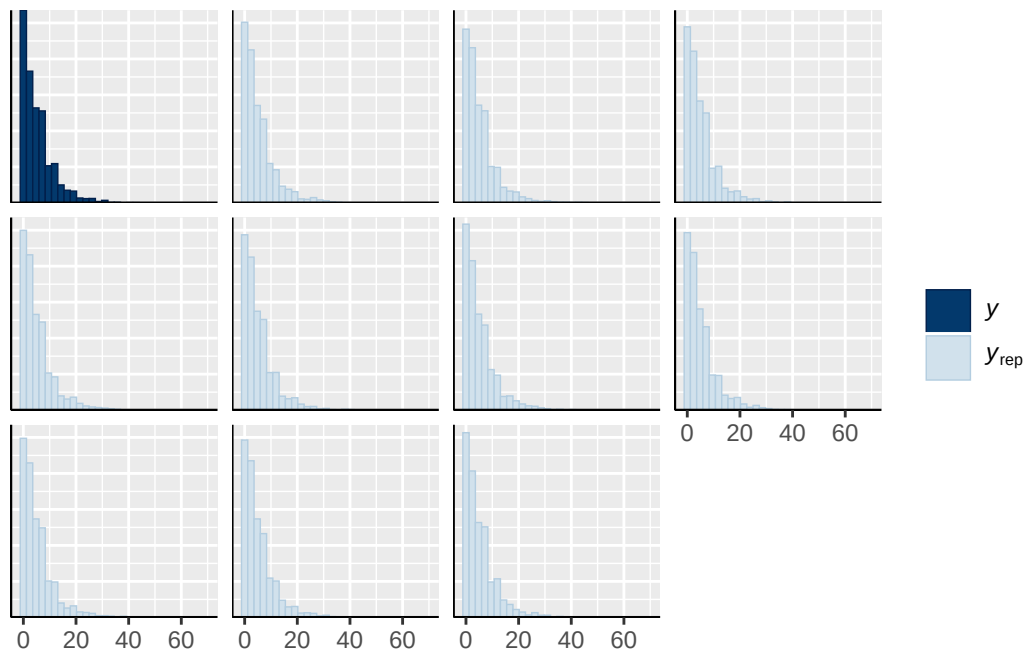
Model 2B longitudinal ADHD present diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_status_presY
	normal(0, 1)	b	age_wave_2_sc
	normal(0, 1)	b	breast_development_p_wave_2
	normal(0, 1)	b	breast_development_p_wave_2:adhd_status_presY
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	inr_wave_0_sc
	normal(0, 1)	b	internalising_wave_0_sc
student_t(3, 1.4, 2.5)	Intercept		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
student_t(3, 0, 2.5)	sd		
student_t(3, 0, 2.5)	sd		Intercept
group resp dpar nlpar lb ub tag		source	
		user	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		(vectorized)	
		default	
	0	default	
family_id	0	(vectorized)	
family_id	0	(vectorized)	
site_id	0	(vectorized)	

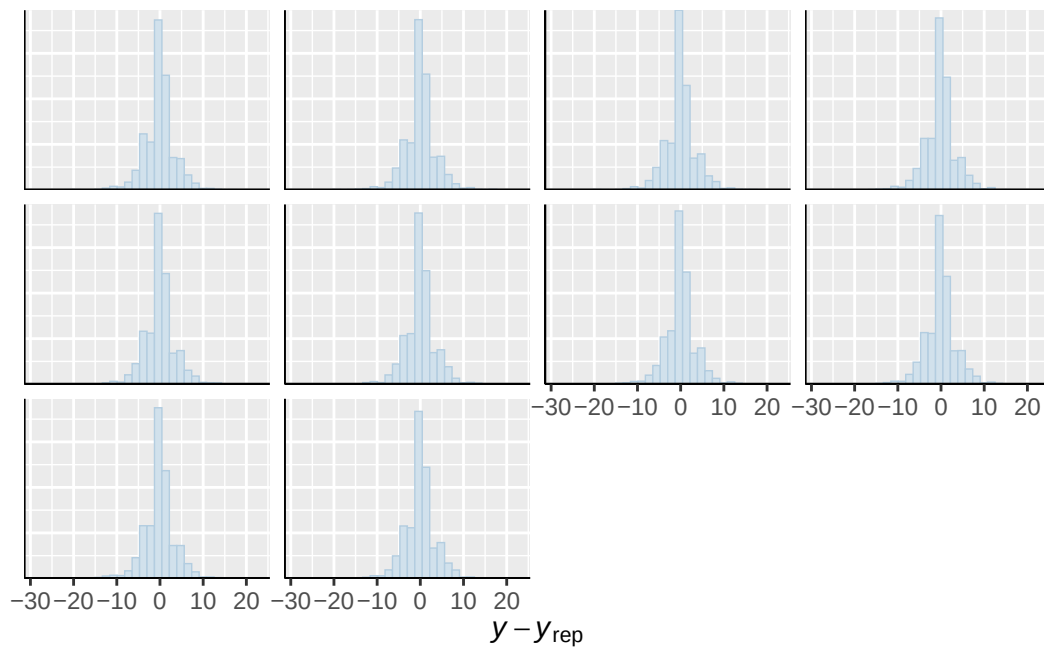




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

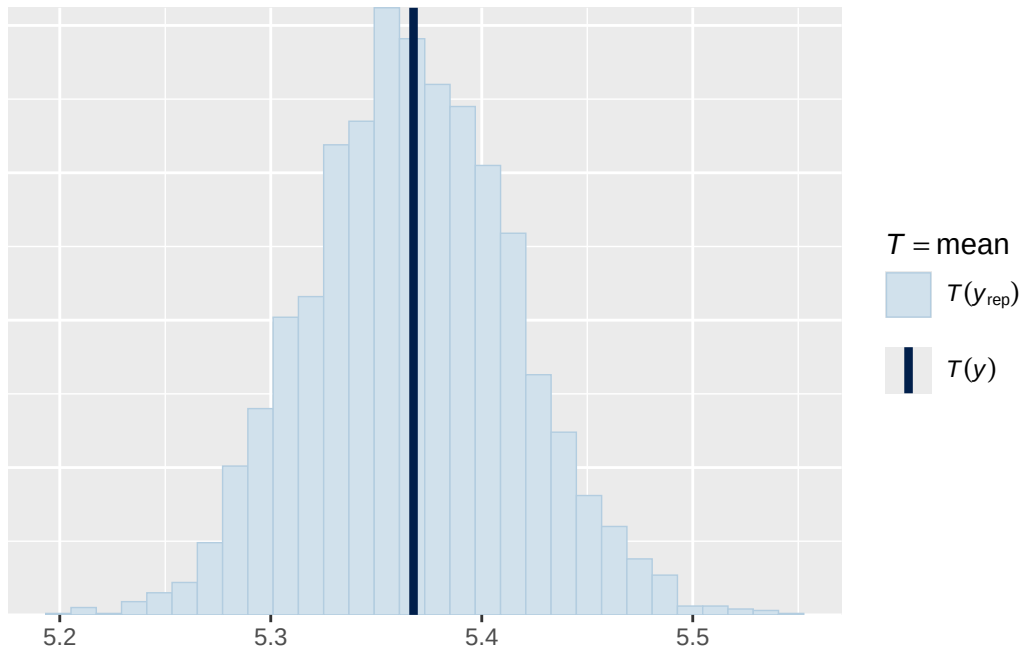


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

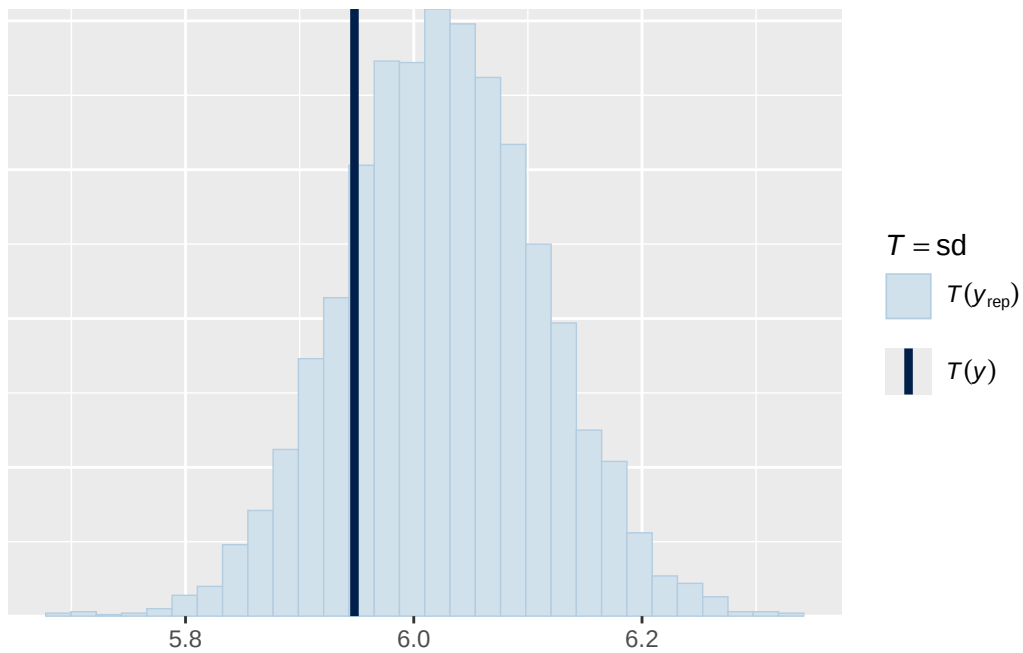


Using all posterior draws for ppc type 'stat' by default.

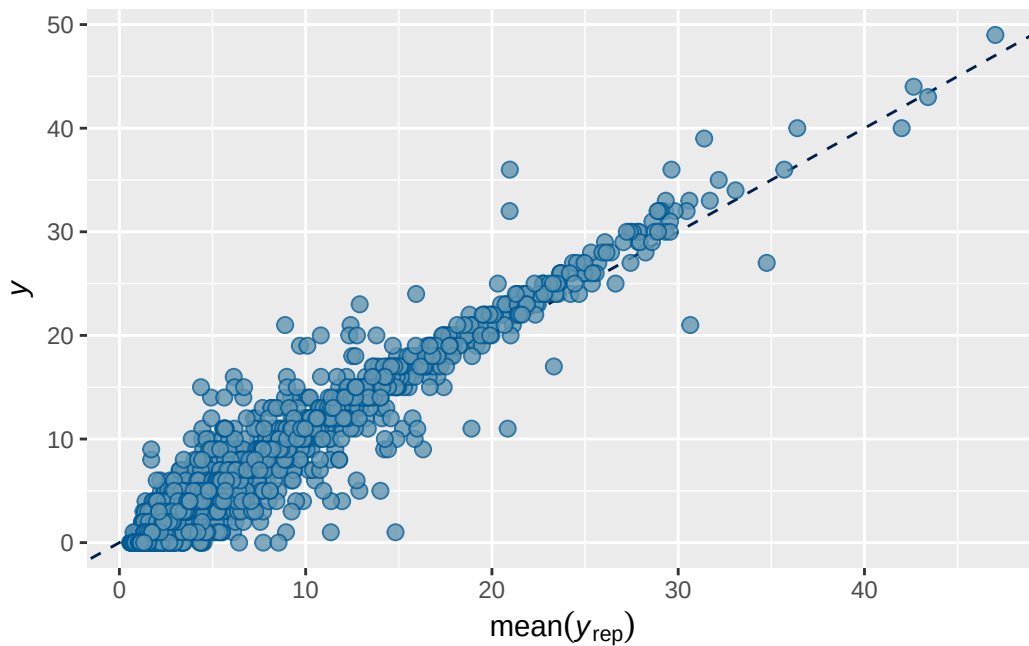
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 3 longitudinal ADHD present output

```
      Estimate Est.Error      Q2.5      Q97.5
R2 0.8212615 0.00533859 0.8102805 0.8313783
```

```
Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ menarche_status_p_wave_2 * adhd_status_pres + +age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.026	0.021	0.986	1.067	1.002	1495	3351

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.070	0.037	0.006	0.151	1.001	602	1363

Regression Coefficients:

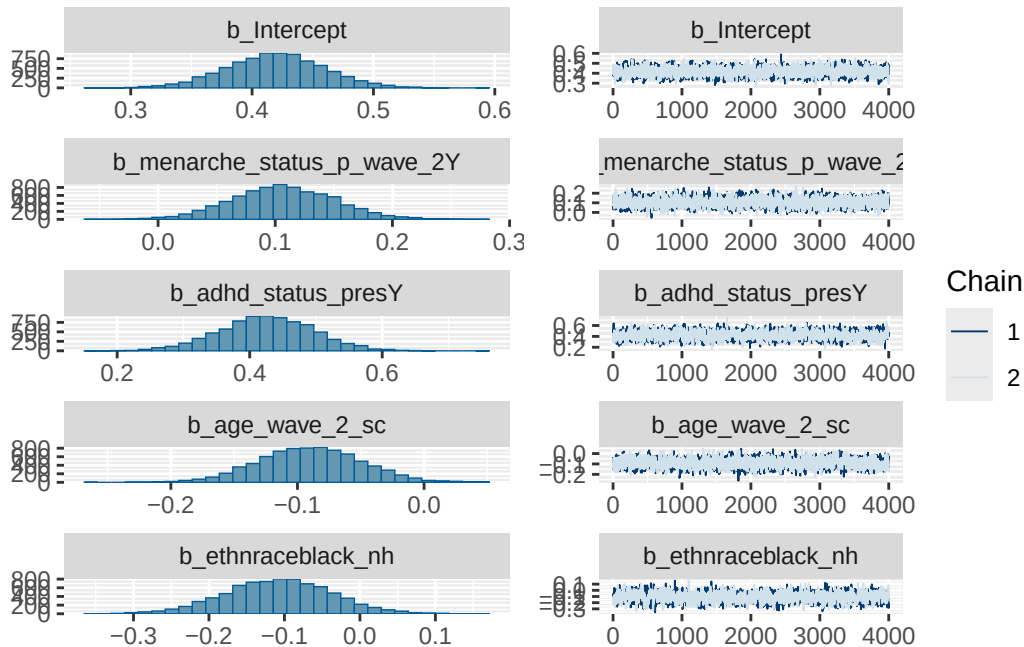
	Estimate	Est.Error	1-95% CI		
Intercept	0.419	0.040	0.339		
menarche_status_p_wave_2Y	0.107	0.044	0.023		
adhd_status_presY	0.425	0.071	0.285		
age_wave_2_sc	-0.090	0.040	-0.169		
ethnraceblack_nh	-0.108	0.071	-0.248		
ethnracehispanic	-0.113	0.059	-0.231		
ethnraceother	-0.038	0.065	-0.167		
inr_wave_0_sc	-0.006	0.045	-0.091		
externalising_wave_0_sc	1.029	0.035	0.961		
menarche_status_p_wave_2Y:adhd_status_presY	0.179	0.108	-0.037		
	u-95% CI	Rhat	Bulk_ESS	Tail_ESS	
Intercept	0.499	1.000	2211	3678	
menarche_status_p_wave_2Y	0.194	1.001	2205	3690	
adhd_status_presY	0.564	1.001	1595	2521	
age_wave_2_sc	-0.012	1.001	2276	3214	
ethnraceblack_nh	0.030	1.003	1493	2467	
ethnracehispanic	-0.001	1.000	2006	3547	
ethnraceother	0.089	1.001	1677	3299	
inr_wave_0_sc	0.085	1.000	1889	3855	
externalising_wave_0_sc	1.097	1.003	1583	2501	
menarche_status_p_wave_2Y:adhd_status_presY	0.384	1.002	1491	2666	

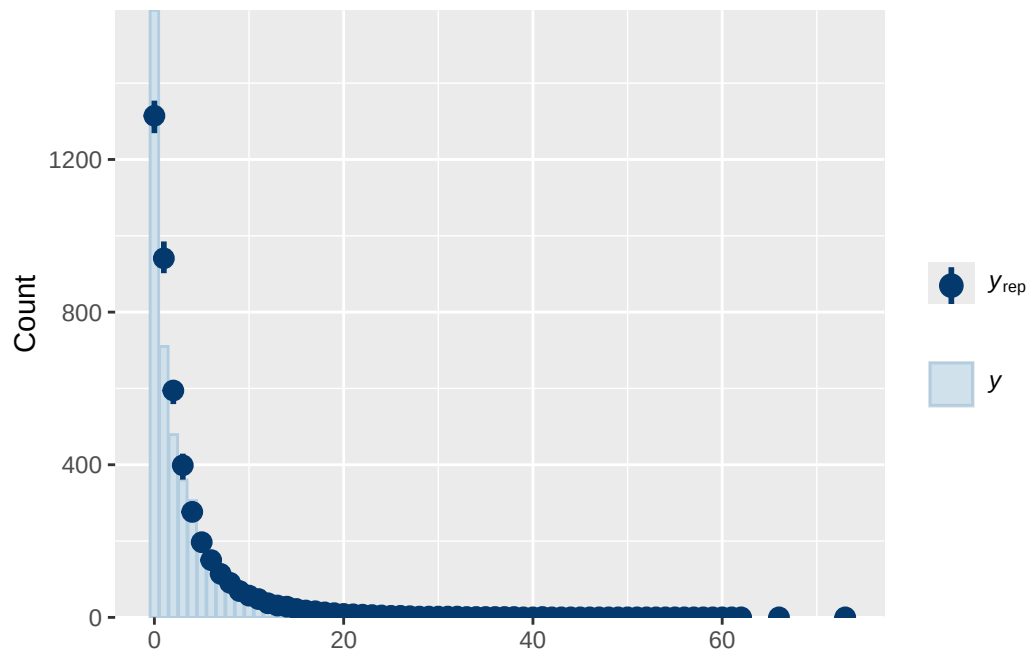
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 3 longitudinal ADHD present diagnostics

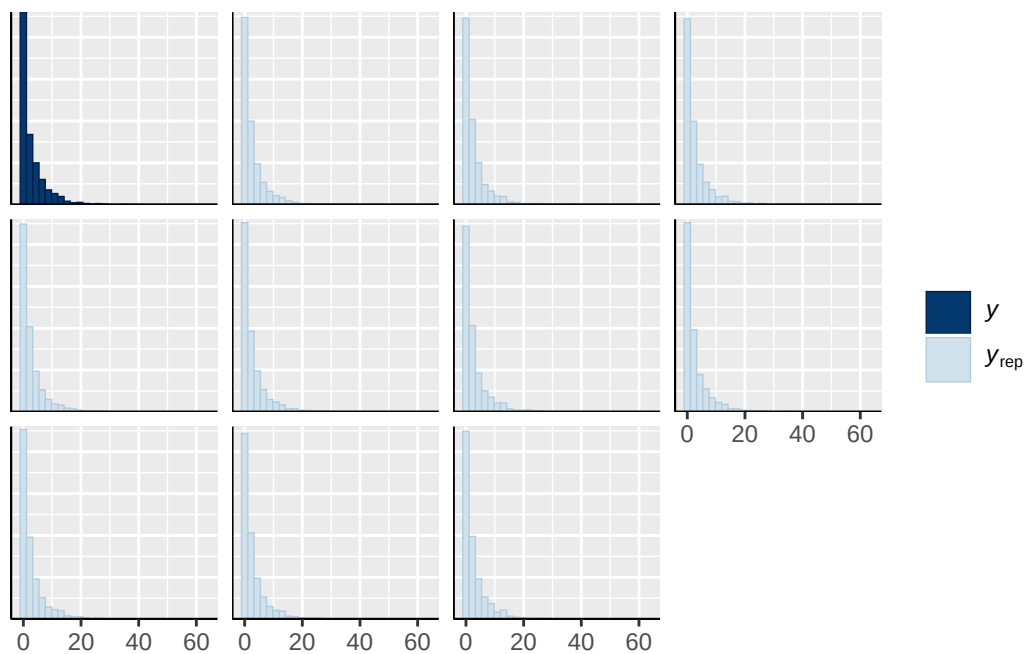
prior	class	coef
normal(0, 1)	b	
normal(0, 1)	b	adhd_status_presY
normal(0, 1)	b	age_wave_2_sc
normal(0, 1)	b	ethnraceblack_nh
normal(0, 1)	b	ethnracehispanic
normal(0, 1)	b	ethnraceother
normal(0, 1)	b	externalising_wave_0_sc
normal(0, 1)	b	inr_wave_0_sc
normal(0, 1)	b	menarche_status_p_wave_2Y
normal(0, 1)	b	menarche_status_p_wave_2Y:adhd_status_presY

student_t(3, 0, 2.9)	Intercept	
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
student_t(3, 0, 2.9)	sd	
student_t(3, 0, 2.9)	sd	Intercept
group resp dpar nlpar lb ub tag	source	
	user	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	(vectorized)	
	default	
	default	
family_id	0	(vectorized)
family_id	0	(vectorized)
site_id	0	(vectorized)
site_id	0	(vectorized)

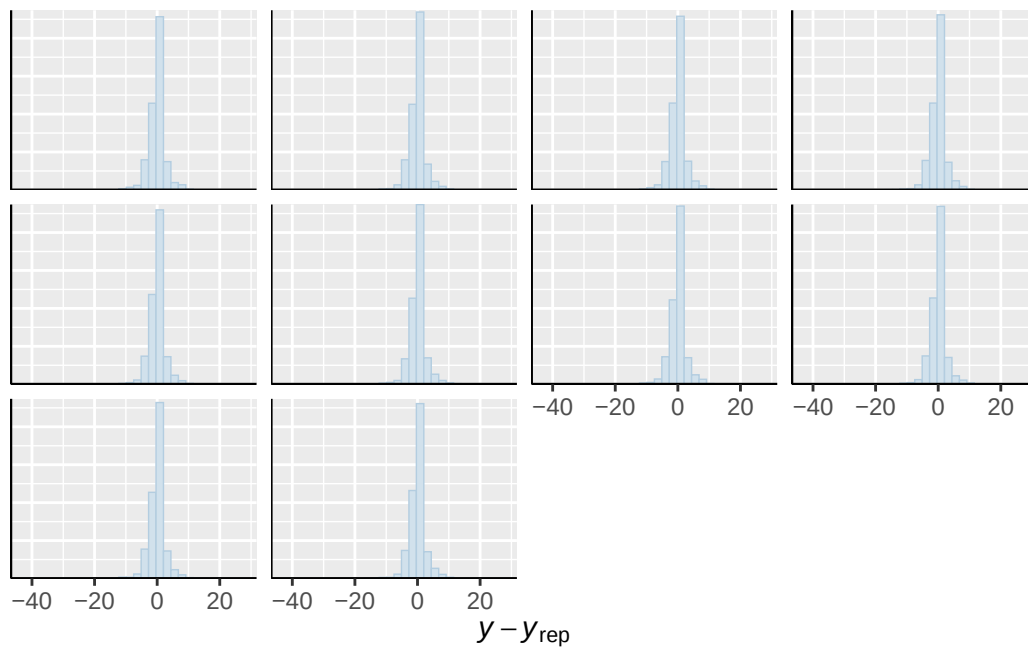




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

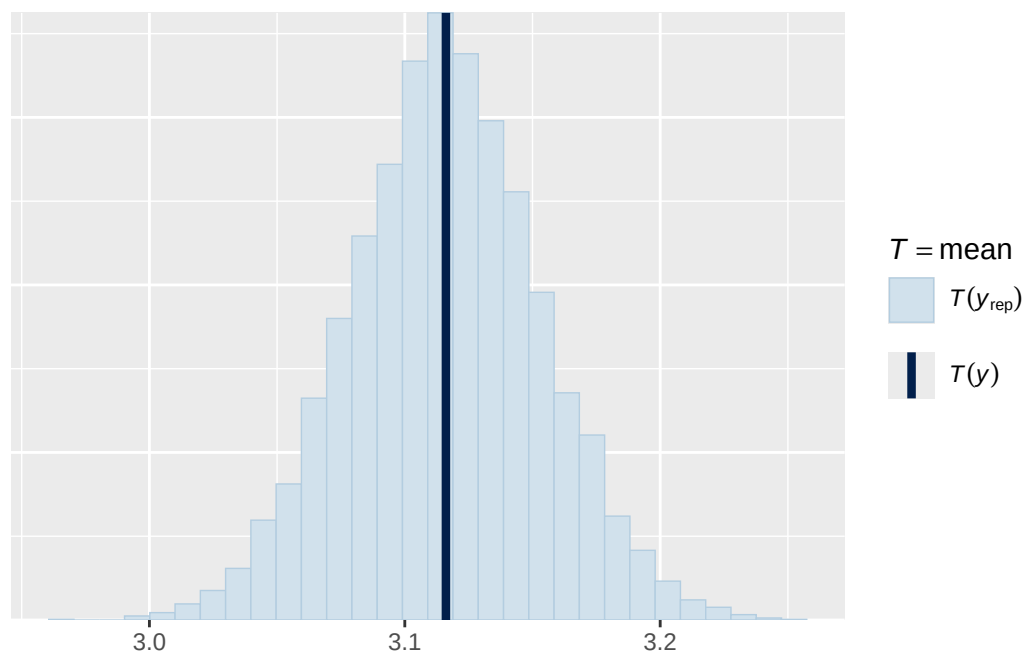


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

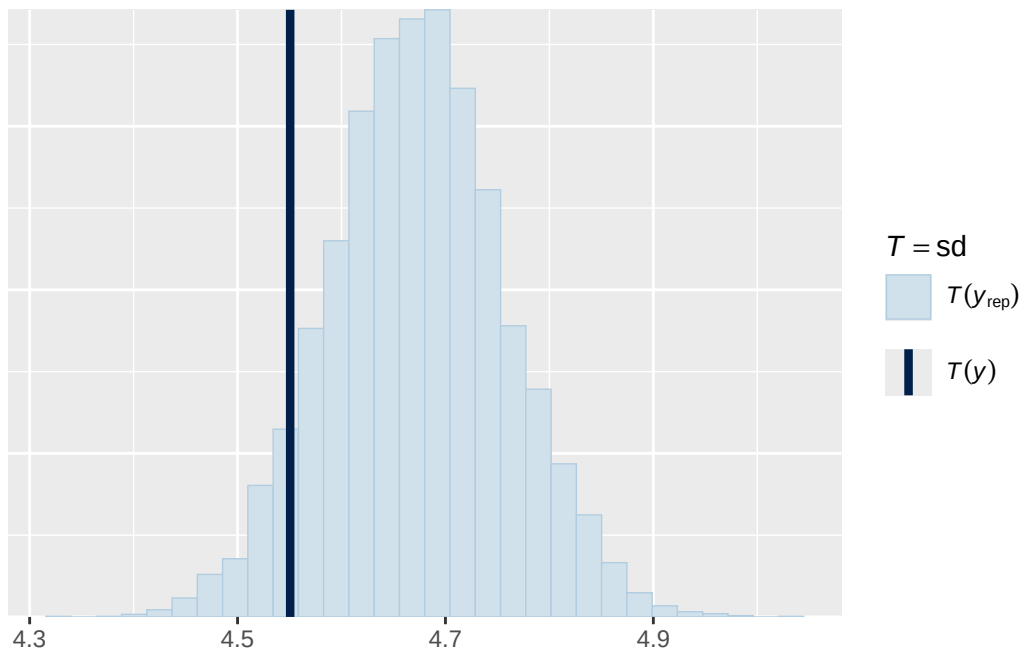


Using all posterior draws for ppc type 'stat' by default.

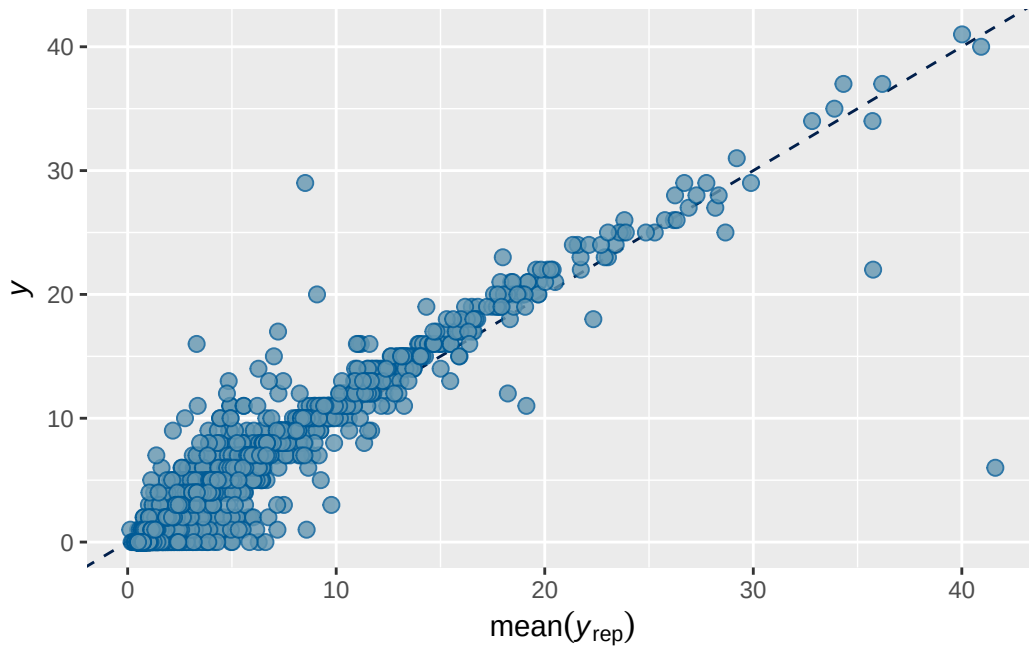
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 3B longitudinal ADHD present output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8238293 0.005235983 0.8129815 0.8334161

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2 * adhd_status_pres + age_wave_2_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    1.034    0.021   0.994   1.074 1.003    1919    3560

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.070    0.038   0.006   0.149 1.003     603    1245

```

Regression Coefficients:

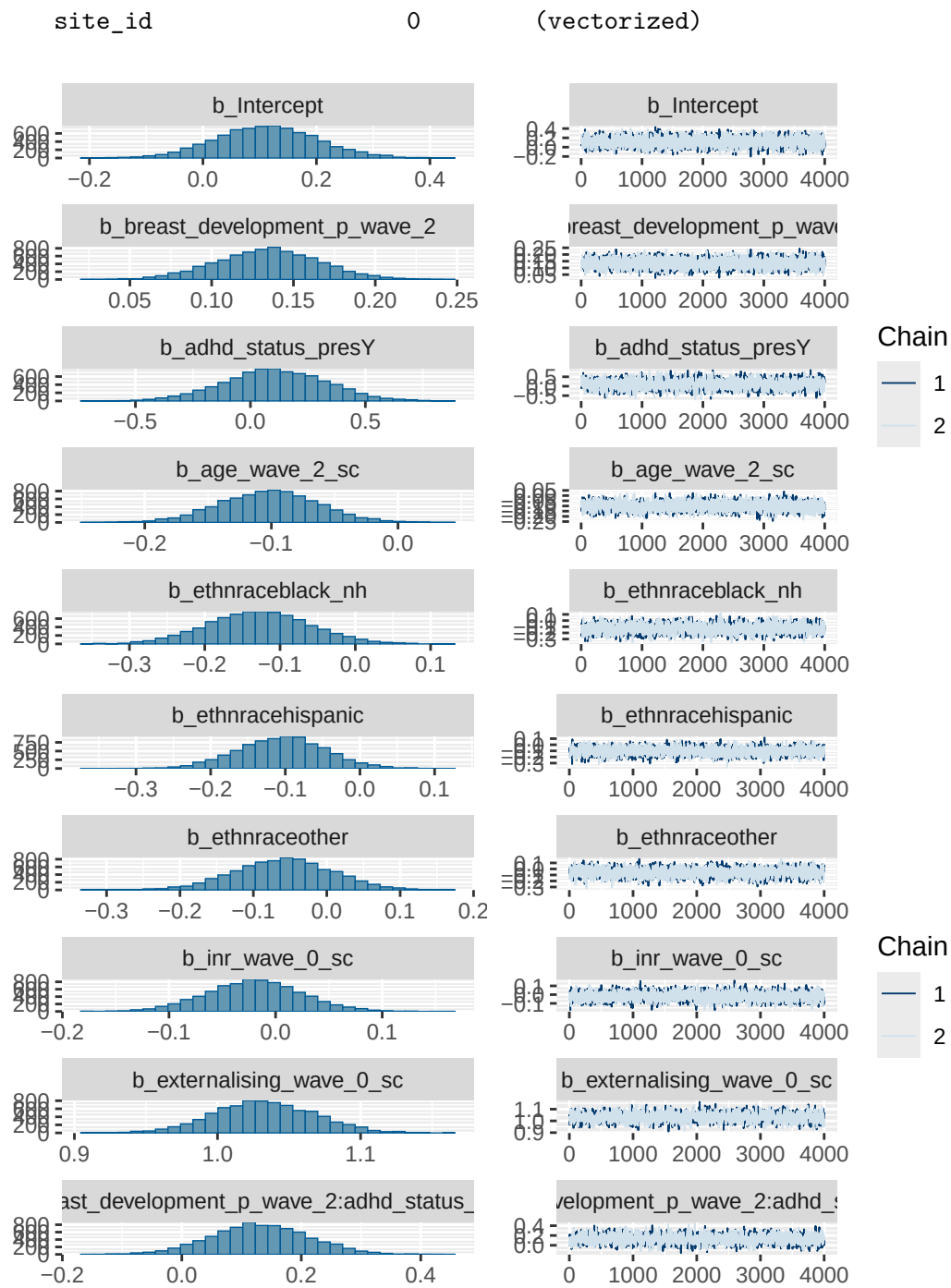
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.112	0.089	-0.060				
breast_development_p_wave_2	0.134	0.031	0.073				
adhd_status_presY	0.107	0.222	-0.331				
age_wave_2_sc	-0.102	0.039	-0.178				
ethnraceblack_nh	-0.128	0.068	-0.260				
ethnracehispanic	-0.105	0.058	-0.219				
ethnraceother	-0.059	0.067	-0.191				
inr_wave_0_sc	-0.020	0.044	-0.107				
externalising_wave_0_sc	1.033	0.035	0.965				
breast_development_p_wave_2:adhd_status_presY	0.131	0.081	-0.028				
				u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.289	1.000	2241	3971			
breast_development_p_wave_2	0.196	1.001	2257	4052			
adhd_status_presY	0.537	1.001	1455	2891			
age_wave_2_sc	-0.025	1.001	1864	3311			
ethnraceblack_nh	0.005	1.001	1444	2701			
ethnracehispanic	0.010	1.002	1820	3539			
ethnraceother	0.069	1.001	1609	3337			
inr_wave_0_sc	0.068	1.001	1776	3324			

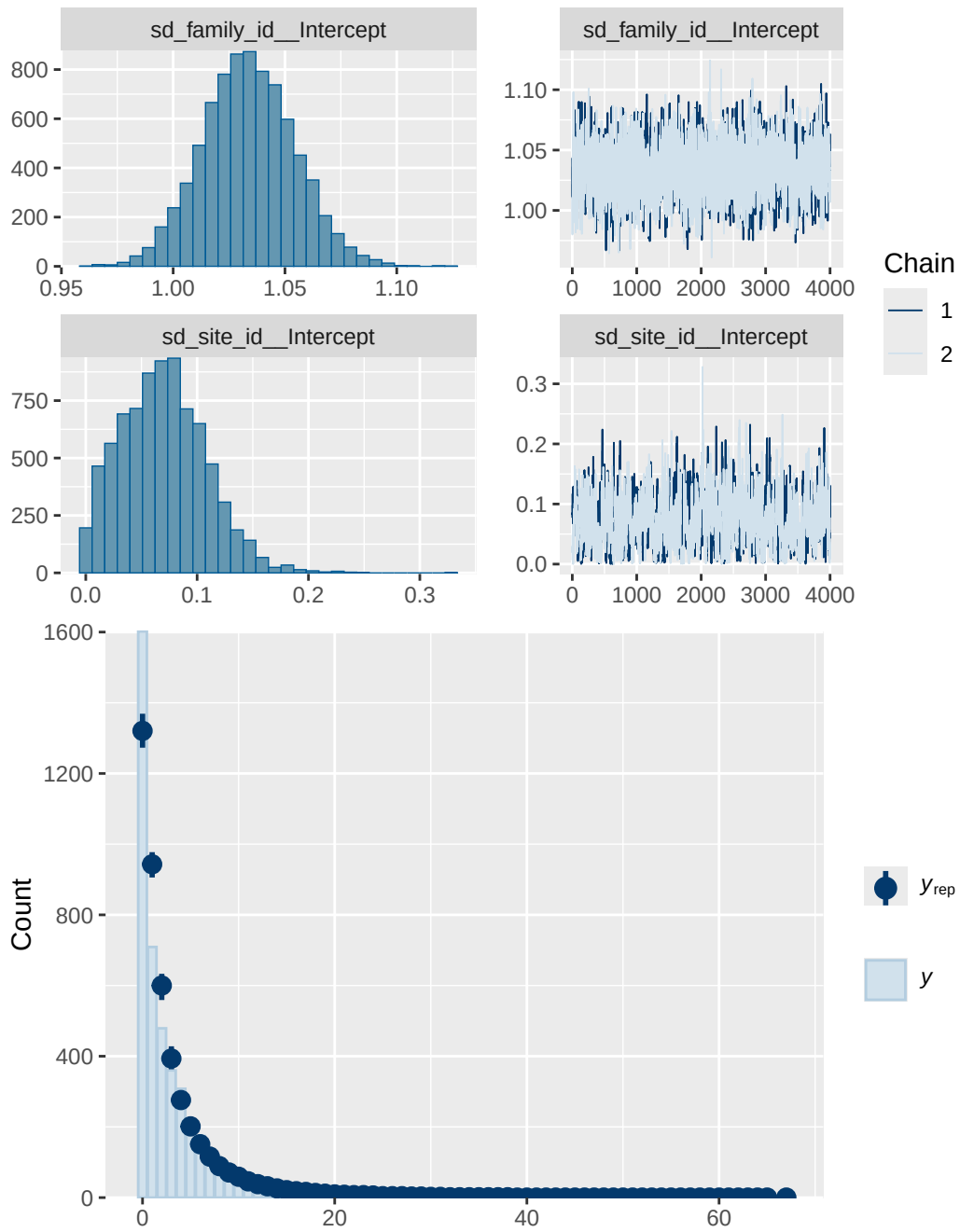
externalising_wave_0_sc	1.100	1.003	1240	2642
breast_development_p_wave_2:adhd_status_presY	0.289	1.001	1403	2937

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

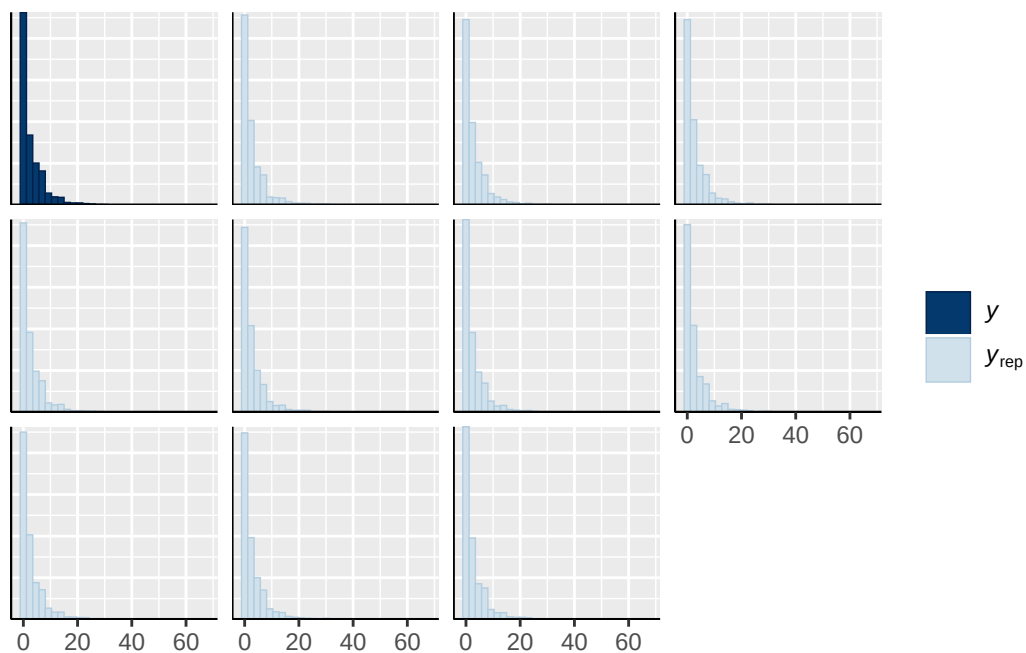
Model 3B longitudinal ADHD present diagnostics

	prior	class	coef
	normal(0, 1)	b	
	normal(0, 1)	b	adhd_status_presY
	normal(0, 1)	b	age_wave_2_sc
	normal(0, 1)	b	breast_development_p_wave_2
	normal(0, 1)	b	breast_development_p_wave_2:adhd_status_presY
	normal(0, 1)	b	ethnraceblack_nh
	normal(0, 1)	b	ethnracehispanic
	normal(0, 1)	b	ethnraceother
	normal(0, 1)	b	externalising_wave_0_sc
	normal(0, 1)	b	inr_wave_0_sc
student_t(3, 0, 2.9)	Intercept		
student_t(3, 0, 2.9)	sd		
student_t(3, 0, 2.9)	sd		
student_t(3, 0, 2.9)	sd		Intercept
student_t(3, 0, 2.9)	sd		
student_t(3, 0, 2.9)	sd		Intercept
group resp dpar nlpar lb ub tag			source
			user
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			(vectorized)
			default
		0	default
family_id		0	(vectorized)
family_id		0	(vectorized)
site_id		0	(vectorized)

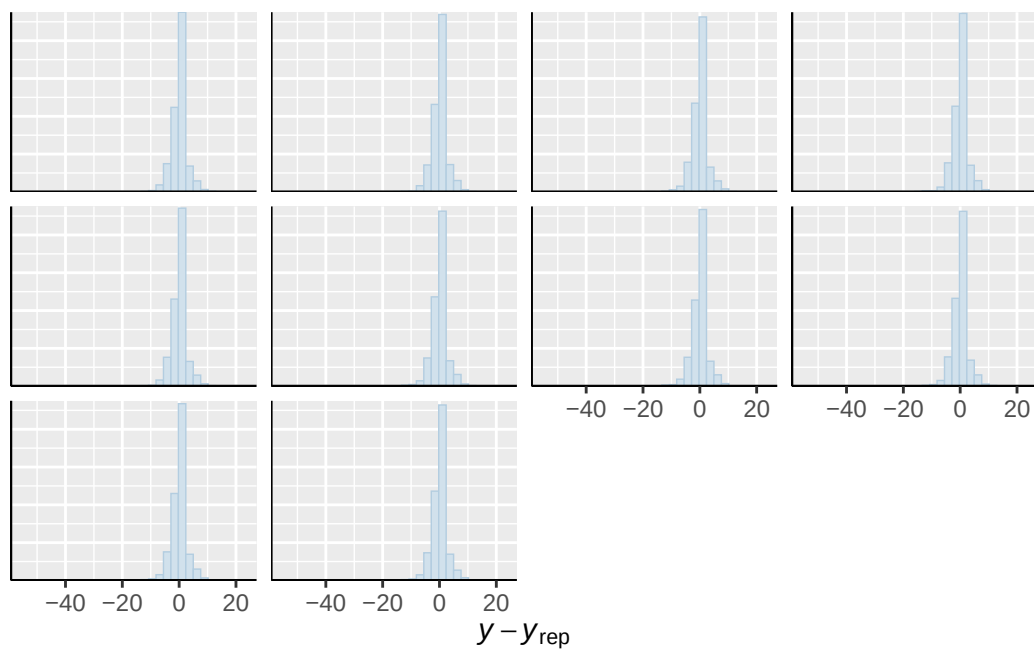




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

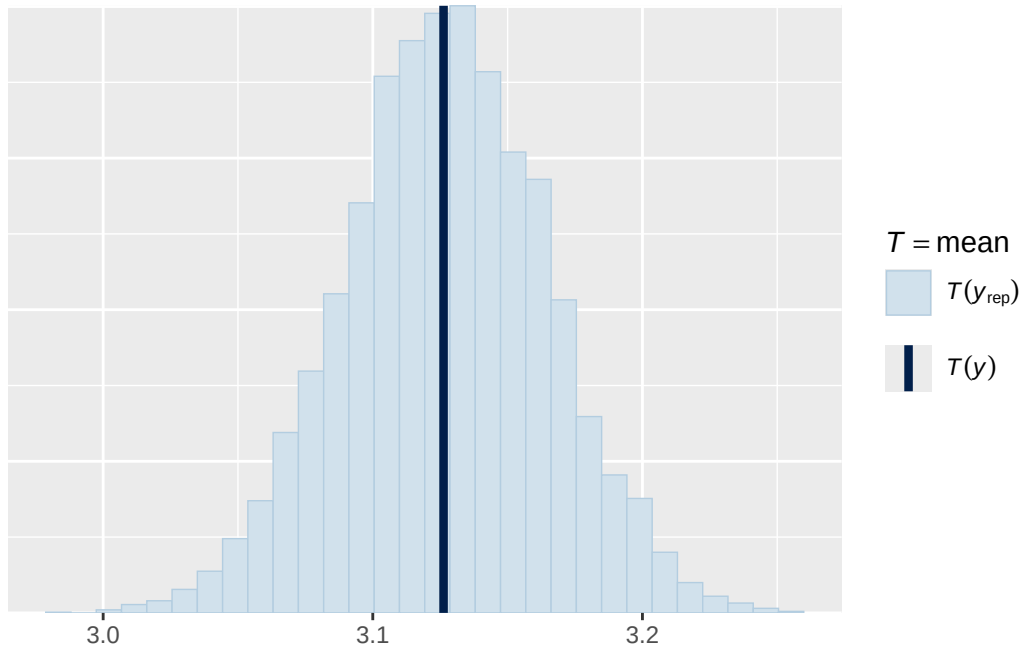


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

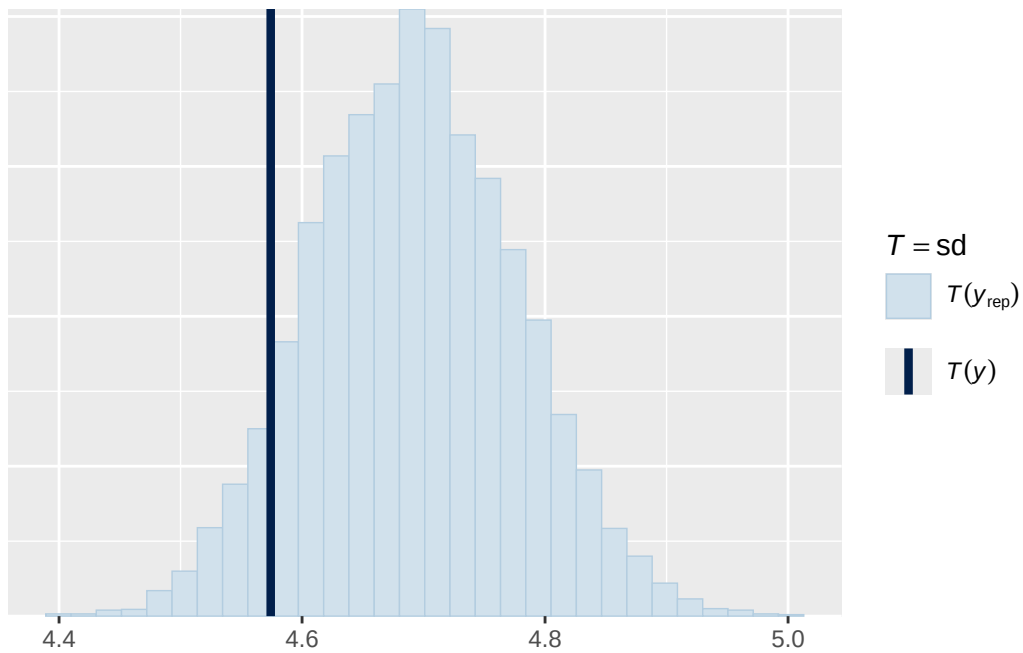


Using all posterior draws for ppc type 'stat' by default.

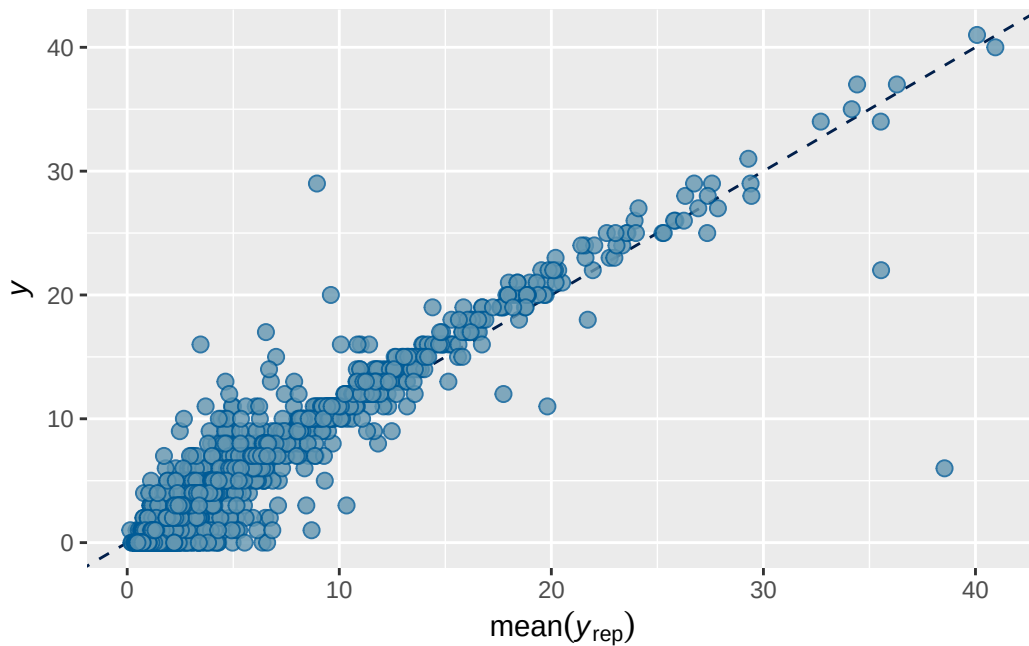
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1 longitudinal observed output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.739703	0.007972974	0.7235204	0.7545925

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: obs_df_wide (Number of observations: 4043)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 3513)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.878	0.022	0.836	0.920	1.001	2291	4325

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.076	0.039	0.007	0.157	1.004	577	872

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	-0.029	0.162	-0.347	0.289	1.000	1338
menarche_status_p_wave_2Y	0.113	0.044	0.027	0.197	1.000	2368
age_wave_2_sc	-0.120	0.043	-0.204	-0.036	1.001	2169
ethnraceblack_nh	0.107	0.171	-0.236	0.442	1.000	1313
ethnracehispanic	0.139	0.167	-0.193	0.464	1.001	1320
ethnraceother	0.210	0.169	-0.124	0.544	1.000	1386
ethnracewhite_nh	0.181	0.161	-0.137	0.495	1.000	1305
inr_wave_0_sc	0.091	0.045	0.002	0.178	1.000	2203
adhd_traits_wave_0_sc	1.347	0.035	1.278	1.418	1.002	1498
	Tail_ESS					
Intercept	2348					
menarche_status_p_wave_2Y	3964					
age_wave_2_sc	3917					
ethnraceblack_nh	2618					
ethnracehispanic	2387					
ethnraceother	2341					
ethnracewhite_nh	2164					
inr_wave_0_sc	3842					
adhd_traits_wave_0_sc	2589					

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

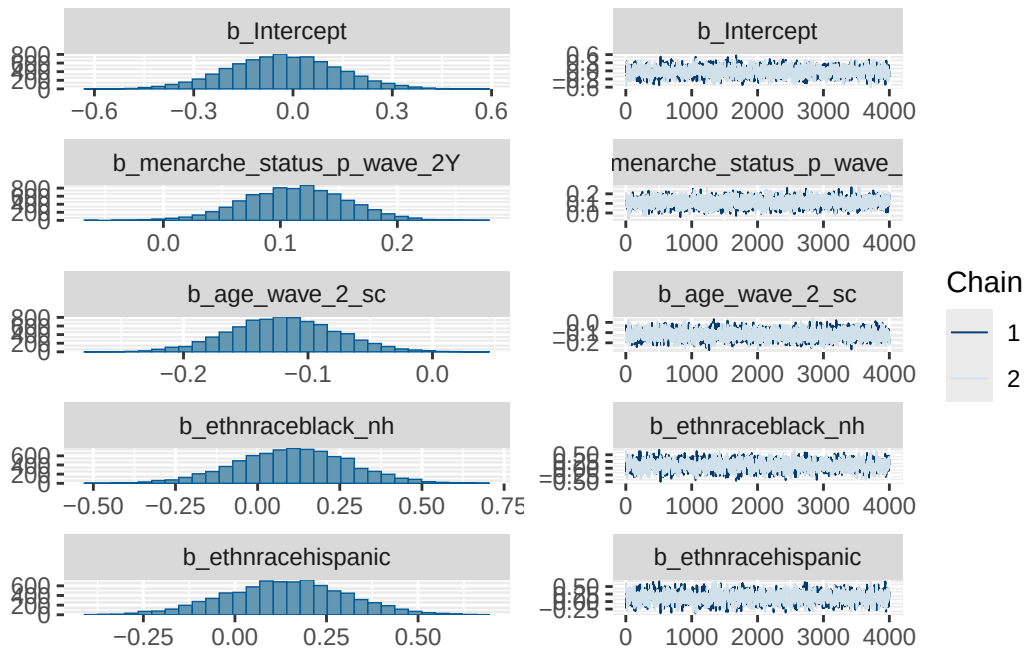
Model 1 longitudinal observed diagnostics

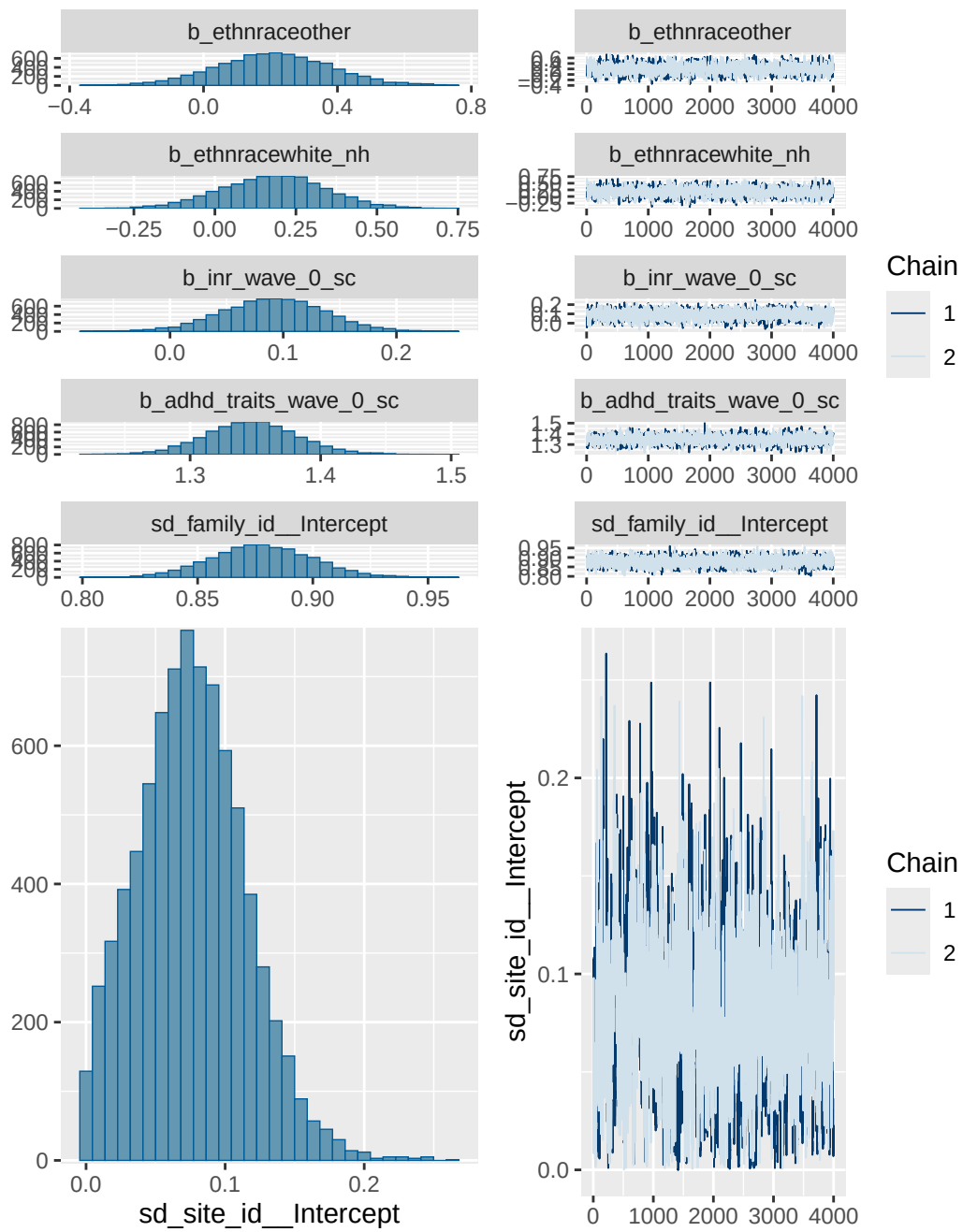
prior	class	coef	group	resp	dpar
normal(0, 1)	b				
normal(0, 1)	b	adhd_traits_wave_0_sc			
normal(0, 1)	b	age_wave_2_sc			
normal(0, 1)	b	ethnraceblack_nh			
normal(0, 1)	b	ethnracehispanic			
normal(0, 1)	b	ethnraceother			
normal(0, 1)	b	ethnracewhite_nh			
normal(0, 1)	b	inr_wave_0_sc			
normal(0, 1)	b	menarche_status_p_wave_2Y			
student_t(3, 0, 2.9)	Intercept				
student_t(3, 0, 2.9)	sd				
student_t(3, 0, 2.9)	sd		family_id		

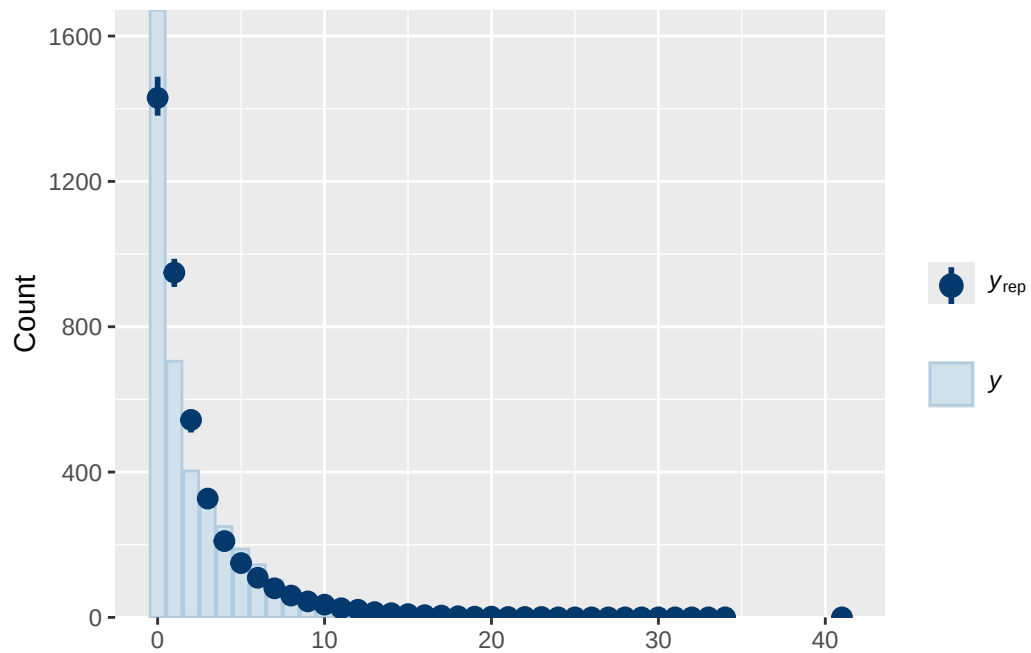

```

student_t(3, 0, 2.9)      sd      Intercept family_id
student_t(3, 0, 2.9)      sd      site_id
student_t(3, 0, 2.9)      sd      Intercept site_id
nlpar lb ub tag          source
                        user
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        default
0                        default
0                        (vectorized)
0                        (vectorized)
0                        (vectorized)
0                        (vectorized)

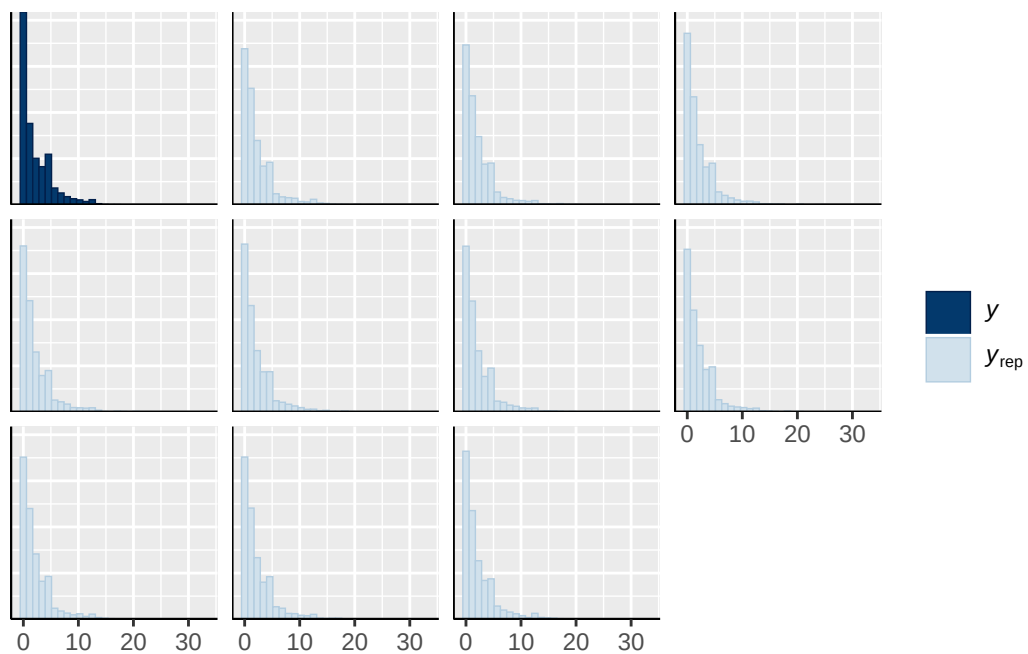
```



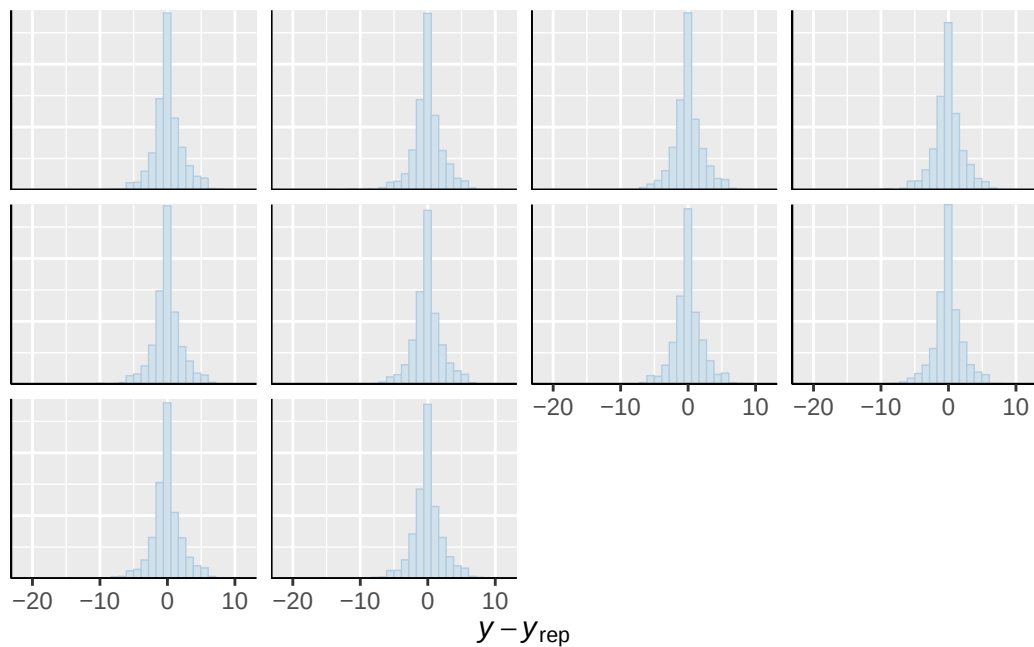




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

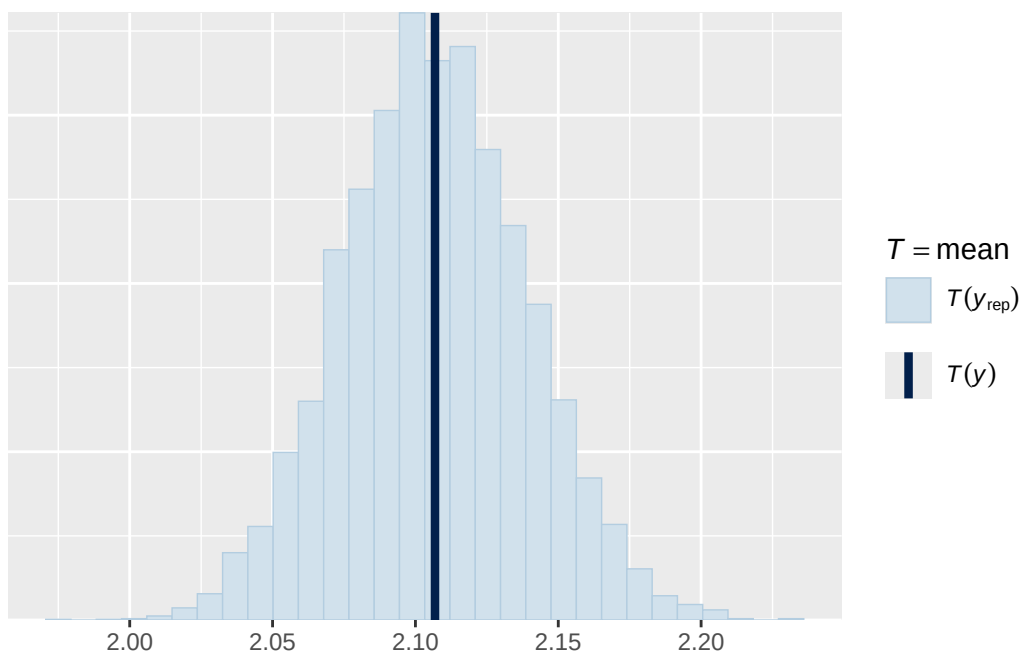


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

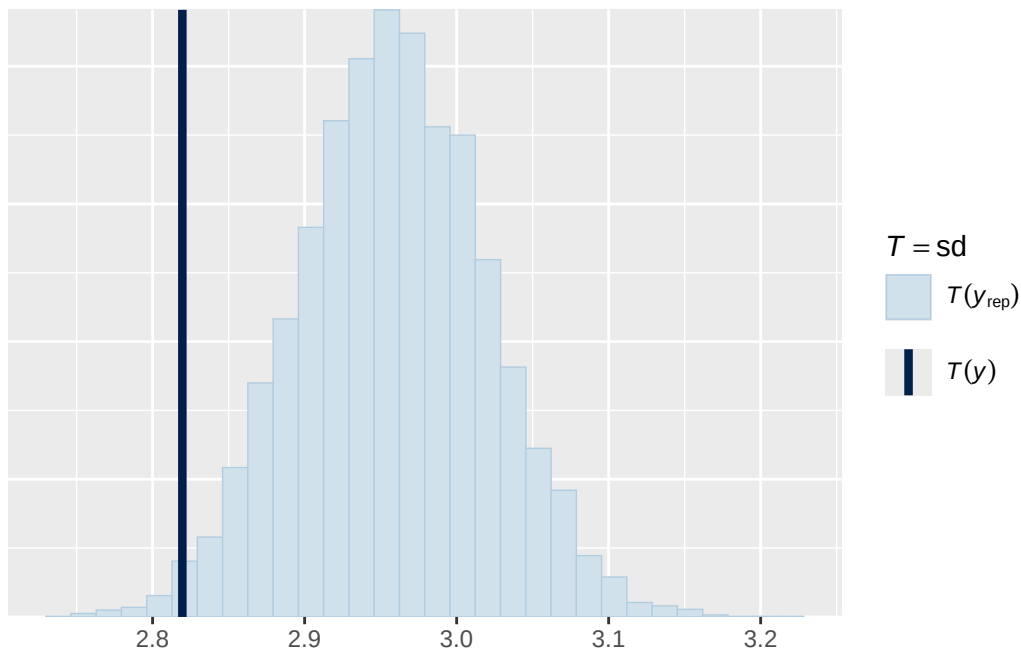


Using all posterior draws for ppc type 'stat' by default.

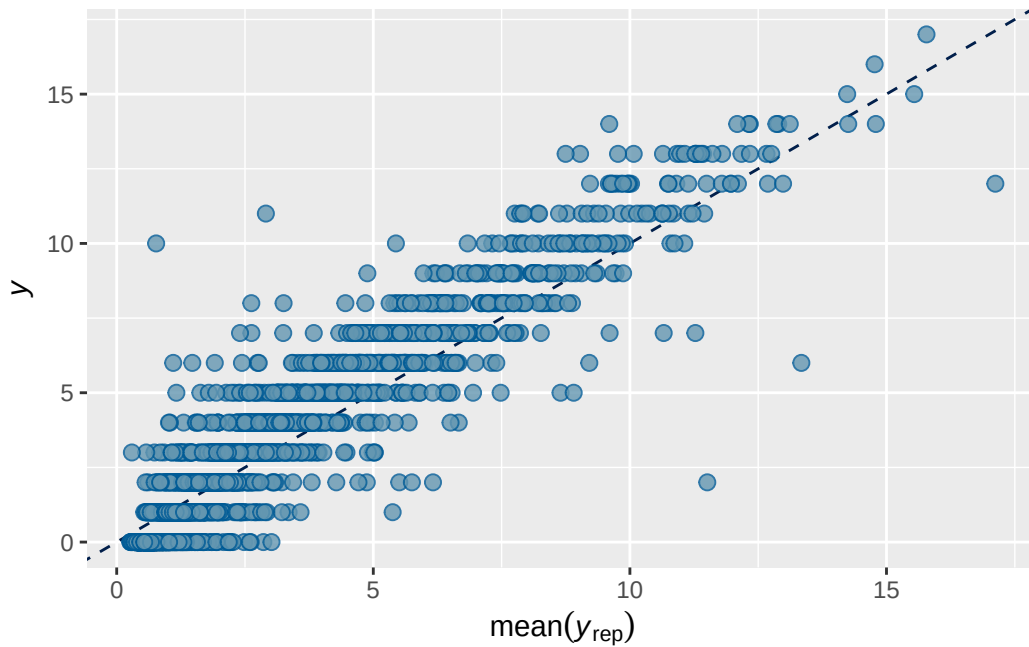
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1B longitudinal observed output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7393405 0.007815485 0.7235745 0.7541467

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: obs_df_wide (Number of observations: 4065)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3531)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.877    0.022   0.835   0.921 1.000   2244   3432

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.083    0.037   0.011   0.158 1.001    839   1246

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat
Intercept          0.017    0.158  -0.295   0.325 1.000
breast_development_p_wave_2_sc  0.135    0.042   0.051   0.217 1.000
age_wave_2_sc      -0.106    0.041  -0.186  -0.027 1.000
ethnraceblack_nh    0.102    0.168  -0.234   0.434 1.000
ethnracehispanic    0.154    0.164  -0.164   0.479 1.000
ethnraceother       0.204    0.168  -0.122   0.534 1.000
ethnracewhite_nh    0.184    0.158  -0.128   0.496 1.000
inr_wave_0_sc       0.079    0.044  -0.008   0.163 1.000
adhd_traits_wave_0_sc 1.342    0.035   1.273   1.411 1.001

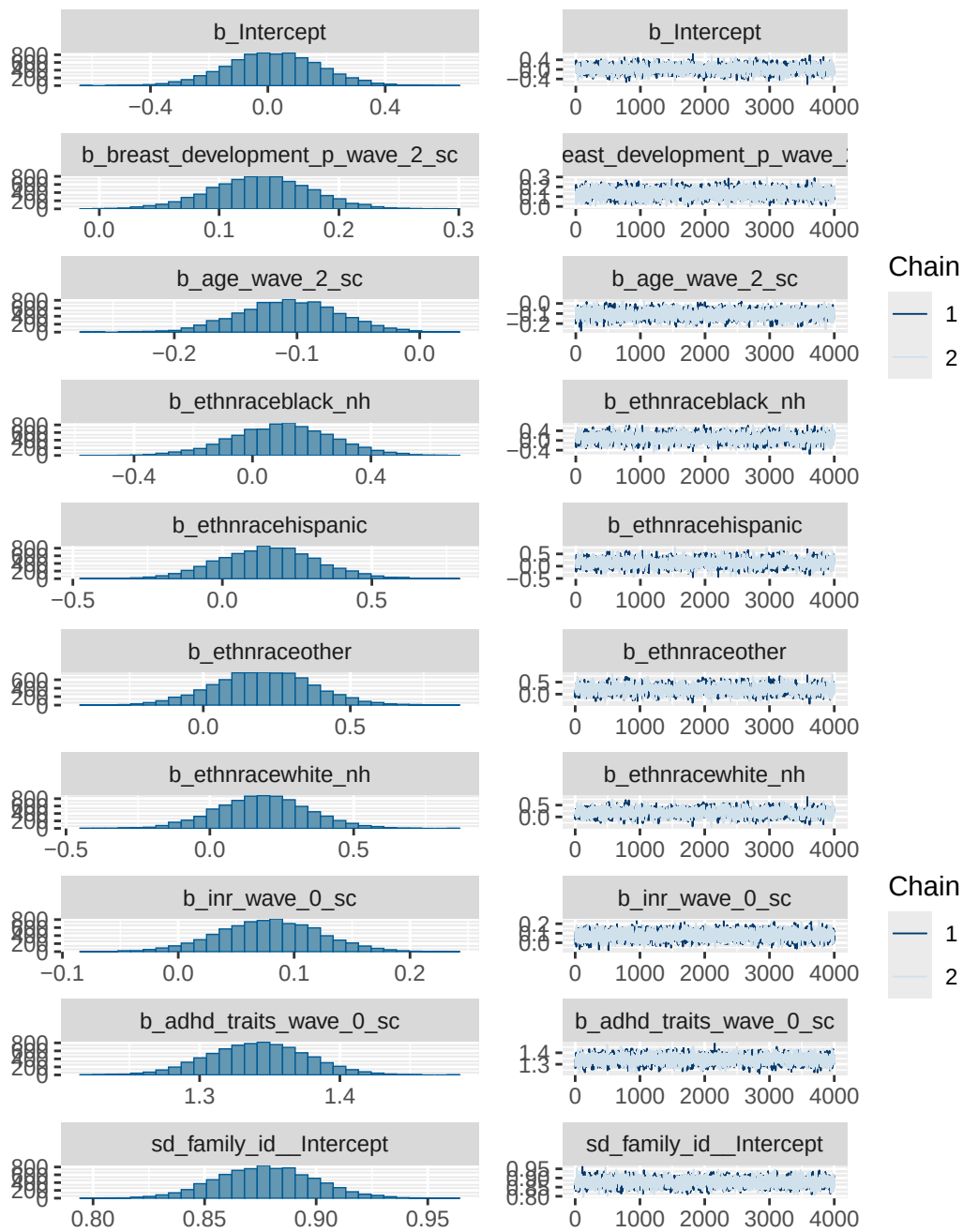
      Bulk_ESS Tail_ESS
Intercept          1379   3095
breast_development_p_wave_2_sc  2735   4430
age_wave_2_sc      2358   4013
ethnraceblack_nh    1320   2801
ethnracehispanic    1386   3053
ethnraceother       1404   2800
ethnracewhite_nh    1360   3056
inr_wave_0_sc       2669   4402
adhd_traits_wave_0_sc 1976   3678

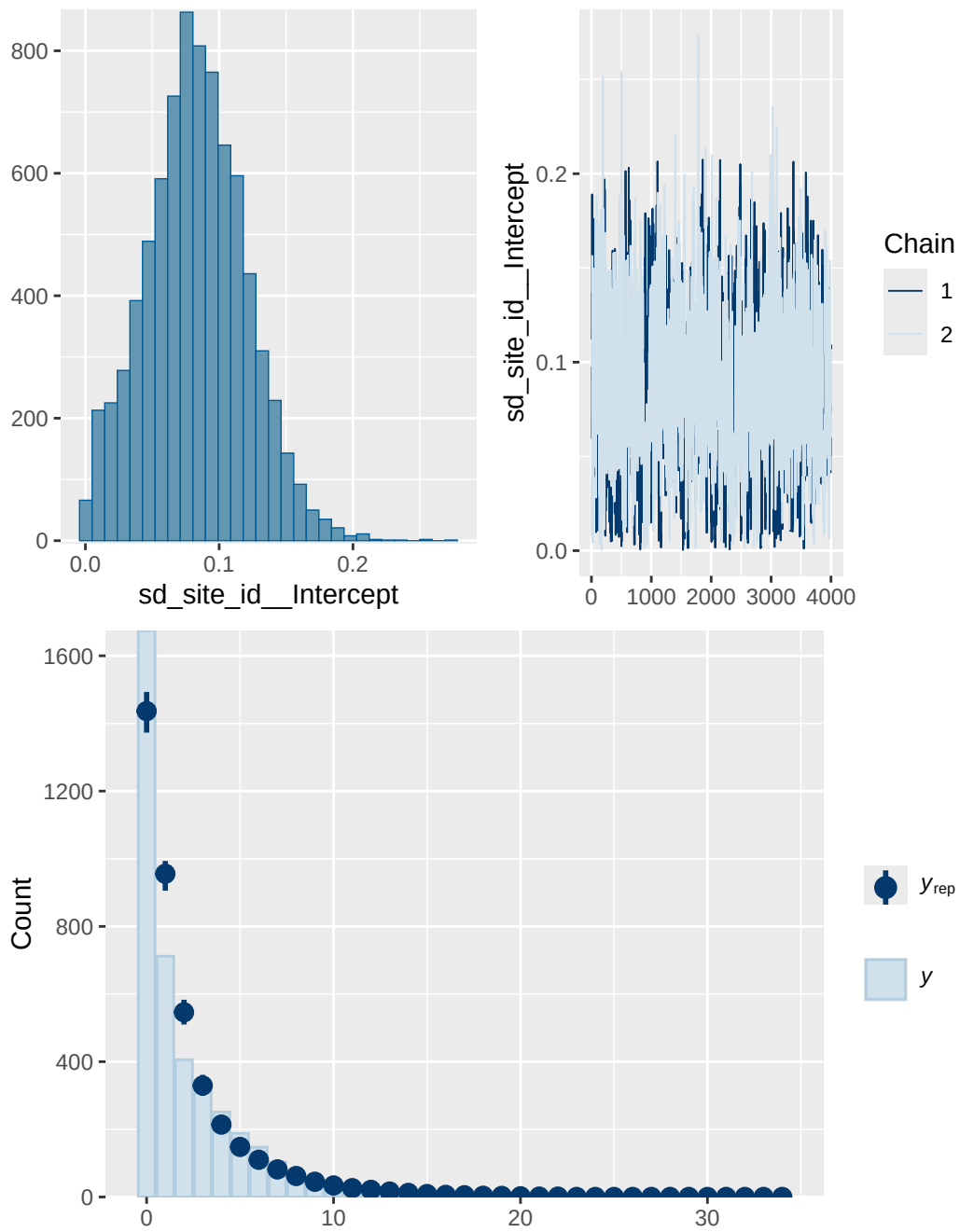
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

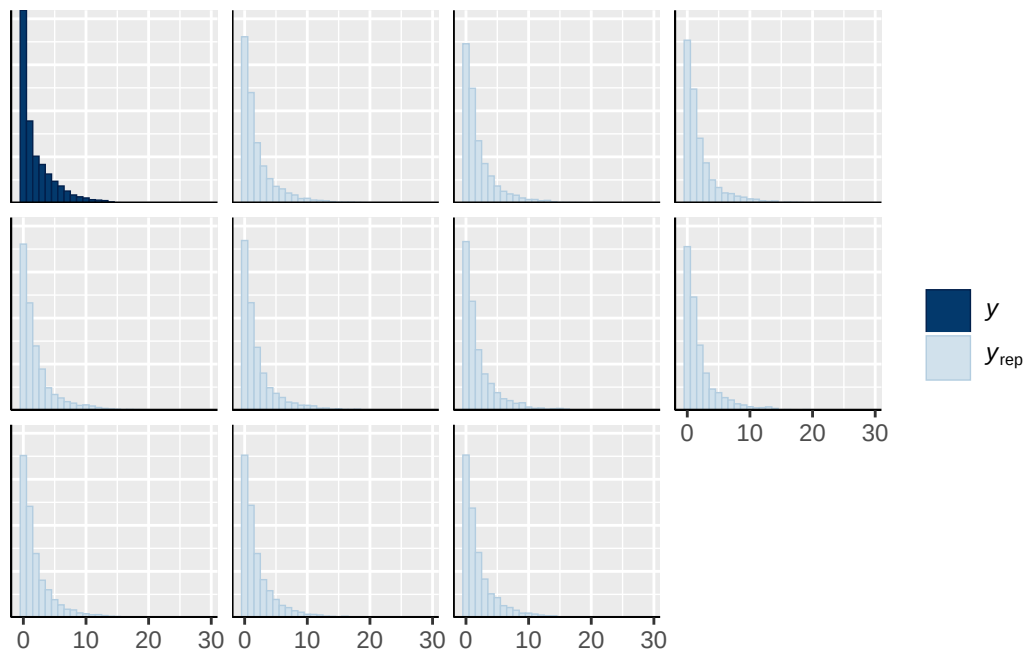
Model 1B longitudinal observed diagnostics

	prior	class	coef	group	resp
	normal(0, 1)	b			
	normal(0, 1)	b	adhd_traits_wave_0_sc		
	normal(0, 1)	b	age_wave_2_sc		
	normal(0, 1)	b	breast_development_p_wave_2_sc		
	normal(0, 1)	b	ethnraceblack_nh		
	normal(0, 1)	b	ethnracehispanic		
	normal(0, 1)	b	ethnraceother		
	normal(0, 1)	b	ethnracewhite_nh		
	normal(0, 1)	b	inr_wave_0_sc		
student_t(3, 0, 2.7)		Intercept			
student_t(3, 0, 2.7)		sd			
student_t(3, 0, 2.7)		sd		family_id	
student_t(3, 0, 2.7)		sd	Intercept	family_id	
student_t(3, 0, 2.7)		sd		site_id	
student_t(3, 0, 2.7)		sd	Intercept	site_id	
dpar nlpar lb ub tag		source			
		user			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		default			
0		default			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			

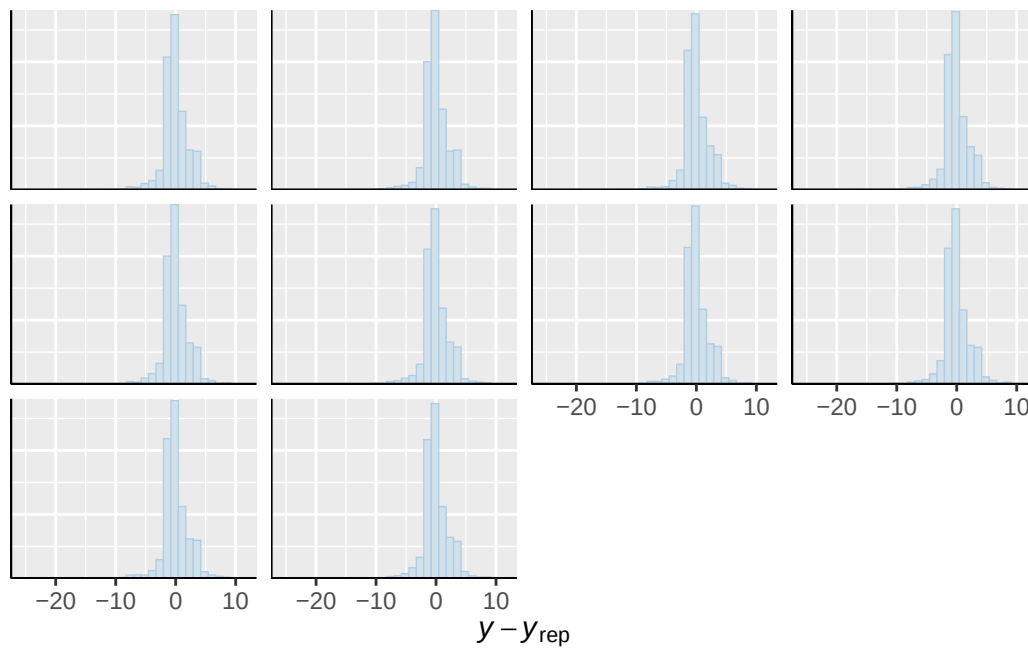




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

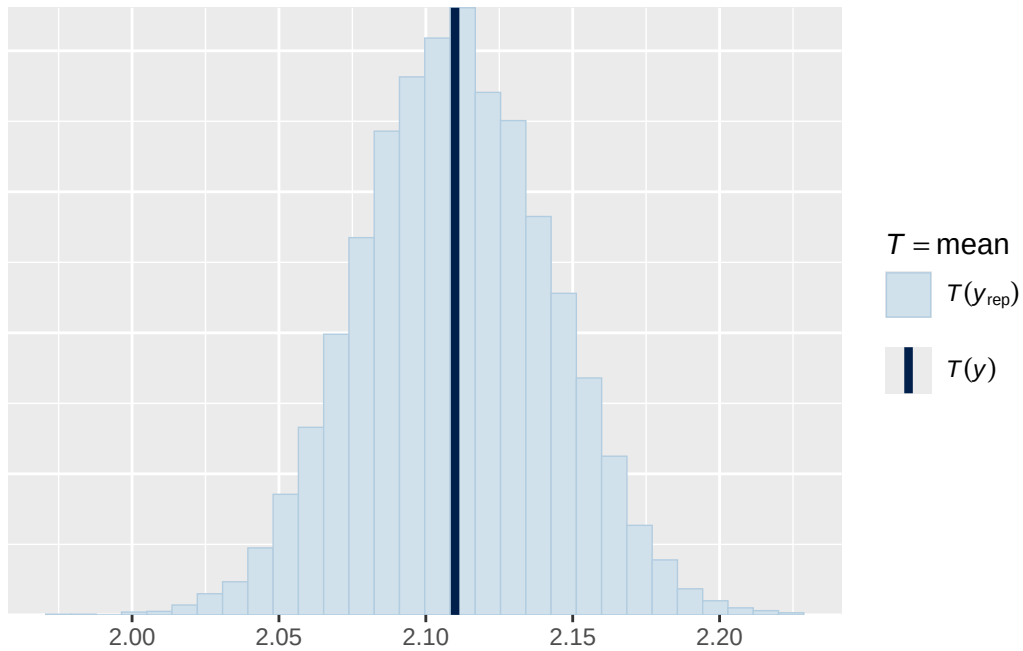


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

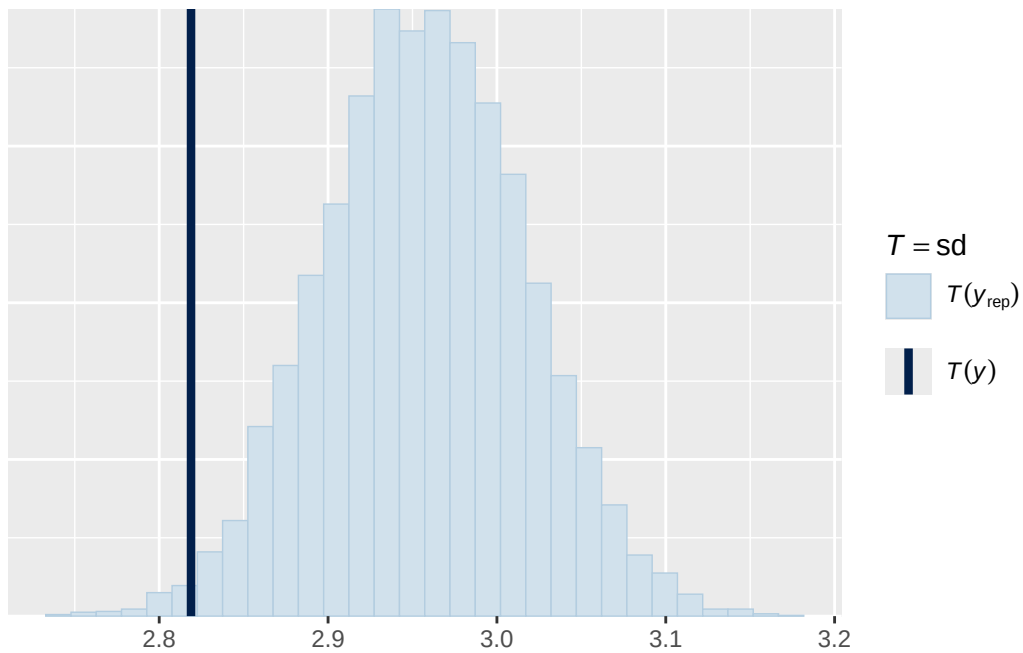


Using all posterior draws for ppc type 'stat' by default.

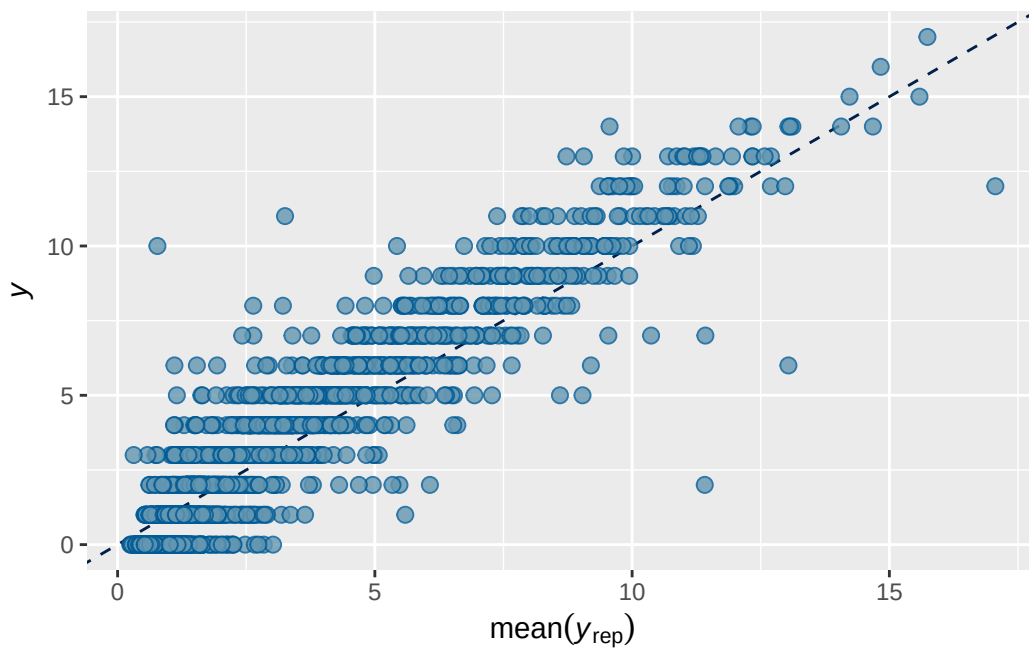
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 5B longitudinal observed output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.8261396	0.005376451	0.8150095	0.836055

Family: poisson
 Links: mu = log
 Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age.
 Data: obs_df_wide (Number of observations: 4064)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 3530)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.956	0.020	0.916	0.995	1.000	1600	2516

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.086	0.039	0.012	0.169	1.002	439	865

Regression Coefficients:

	Estimate	Est.Error	
Intercept	0.188	0.165	
breast_development_p_wave_2_sc	0.207	0.040	
adhd_traits_wave_2_sc	0.596	0.036	
age_wave_2_sc	-0.104	0.039	
ethnraceblack_nh	0.206	0.175	
ethnracehispanic	0.256	0.171	
ethnraceother	0.271	0.173	
ethnracewhite_nh	0.341	0.165	
inr_wave_0_sc	0.023	0.046	
externalising_wave_0_sc	0.992	0.036	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.144	0.069	
	1-95% CI	u-95% CI	Rhat
Intercept	-0.133	0.512	1.001
breast_development_p_wave_2_sc	0.131	0.284	1.000
adhd_traits_wave_2_sc	0.528	0.671	1.002
age_wave_2_sc	-0.182	-0.028	1.001
ethnraceblack_nh	-0.143	0.546	1.001
ethnracehispanic	-0.075	0.582	1.001
ethnraceother	-0.059	0.614	1.000
ethnracewhite_nh	0.018	0.663	1.000
inr_wave_0_sc	-0.064	0.114	1.000
externalising_wave_0_sc	0.920	1.062	1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.009	0.280	1.000
	Bulk_ESS	Tail_ESS	
Intercept	1994	2260	
breast_development_p_wave_2_sc	2797	3168	
adhd_traits_wave_2_sc	2173	2524	
age_wave_2_sc	2560	2998	
ethnraceblack_nh	1813	2078	
ethnracehispanic	1981	2417	
ethnraceother	1907	1991	
ethnracewhite_nh	1985	2262	
inr_wave_0_sc	2291	2922	
externalising_wave_0_sc	1832	2599	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	2069	2820	

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5B longitudinal observed diagnostics

```

prior      class
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd

coef      group resp dpar nlpar

adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
ethnraceblack_nh
ethnracehispanic
ethnraceother
ethnracewhite_nh
externalising_wave_0_sc
inr_wave_0_sc

family_id
Intercept family_id
site_id
Intercept site_id

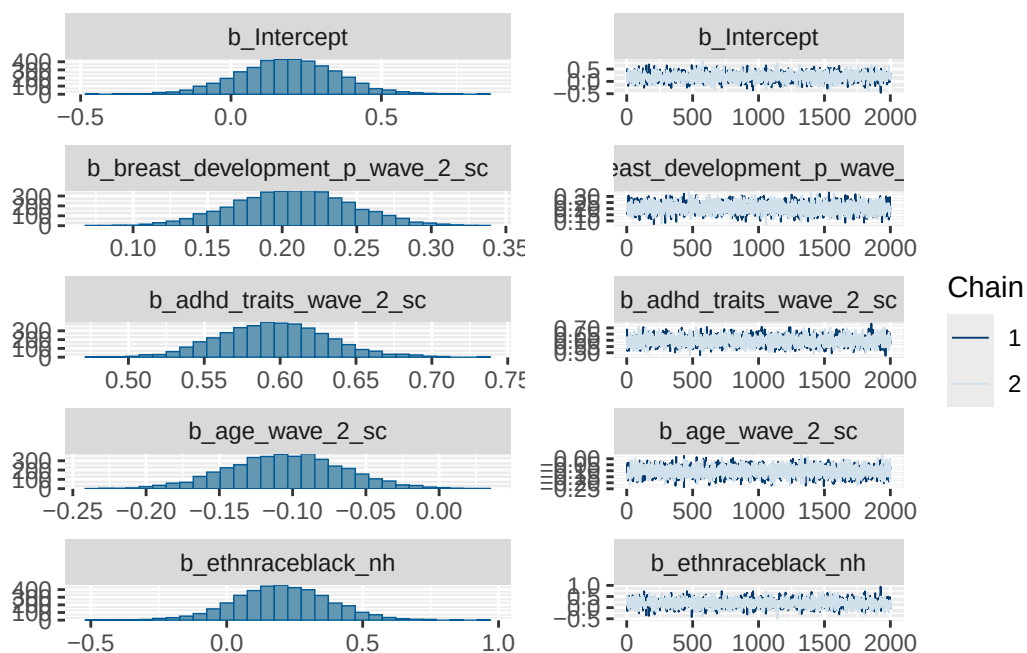
lb ub tag      source
user
(vectorized)
(vectorized)

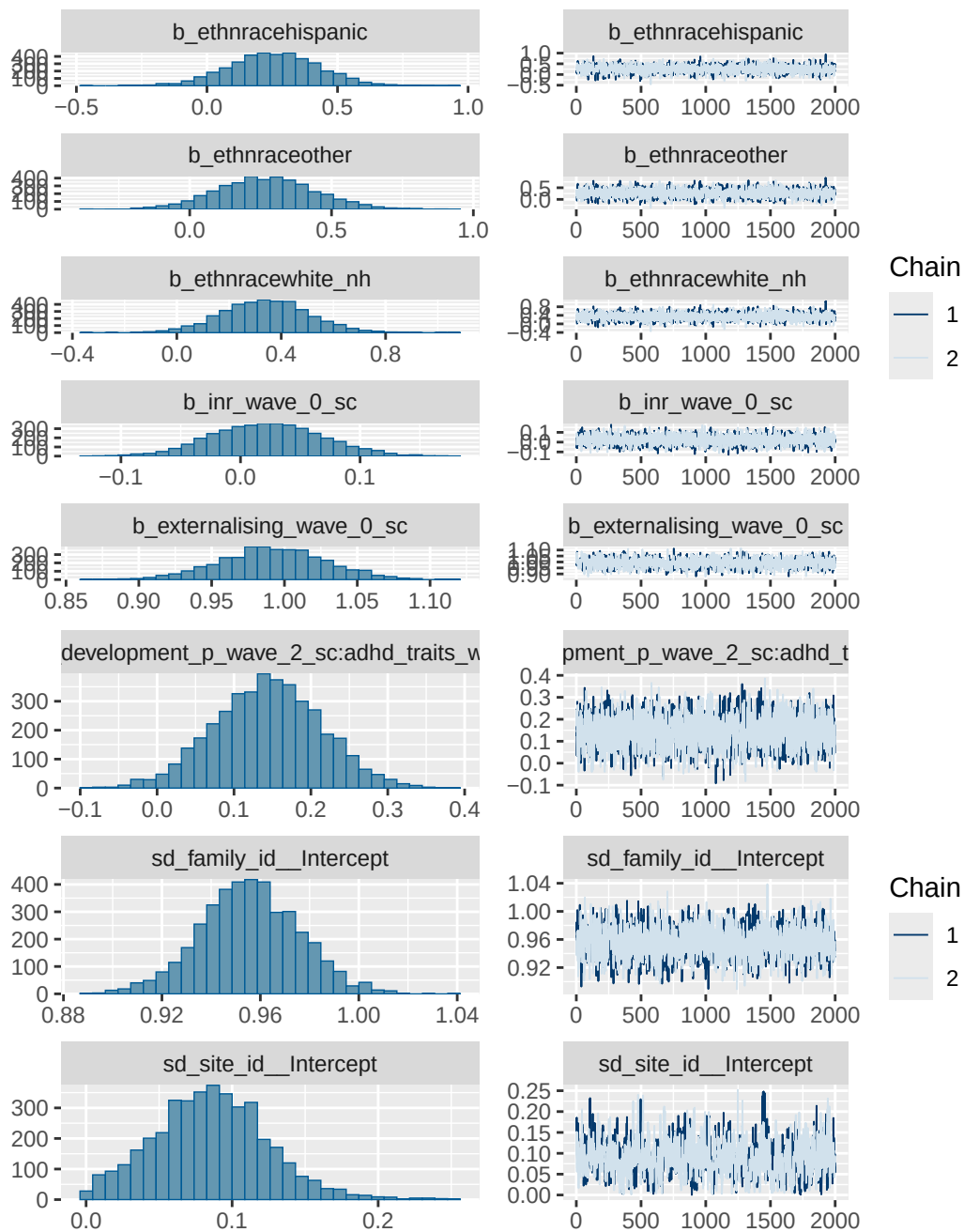
```

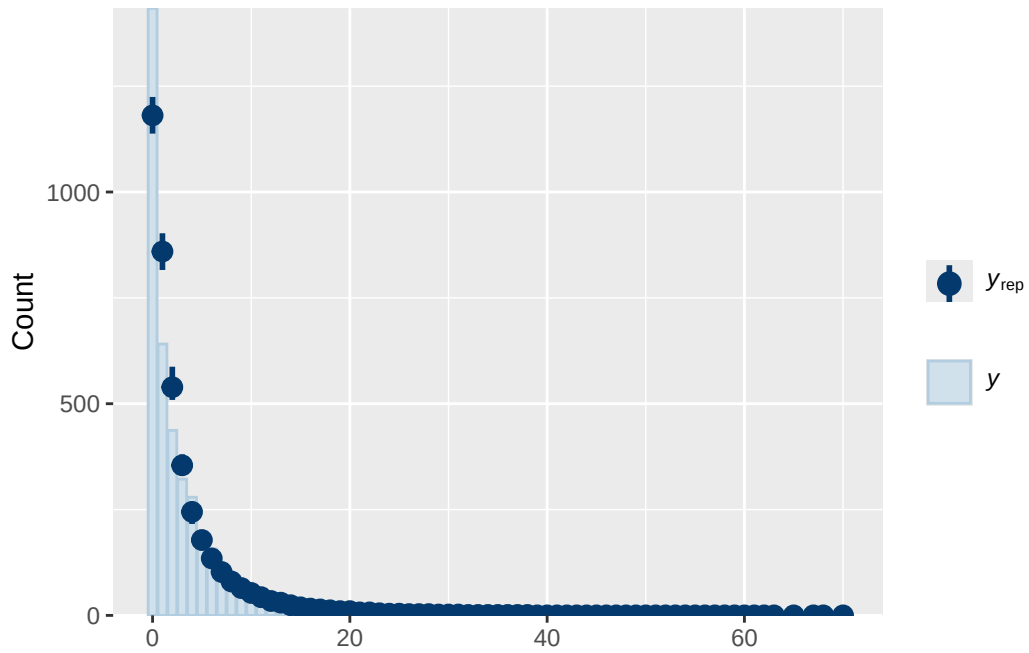
```

(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

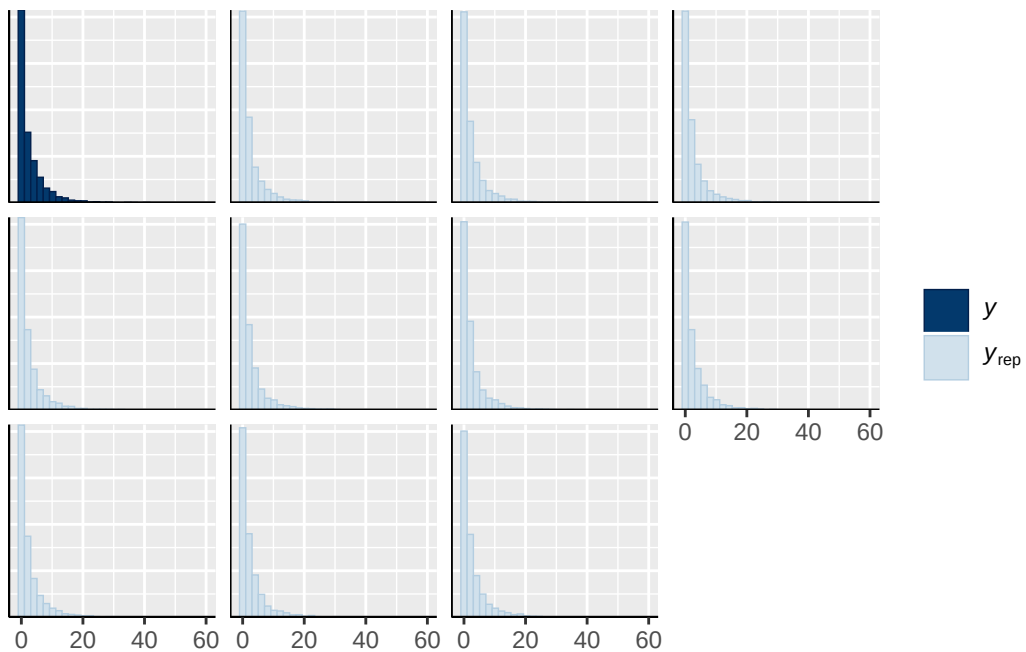
```



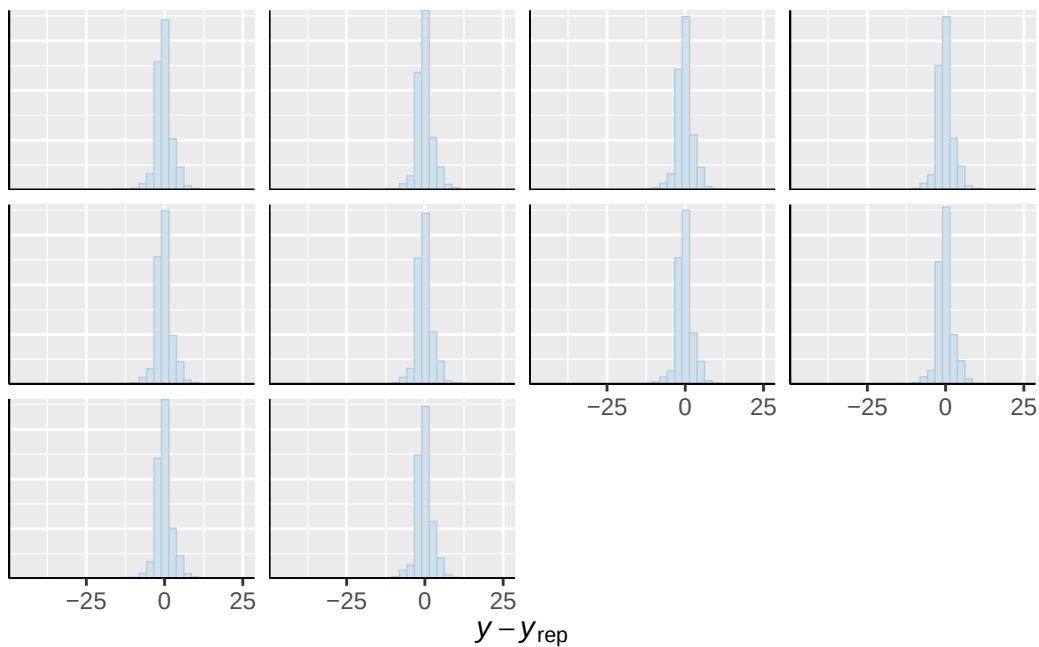




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

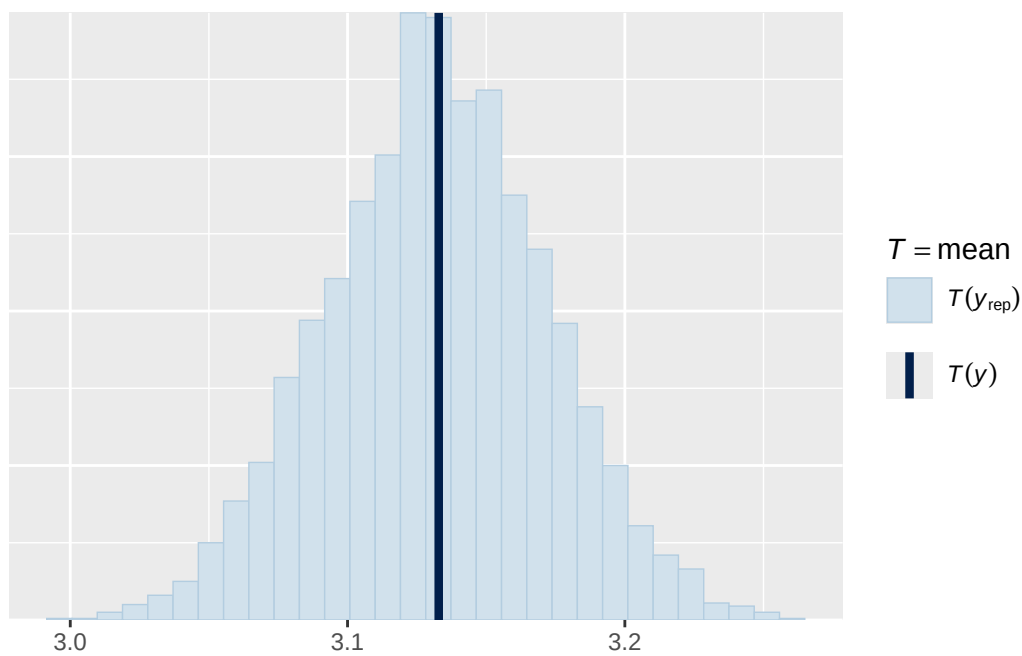


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

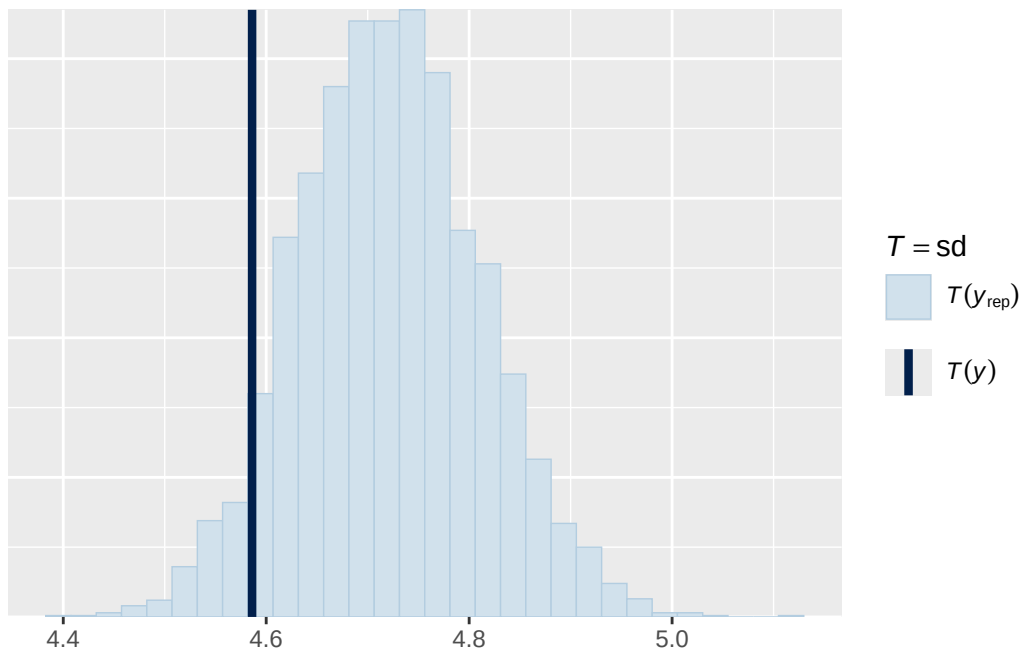


Using all posterior draws for ppc type 'stat' by default.

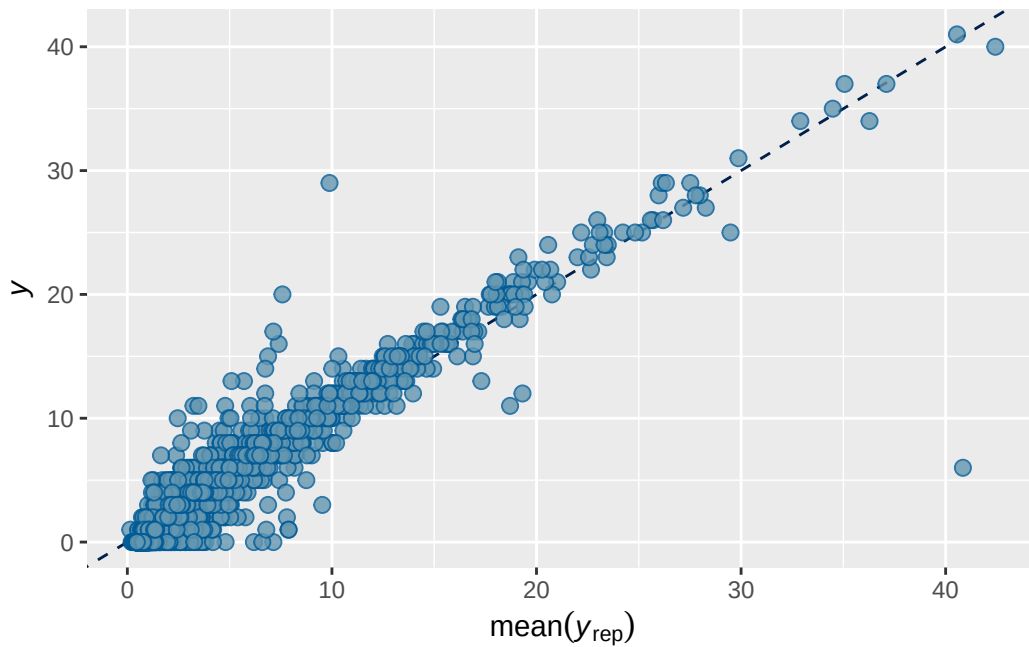
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1 longitudinal BMI output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7360488 0.007489116 0.7209785 0.7505381

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3915)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.885    0.021   0.844   0.927 1.002    1412    2182

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.089    0.036   0.019   0.164 1.001     555     595

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.147    0.040   0.067   0.224 1.001    1879
menarche_status_p_wave_2Y  0.099    0.042   0.017   0.181 1.001    1900
age_wave_2_sc       -0.120    0.041  -0.200  -0.040 1.001    1712
ethnraceblack_nh    -0.088    0.064  -0.213   0.041 1.000    1642
ethnracehispanic    -0.047    0.057  -0.157   0.064 1.000    1851
ethnraceother        0.002    0.063  -0.119   0.133 1.001    1562
inr_wave_0_sc        0.109    0.042   0.025   0.191 1.001    1573
adhd_traits_wave_0_sc   1.331    0.033   1.266   1.395 1.006    1239
bmi_wave_0_sc         0.028    0.041  -0.050   0.107 1.002    1879

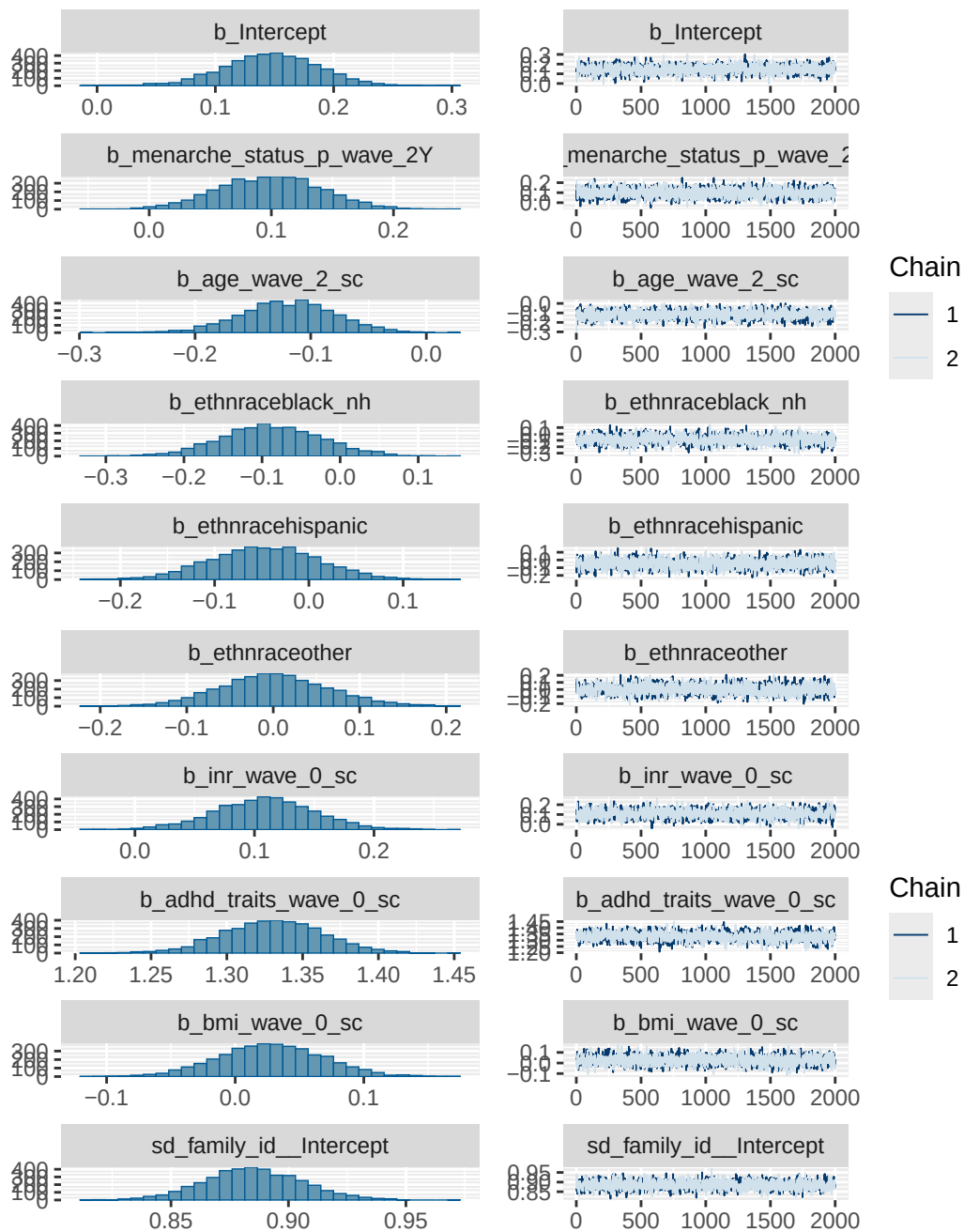
      Tail_ESS
Intercept          2160
menarche_status_p_wave_2Y  2718
age_wave_2_sc       2418
ethnraceblack_nh    2631
ethnracehispanic    2863
ethnraceother       2195
inr_wave_0_sc       2221
adhd_traits_wave_0_sc  2174
bmi_wave_0_sc       2422

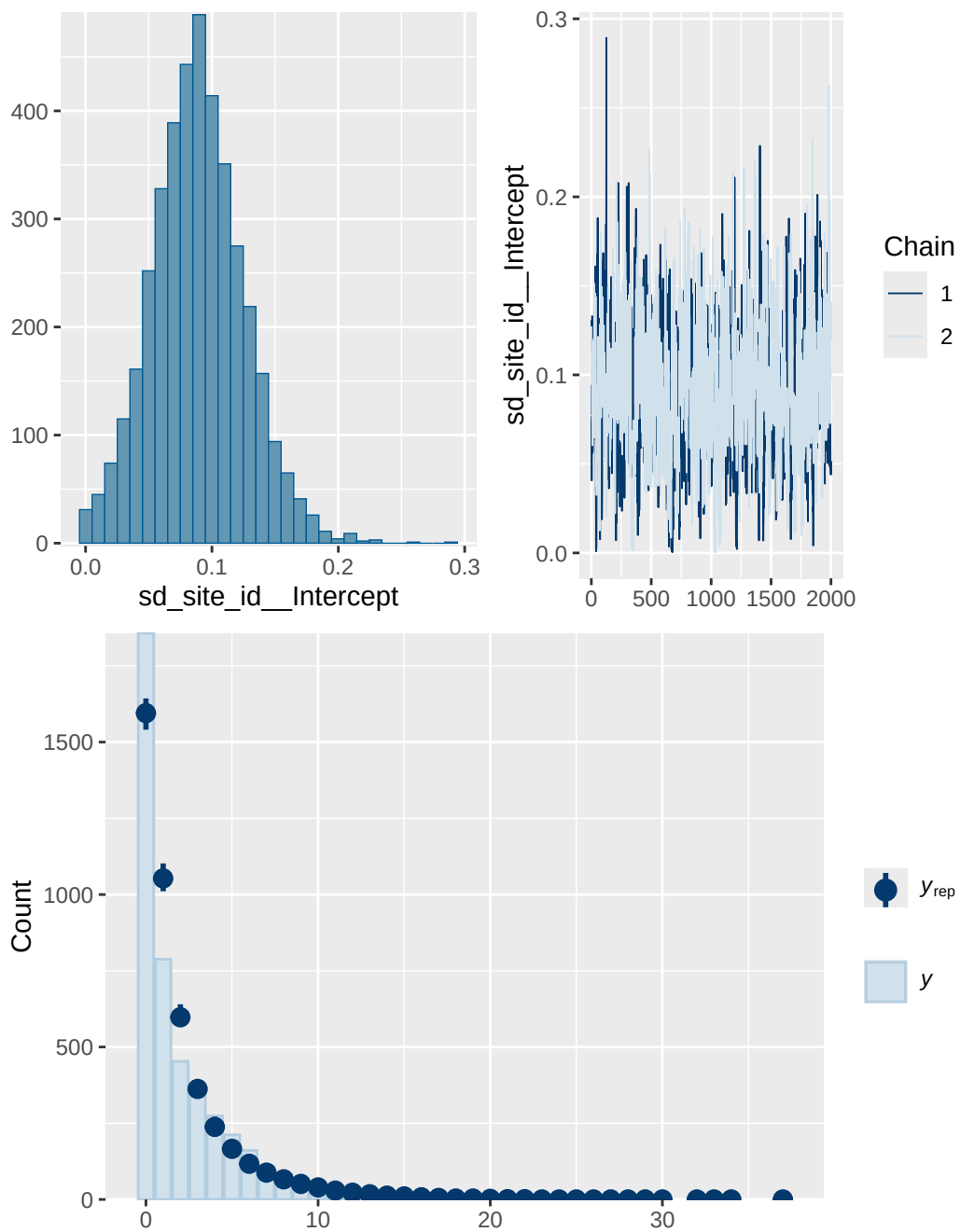
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

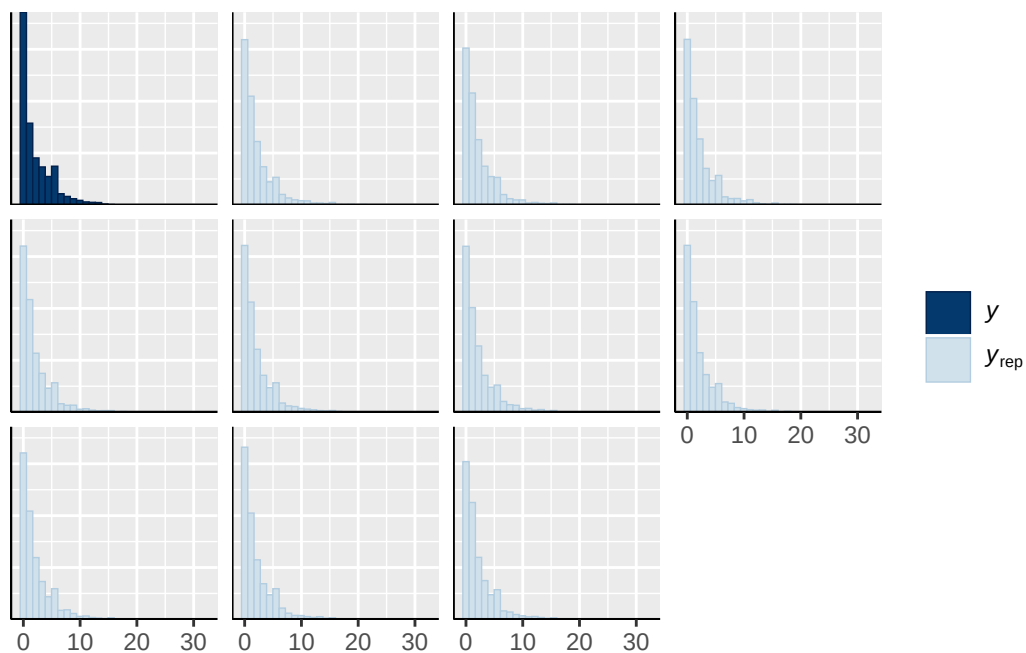
Model 1 longitudinal BMI diagnostics

	prior	class	coef	group	resp	dpar
	normal(0, 1)	b				
	normal(0, 1)	b	adhd_traits_wave_0_sc			
	normal(0, 1)	b	age_wave_2_sc			
	normal(0, 1)	b	bmi_wave_0_sc			
	normal(0, 1)	b	ethnraceblack_nh			
	normal(0, 1)	b	ethnracehispanic			
	normal(0, 1)	b	ethnraceother			
	normal(0, 1)	b	inr_wave_0_sc			
	normal(0, 1)	b	menarche_status_p_wave_2Y			
student_t(3, 0, 2.7)	Intercept					
student_t(3, 0, 2.7)	sd					
student_t(3, 0, 2.7)	sd			family_id		
student_t(3, 0, 2.7)	sd		Intercept	family_id		
student_t(3, 0, 2.7)	sd			site_id		
student_t(3, 0, 2.7)	sd		Intercept	site_id		
nlpar lb ub tag		source				
		user				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		default				
0		default				
0		(vectorized)				
0		(vectorized)				
0		(vectorized)				
0		(vectorized)				

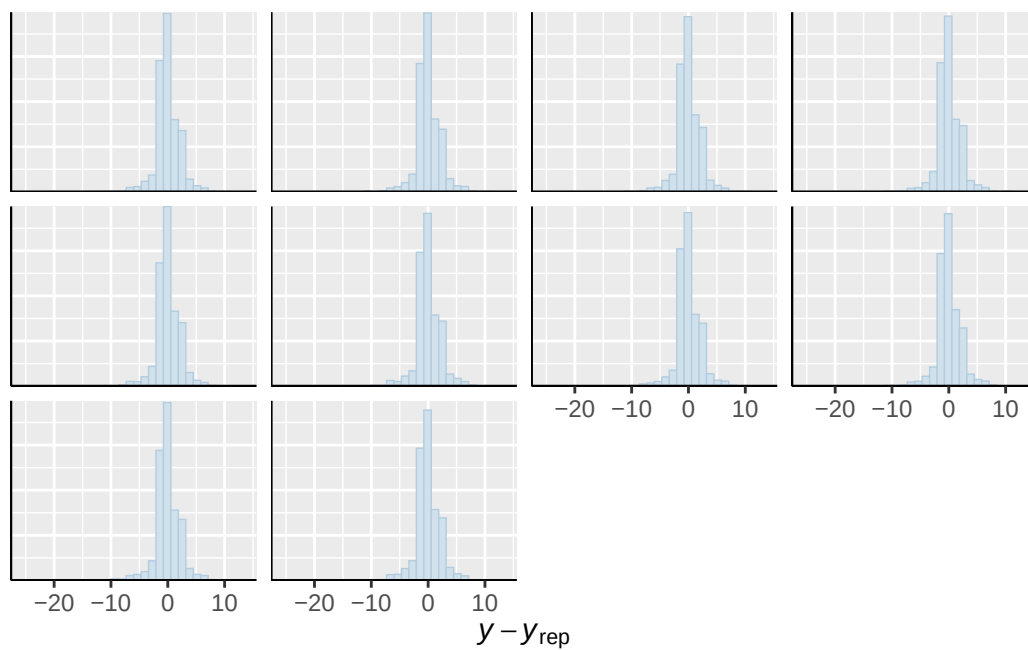




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

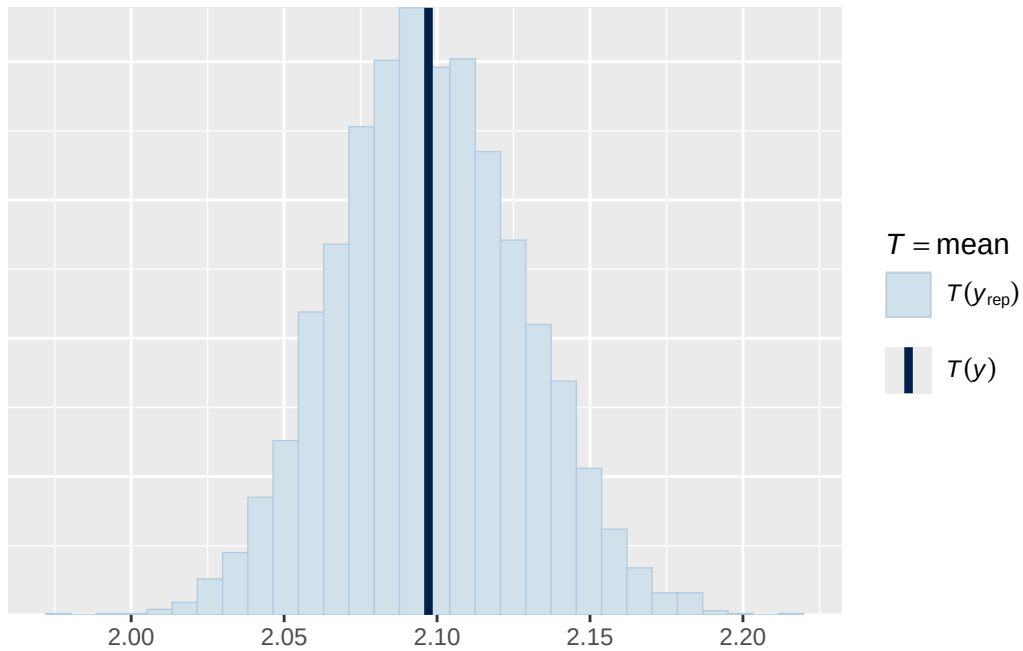


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

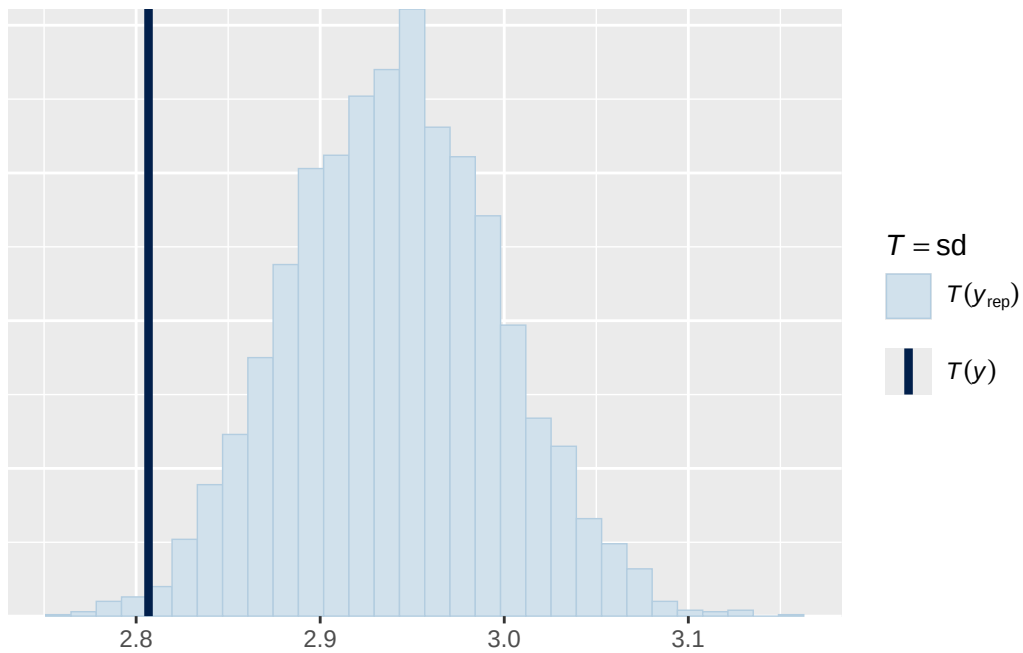


Using all posterior draws for ppc type 'stat' by default.

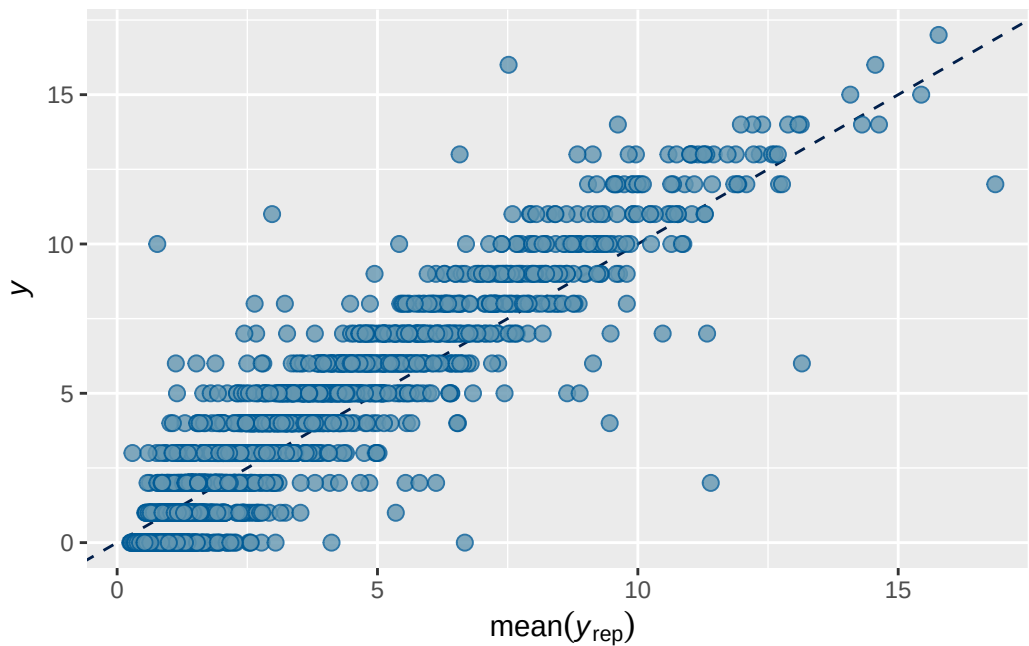
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1B longitudinal BMI output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.7358724	0.007396846	0.721296	0.7497154

Family: poisson
 Links: mu = log
 Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + in
 Data: analytic_df_wide (Number of observations: 4515)
 Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
 total post-warmup draws = 4000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.885	0.021	0.846	0.929	1.000	1313	1736

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.092	0.037	0.021	0.169	1.009	609	679

Regression Coefficients:

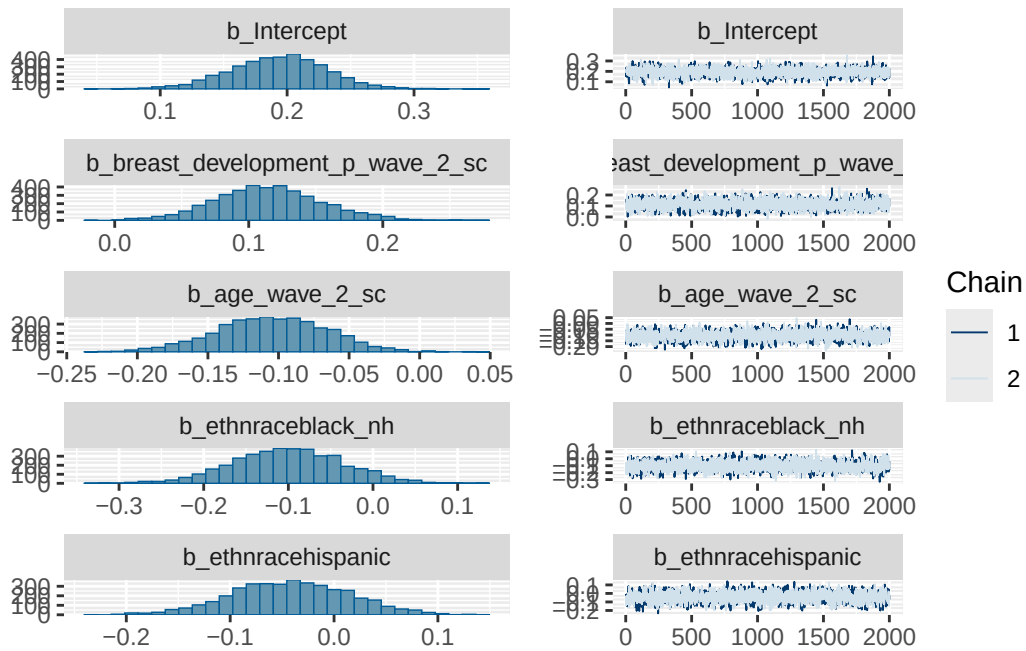
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	0.193	0.038	0.116	0.265	1.000
breast_development_p_wave_2_sc	0.115	0.040	0.038	0.196	1.001
age_wave_2_sc	-0.105	0.039	-0.183	-0.031	1.000
ethnraceblack_nh	-0.100	0.065	-0.227	0.025	1.001
ethnracehispanic	-0.042	0.057	-0.156	0.071	1.001
ethnraceother	-0.011	0.062	-0.131	0.112	1.001
inr_wave_0_sc	0.092	0.042	0.011	0.174	1.000
adhd_traits_wave_0_sc	1.322	0.032	1.258	1.386	1.001
bmi_wave_0_sc	0.042	0.041	-0.041	0.121	1.000
	Bulk_ESS	Tail_ESS			
Intercept	2353	2745			
breast_development_p_wave_2_sc	2280	2800			
age_wave_2_sc	1932	2678			
ethnraceblack_nh	1930	2513			
ethnracehispanic	1778	2742			
ethnraceother	1938	2569			
inr_wave_0_sc	1850	2596			
adhd_traits_wave_0_sc	1449	2200			
bmi_wave_0_sc	1985	2574			

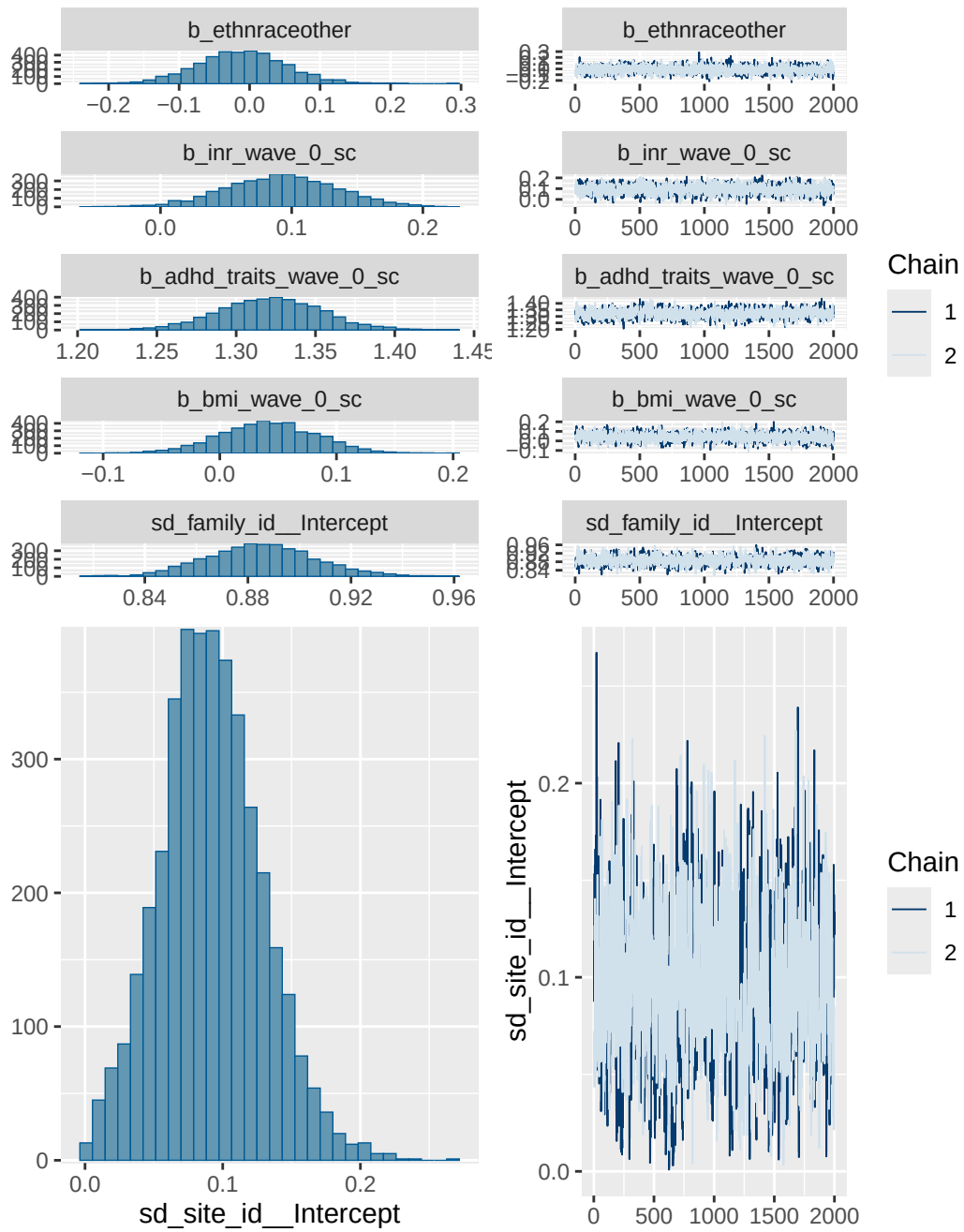
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

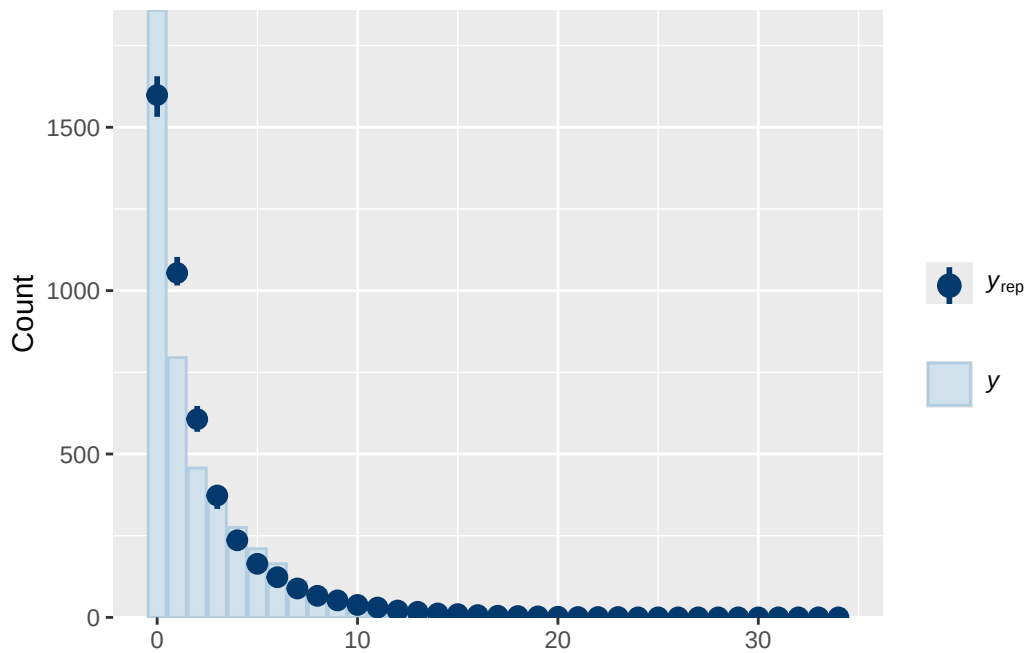
Model 1B longitudinal BMI diagnostics

prior	class	coef	group resp
normal(0, 1)	b		
normal(0, 1)	b	adhd_traits_wave_0_sc	
normal(0, 1)	b	age_wave_2_sc	
normal(0, 1)	b	bmi_wave_0_sc	
normal(0, 1)	b	breast_development_p_wave_2_sc	
normal(0, 1)	b	ethnraceblack_nh	
normal(0, 1)	b	ethnracehispanic	
normal(0, 1)	b	ethnraceother	
normal(0, 1)	b	inr_wave_0_sc	
student_t(3, 0, 2.7)	Intercept		
student_t(3, 0, 2.7)	sd		
student_t(3, 0, 2.7)	sd		family_id

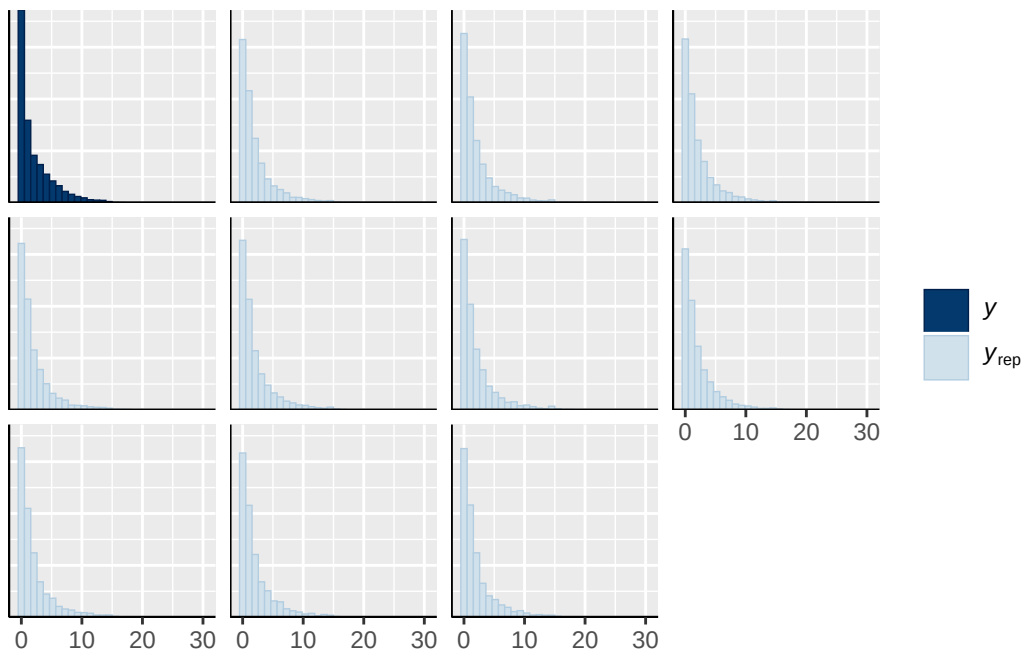
student_t(3, 0, 2.7)	sd	Intercept	family_id
student_t(3, 0, 2.7)	sd		site_id
student_t(3, 0, 2.7)	sd	Intercept	site_id
dpar nlpar lb ub tag	source		
	user		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	(vectorized)		
	default		
0	default		
0	(vectorized)		
0	(vectorized)		
0	(vectorized)		
0	(vectorized)		



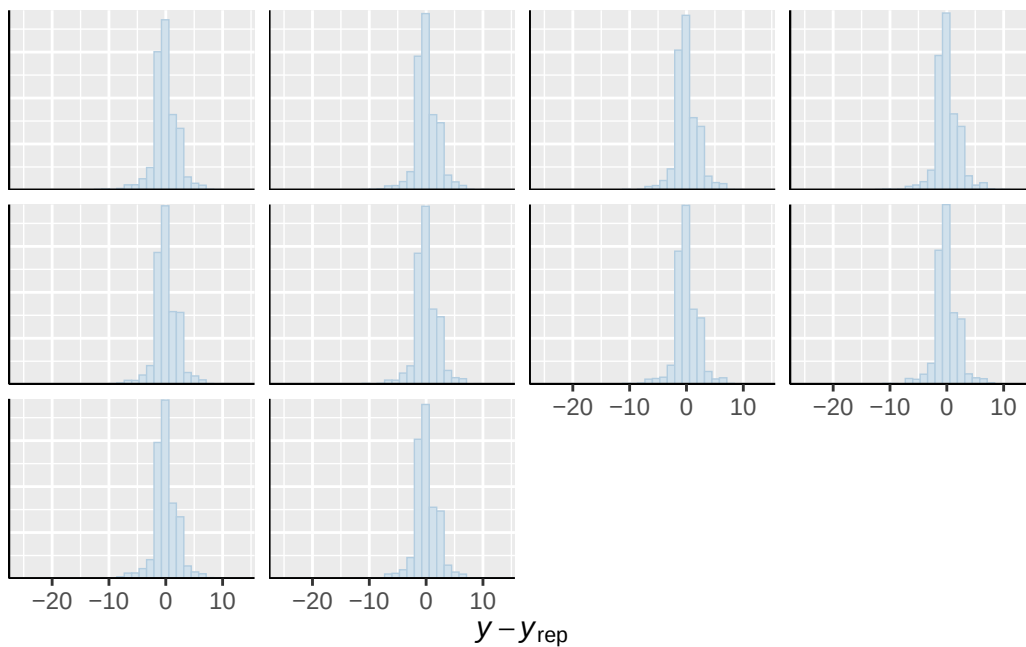




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

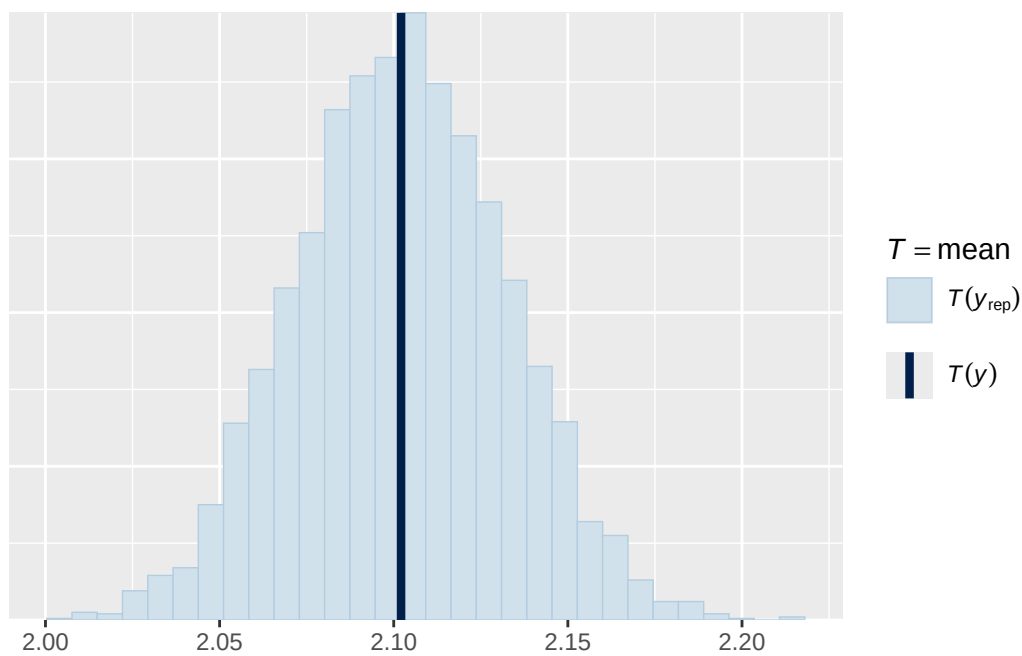


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

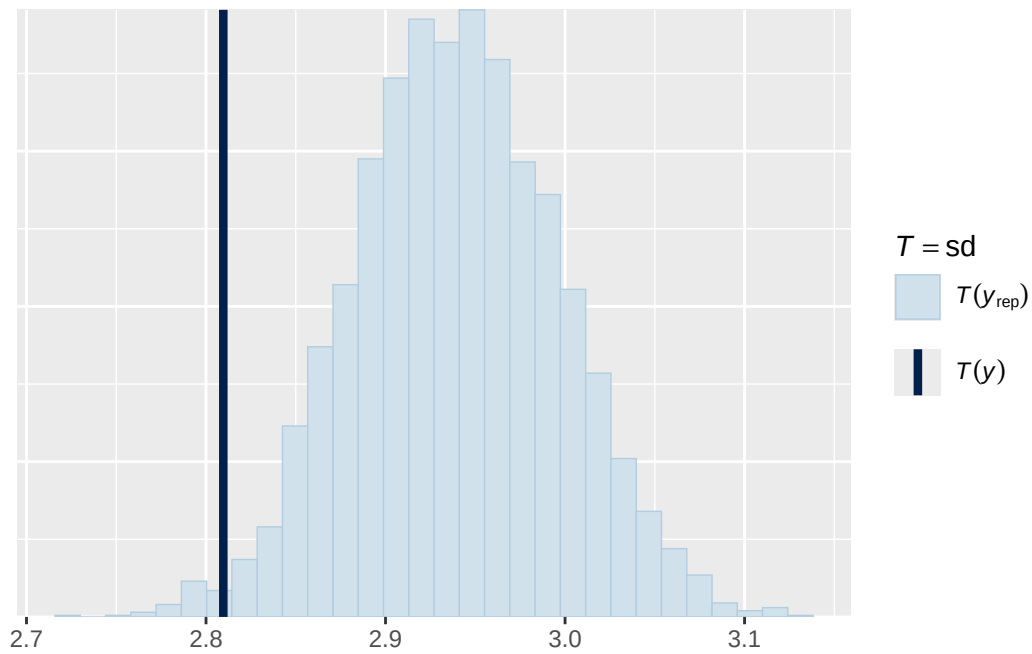


Using all posterior draws for ppc type 'stat' by default.

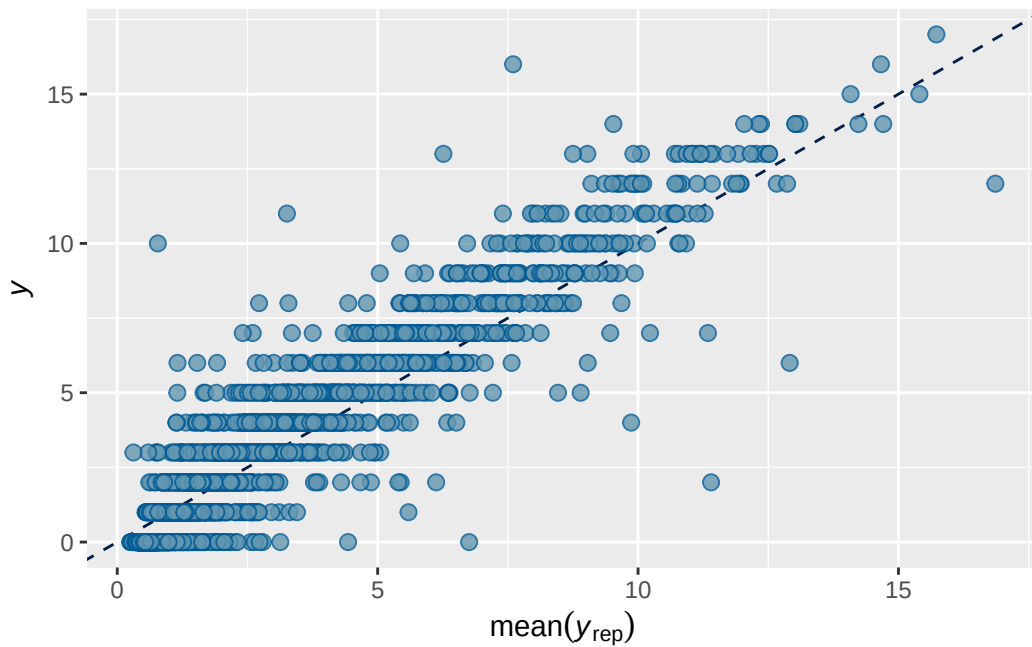
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 5B longitudinal BMI output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8235553 0.005162047 0.8131239 0.8331237

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.986    0.020   0.948   1.025 1.000    2542    3803

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.067    0.038   0.004   0.146 1.000     717    1701

```

Regression Coefficients:

	Estimate	Est.Error			
Intercept	0.520	0.035			
breast_development_p_wave_2_sc	0.154	0.039			
adhd_traits_wave_2_sc	0.591	0.034			
age_wave_2_sc	-0.118	0.038			
ethnraceblack_nh	-0.173	0.067			
ethnracehispanic	-0.118	0.058			
ethnraceother	-0.087	0.063			
inr_wave_0_sc	0.017	0.043			
externalising_wave_0_sc	0.908	0.034			
bmi_wave_0_sc	0.159	0.040			
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.170	0.066			
	1-95% CI	u-95% CI	Rhat		
Intercept	0.450	0.587	1.000		
breast_development_p_wave_2_sc	0.078	0.231	1.000		
adhd_traits_wave_2_sc	0.525	0.657	1.000		
age_wave_2_sc	-0.190	-0.042	1.000		
ethnraceblack_nh	-0.305	-0.042	1.000		
ethnracehispanic	-0.229	-0.003	1.000		
ethnraceother	-0.209	0.035	1.001		

inr_wave_0_sc	-0.071	0.102	1.000
externalising_wave_0_sc	0.841	0.975	1.000
bmi_wave_0_sc	0.082	0.239	1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.043	0.302	1.000
	Bulk_ESS	Tail_ESS	
Intercept	4214	5044	
breast_development_p_wave_2_sc	3976	5238	
adhd_traits_wave_2_sc	3207	4994	
age_wave_2_sc	4181	5455	
ethnraceblack_nh	3215	4931	
ethnracehispanic	3137	4979	
ethnraceother	3458	4916	
inr_wave_0_sc	3600	4754	
externalising_wave_0_sc	2728	4140	
bmi_wave_0_sc	3302	4481	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	2872	4850	

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5B longitudinal BMI diagnostics

prior	class				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
normal(0, 1)	b				
student_t(3, 0, 2.9)	Intercept				
student_t(3, 0, 2.9)	sd				
student_t(3, 0, 2.9)	sd				
student_t(3, 0, 2.9)	sd				
student_t(3, 0, 2.9)	sd				
student_t(3, 0, 2.9)	sd				
		coef	group	resp	dpar nlpar

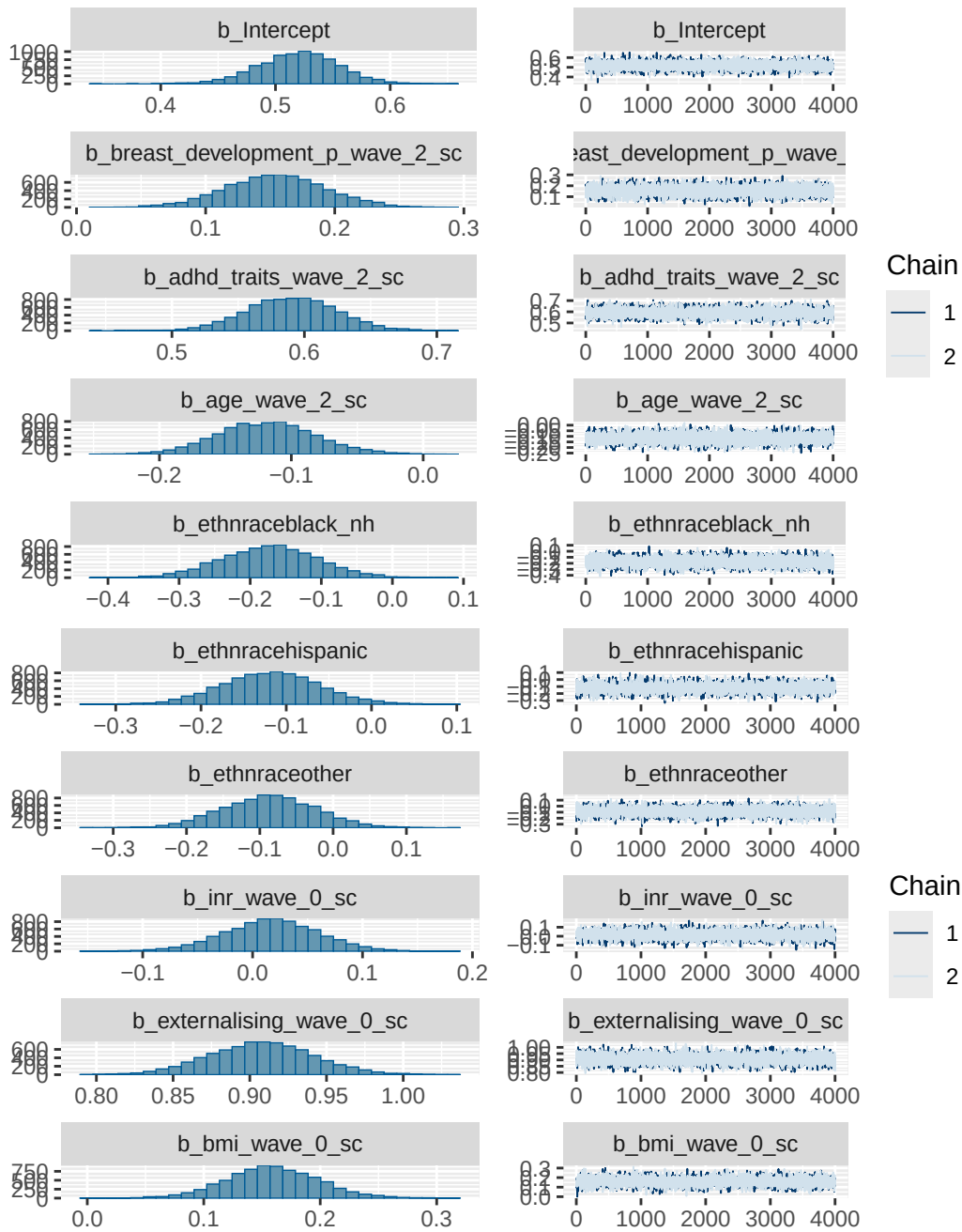
```

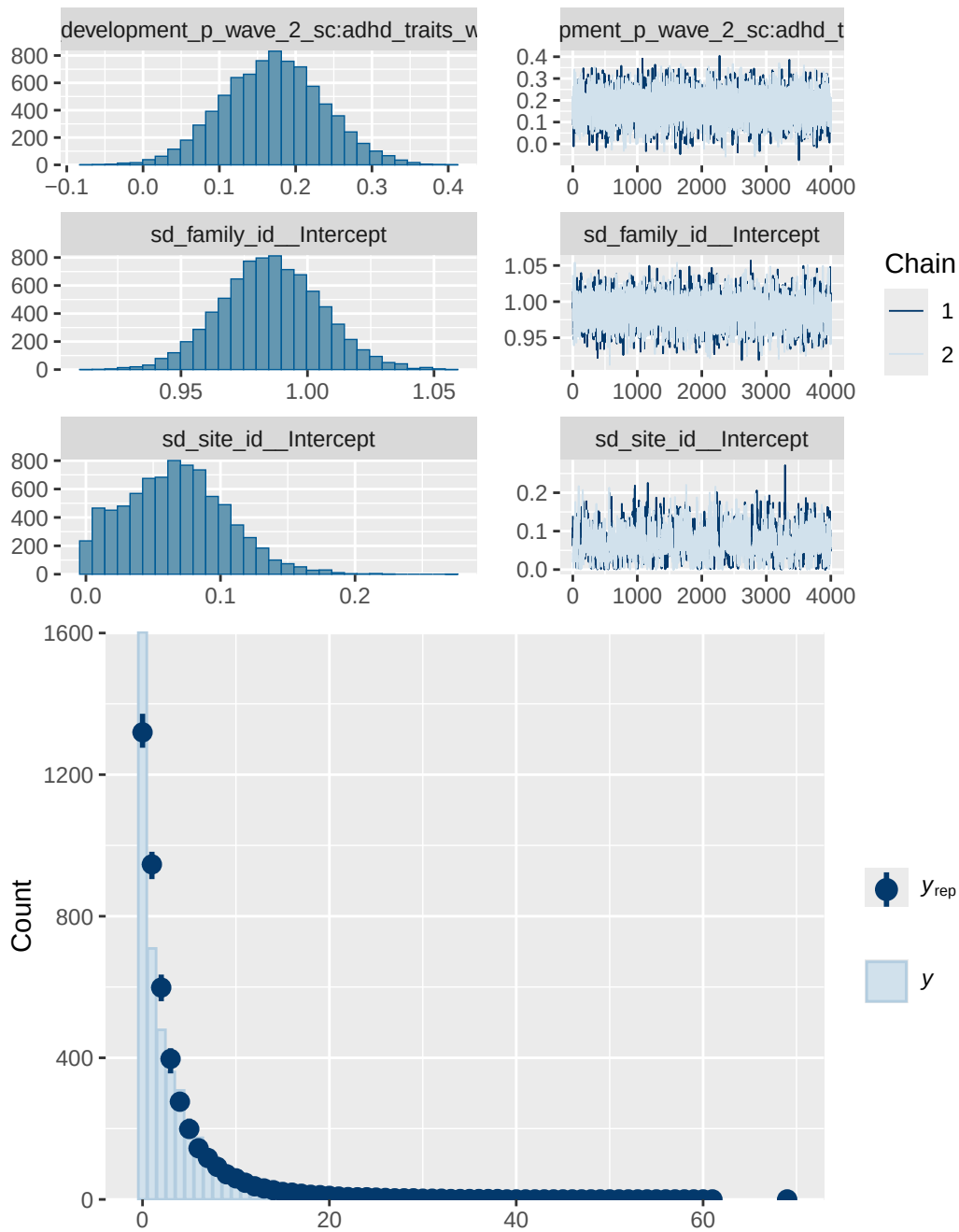
                                adhd_traits_wave_2_sc
                                age_wave_2_sc
                                bmi_wave_0_sc
                                breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
                                ethnraceblack_nh
                                ethnracehispanic
                                ethnraceother
                                externalising_wave_0_sc
                                inr_wave_0_sc

                                family_id
                                Intercept family_id
                                site_id
                                Intercept site_id

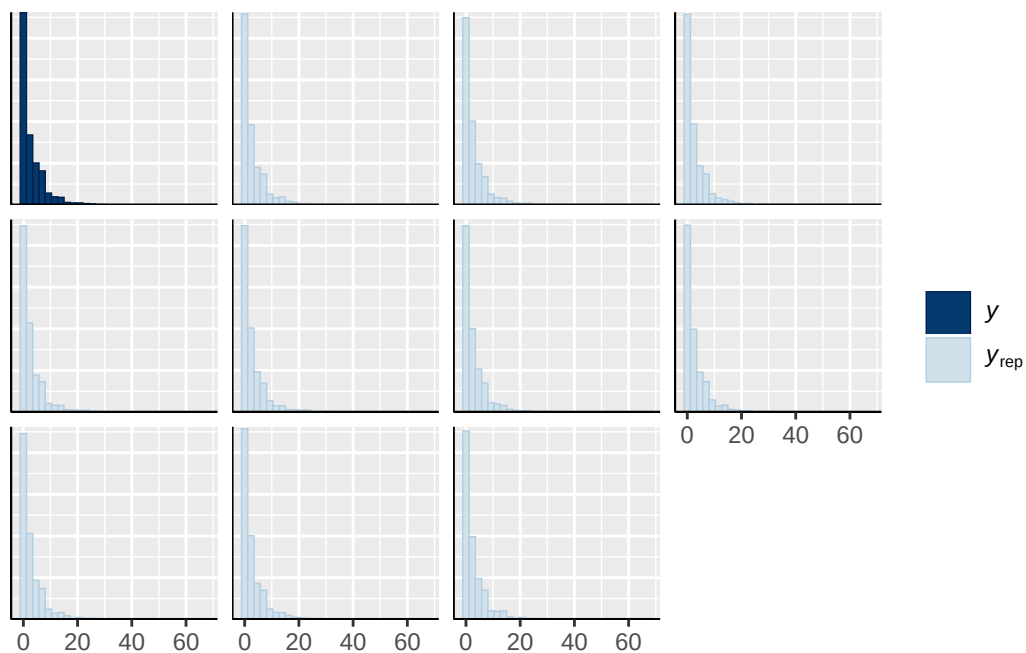
lb ub tag      source
              user
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              (vectorized)
              default
0          default
0          (vectorized)
0          (vectorized)
0          (vectorized)
0          (vectorized)

```

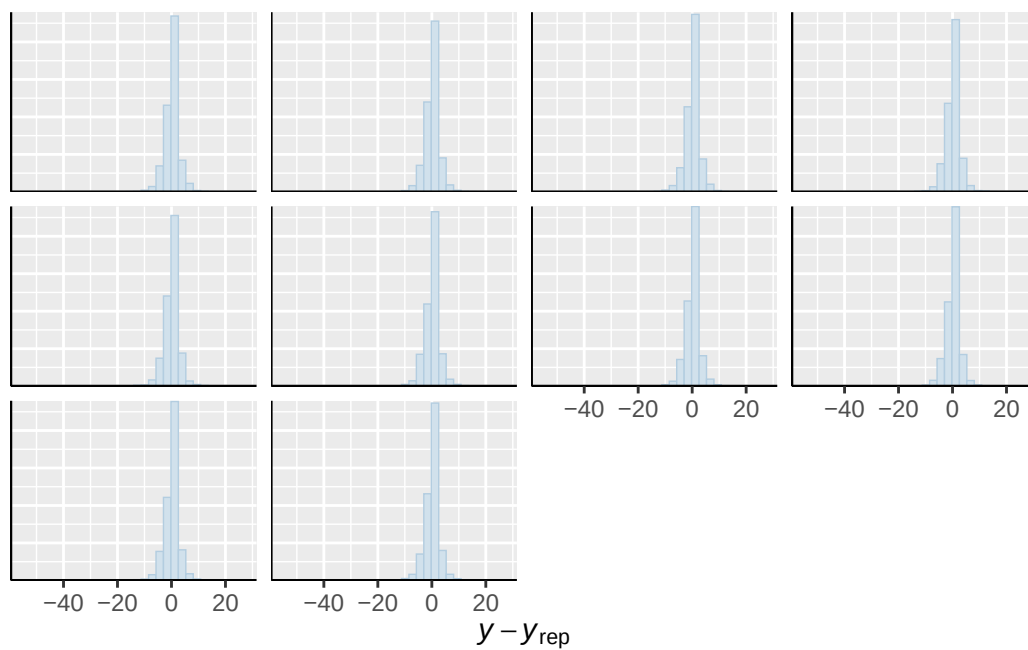




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

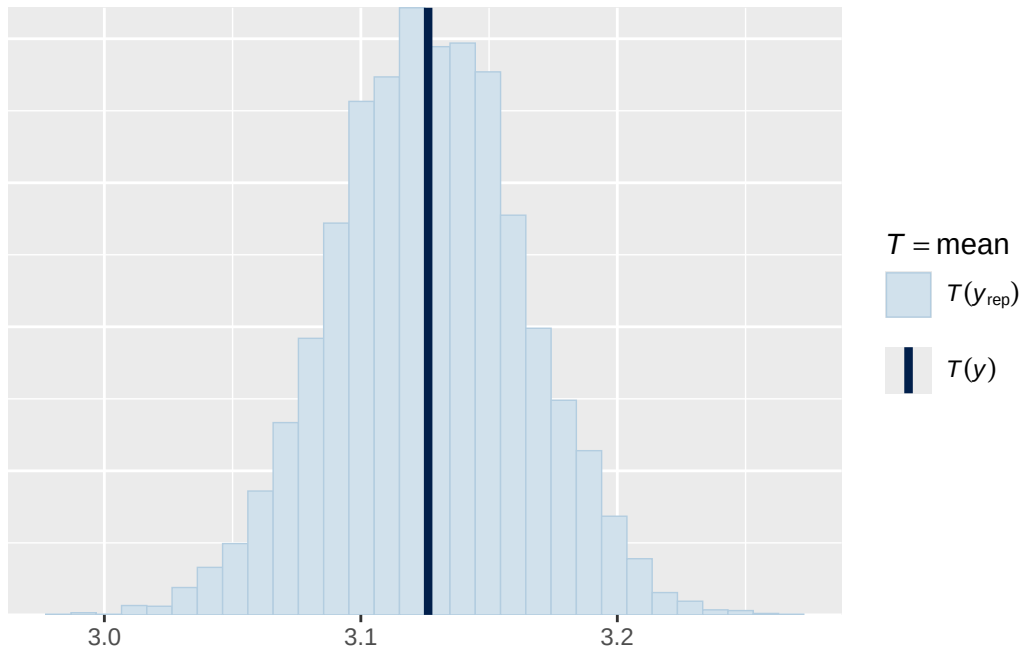


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

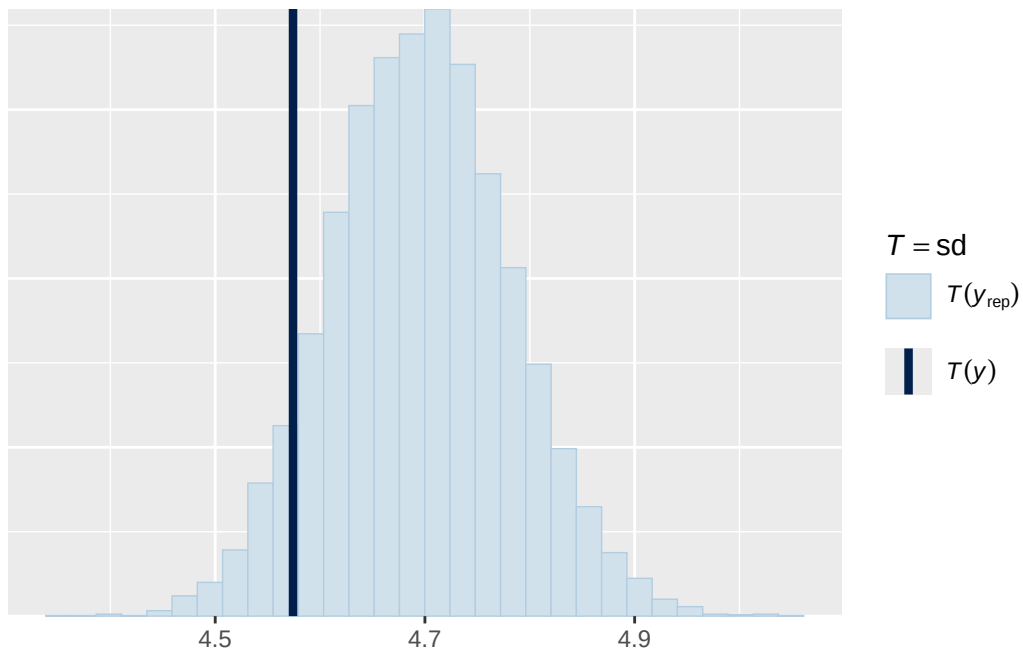


Using all posterior draws for ppc type 'stat' by default.

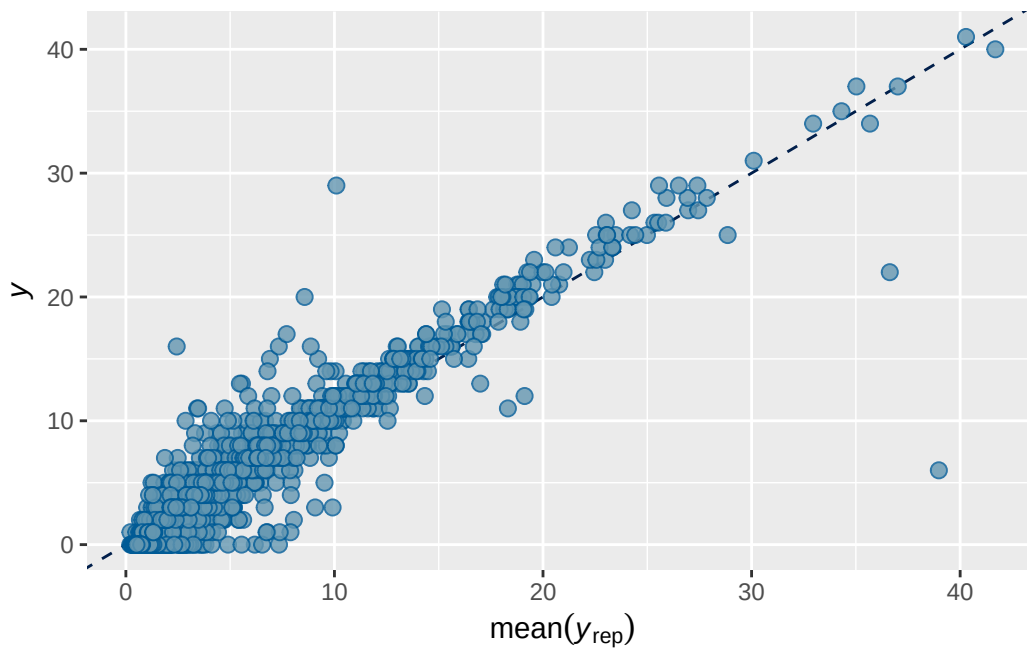
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1 longitudinal stimulant output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.7394205	0.007405415	0.7247865	0.7533296

```
Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

~family_id (Number of levels: 3915)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.871	0.021	0.832	0.913	1.001	1580	2613

~site_id (Number of levels: 22)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.082	0.036	0.012	0.158	1.001	663	964

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
Intercept	0.112	0.040	0.033	0.191	1.001	3038
menarche_status_p_wave_2Y	0.101	0.041	0.021	0.182	1.000	3011
age_wave_2_sc	-0.119	0.041	-0.201	-0.038	1.000	3452
ethnraceblack_nh	-0.060	0.063	-0.183	0.062	1.001	2584
ethnracehispanic	-0.026	0.059	-0.140	0.091	1.000	2499
ethnraceother	-0.003	0.061	-0.121	0.121	1.000	3267
inr_wave_0_sc	0.080	0.041	-0.001	0.161	1.001	2776
adhd_traits_wave_0_sc	1.229	0.034	1.164	1.298	1.001	2386
stimulant_status_wave_3Y	0.592	0.068	0.458	0.722	1.001	2203
	Tail_ESS					
Intercept	3069					
menarche_status_p_wave_2Y	3263					
age_wave_2_sc	3191					
ethnraceblack_nh	3439					
ethnracehispanic	2786					
ethnraceother	3443					
inr_wave_0_sc	3077					
adhd_traits_wave_0_sc	2675					
stimulant_status_wave_3Y	2754					

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

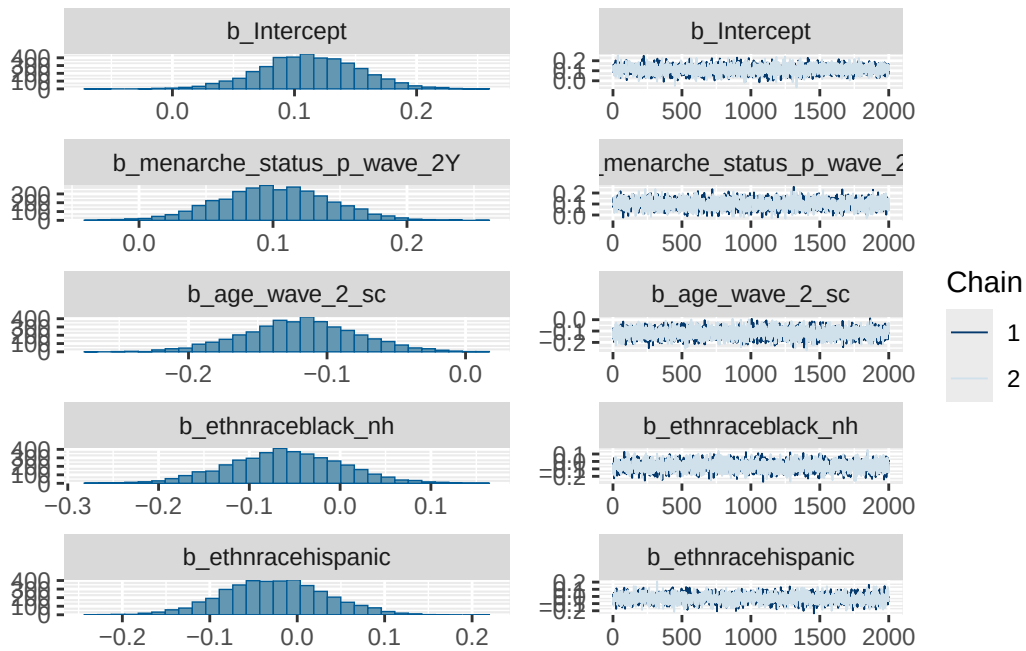
Model 1 stimulant diagnostics

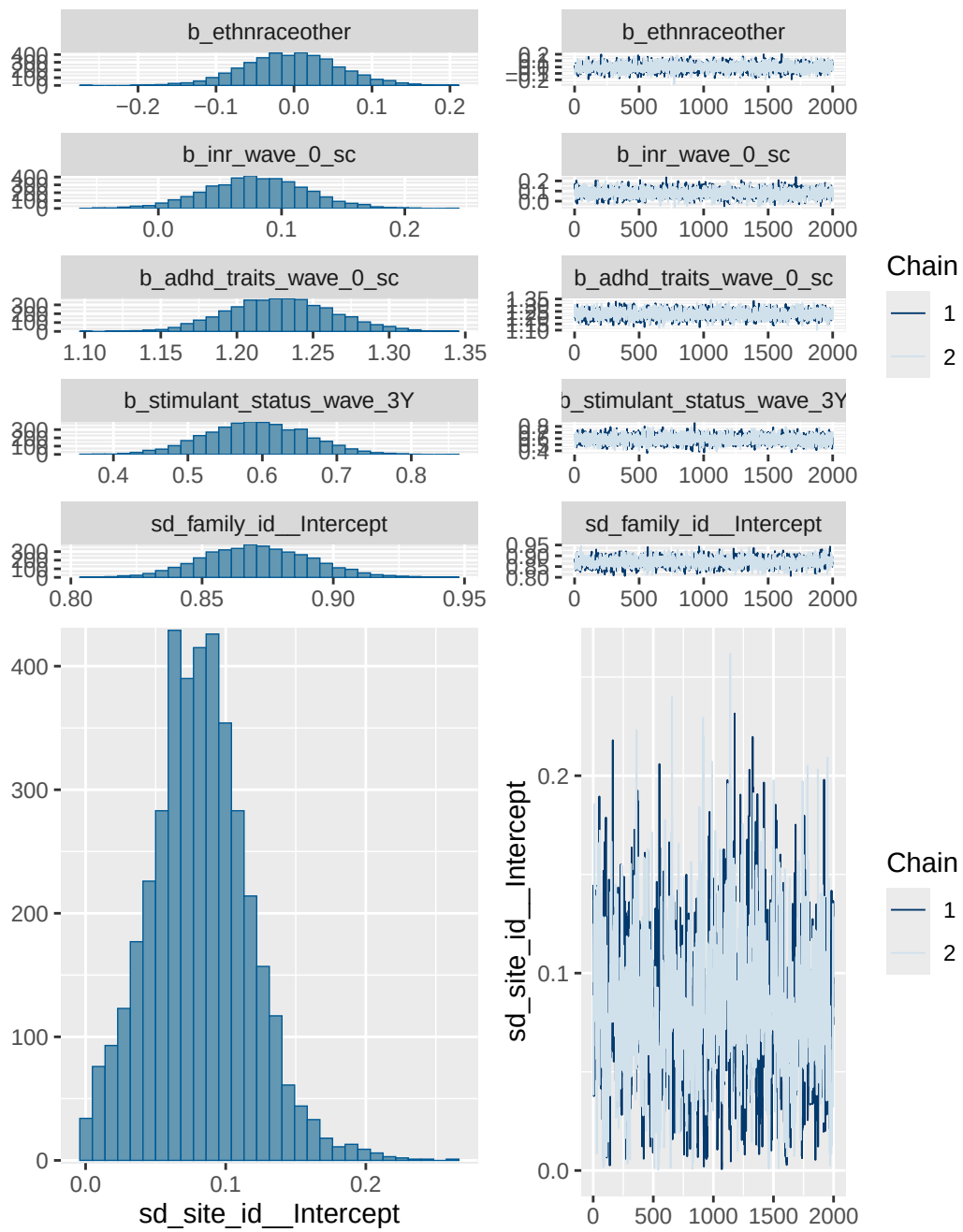
prior	class	coef	group	resp	dpar
normal(0, 1)	b				
normal(0, 1)	b	adhd_traits_wave_0_sc			
normal(0, 1)	b	age_wave_2_sc			
normal(0, 1)	b	ethnraceblack_nh			
normal(0, 1)	b	ethnracehispanic			
normal(0, 1)	b	ethnraceother			
normal(0, 1)	b	inr_wave_0_sc			
normal(0, 1)	b	menarche_status_p_wave_2Y			
normal(0, 1)	b	stimulant_status_wave_3Y			
student_t(3, 0, 2.7)	Intercept				
student_t(3, 0, 2.7)	sd				
student_t(3, 0, 2.7)	sd		family_id		

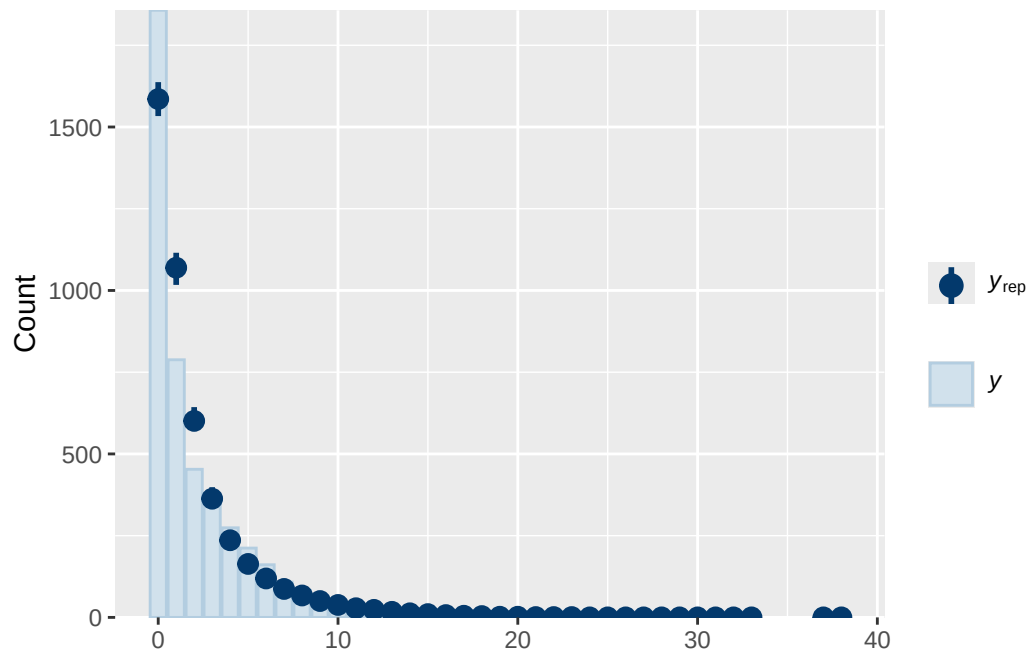
```

student_t(3, 0, 2.7)      sd      Intercept family_id
student_t(3, 0, 2.7)      sd      site_id
student_t(3, 0, 2.7)      sd      Intercept    site_id
nlpar lb ub tag          source
                        user
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        (vectorized)
                        default
0                        default
0                        (vectorized)
0                        (vectorized)
0                        (vectorized)
0                        (vectorized)

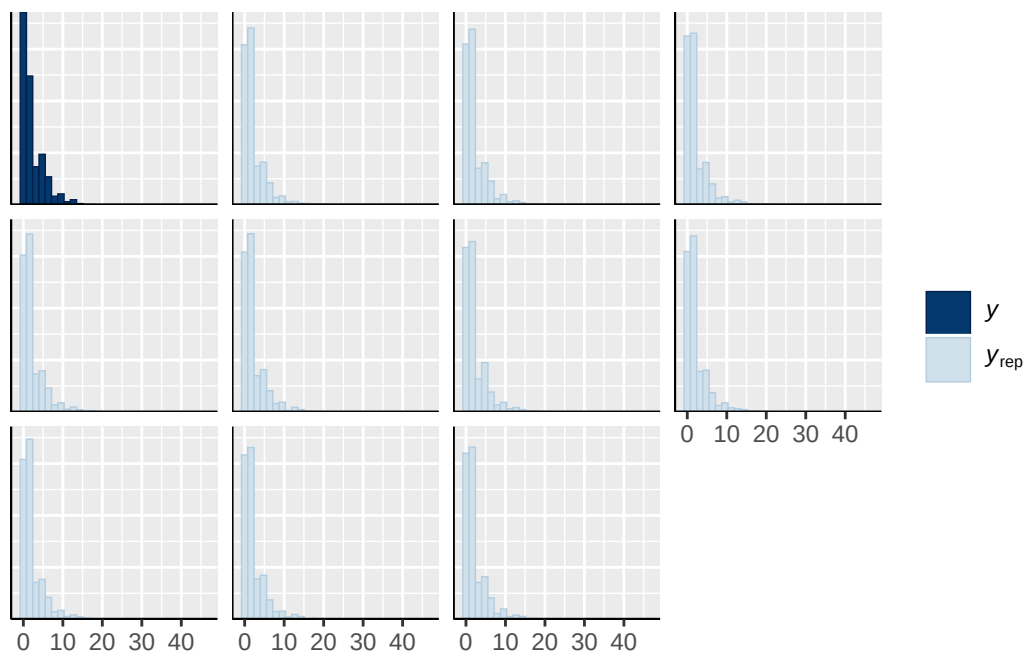
```



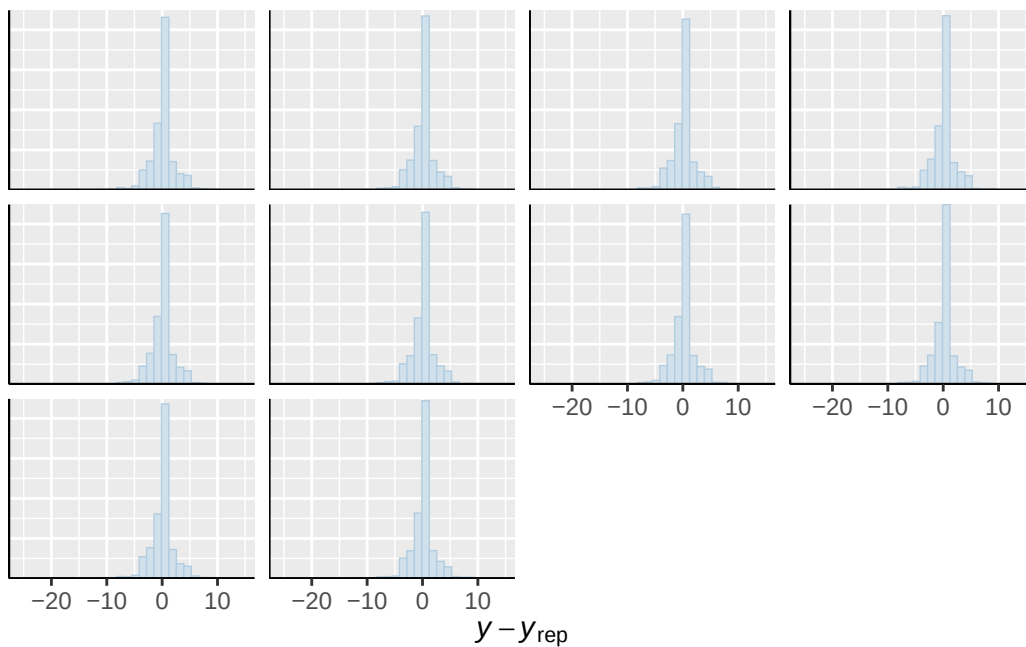




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

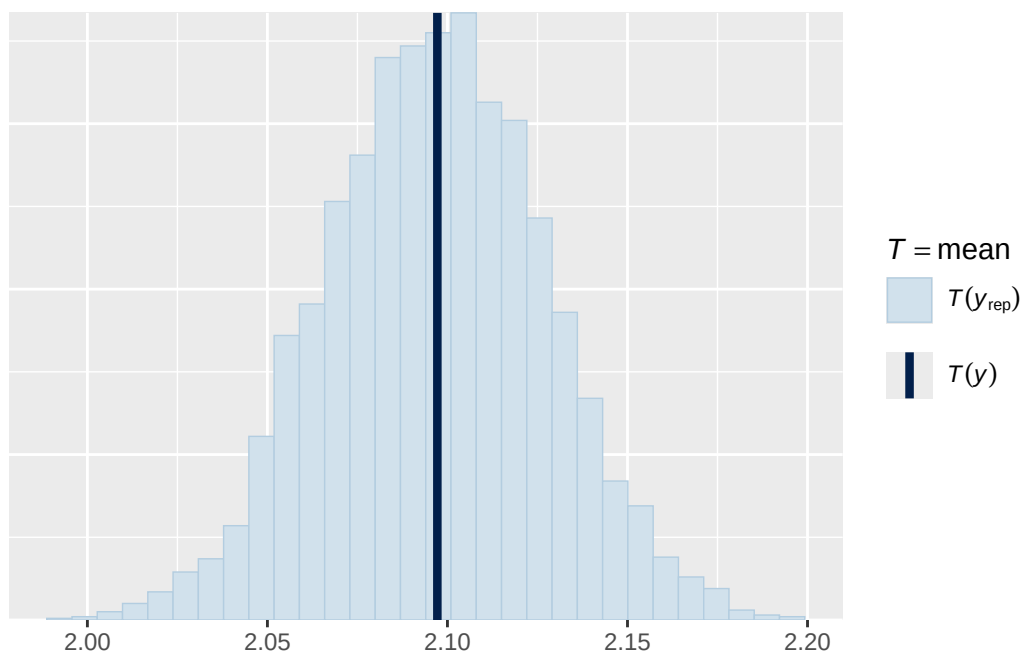


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

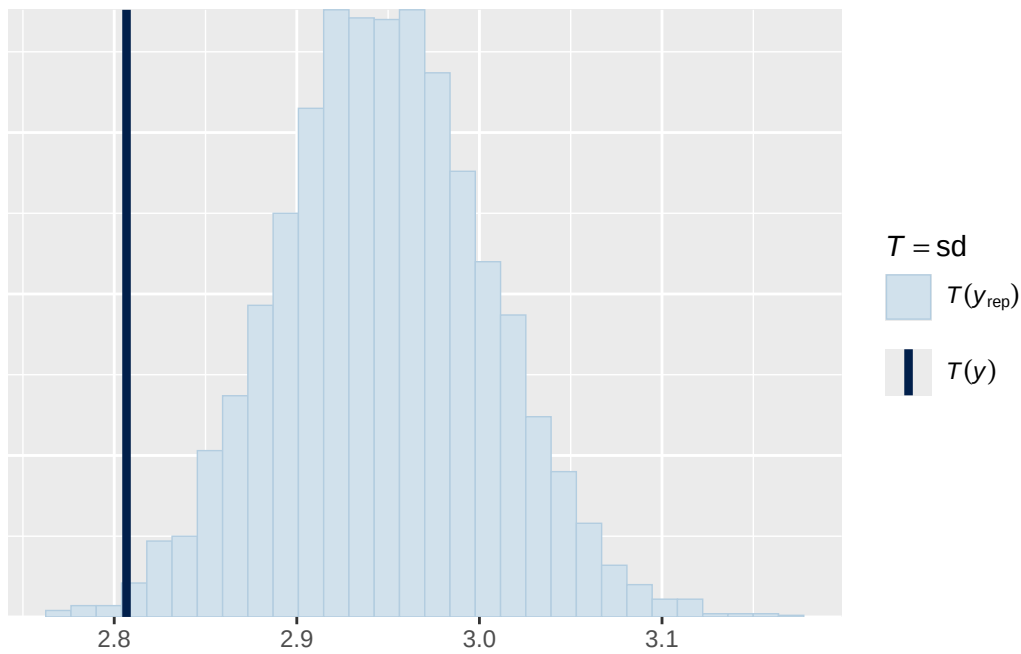


Using all posterior draws for ppc type 'stat' by default.

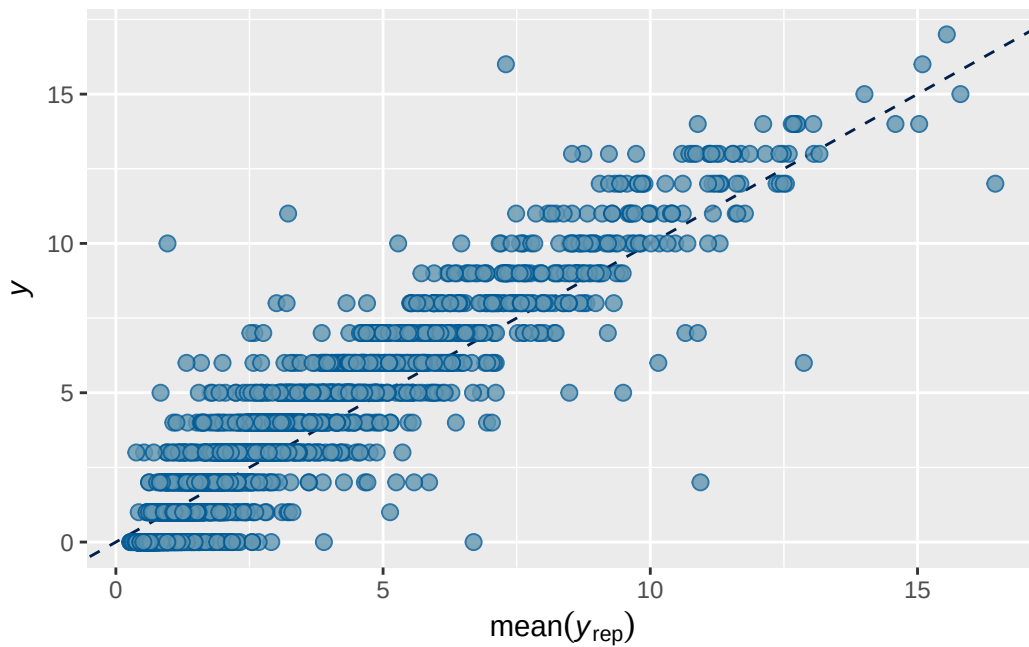
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use `stat_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'stat' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1B longitudinal stimulant output

```

      Estimate   Est.Error      Q2.5    Q97.5
R2 0.7392941 0.007295095 0.7243178 0.752978

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.871    0.021   0.831   0.913 1.001    1124    2045

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.086    0.036   0.017   0.162 1.004     473     604

```

Regression Coefficients:

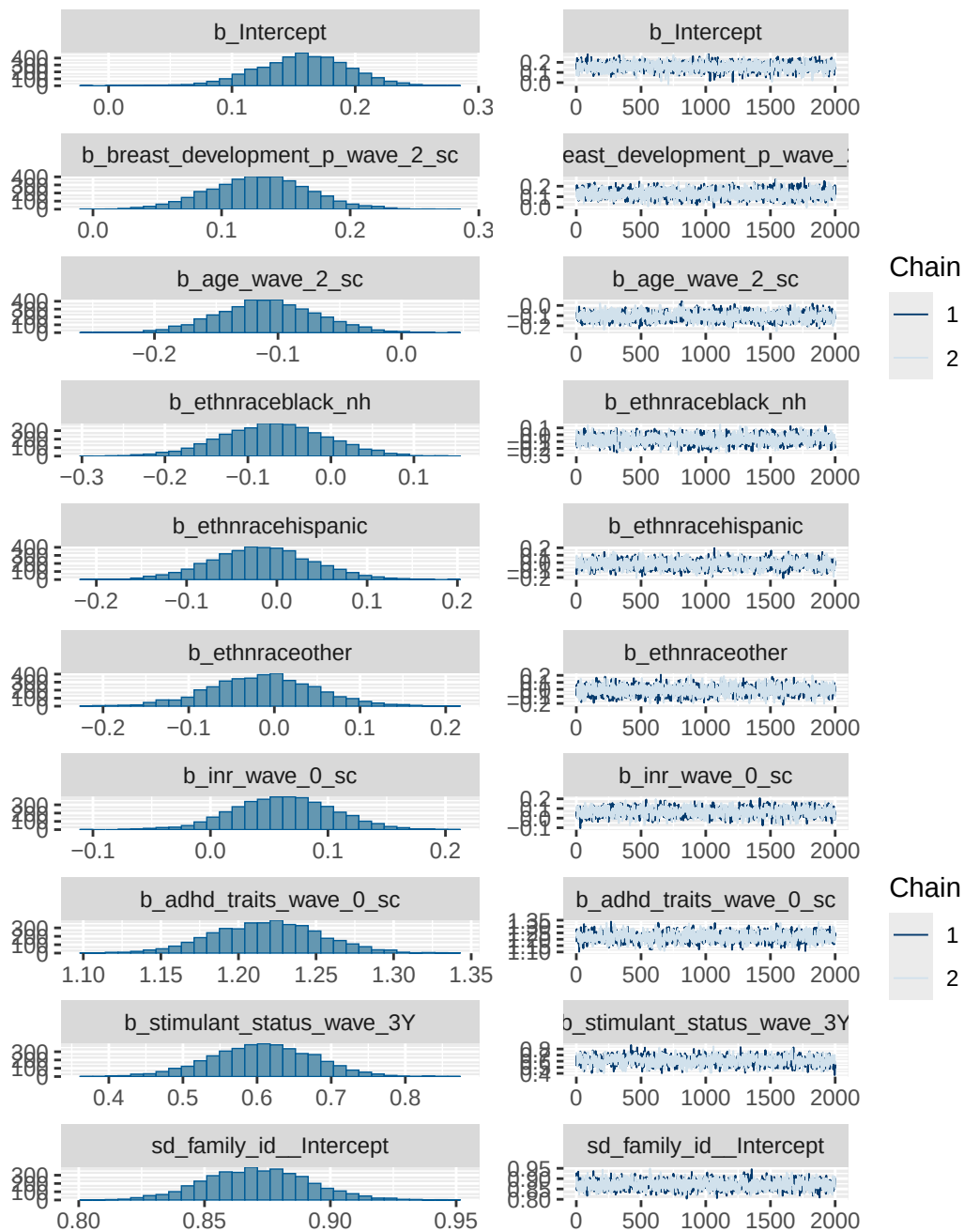
	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat
Intercept	0.158	0.037	0.083	0.229	1.001
breast_development_p_wave_2_sc	0.130	0.039	0.055	0.207	1.001
age_wave_2_sc	-0.108	0.040	-0.185	-0.029	1.002
ethnraceblack_nh	-0.068	0.063	-0.194	0.058	1.000
ethnracehispanic	-0.017	0.056	-0.127	0.094	1.002
ethnraceother	-0.013	0.062	-0.136	0.112	1.002
inr_wave_0_sc	0.062	0.042	-0.021	0.143	1.000
adhd_traits_wave_0_sc	1.218	0.035	1.151	1.286	1.000
stimulant_status_wave_3Y	0.607	0.069	0.468	0.740	1.003

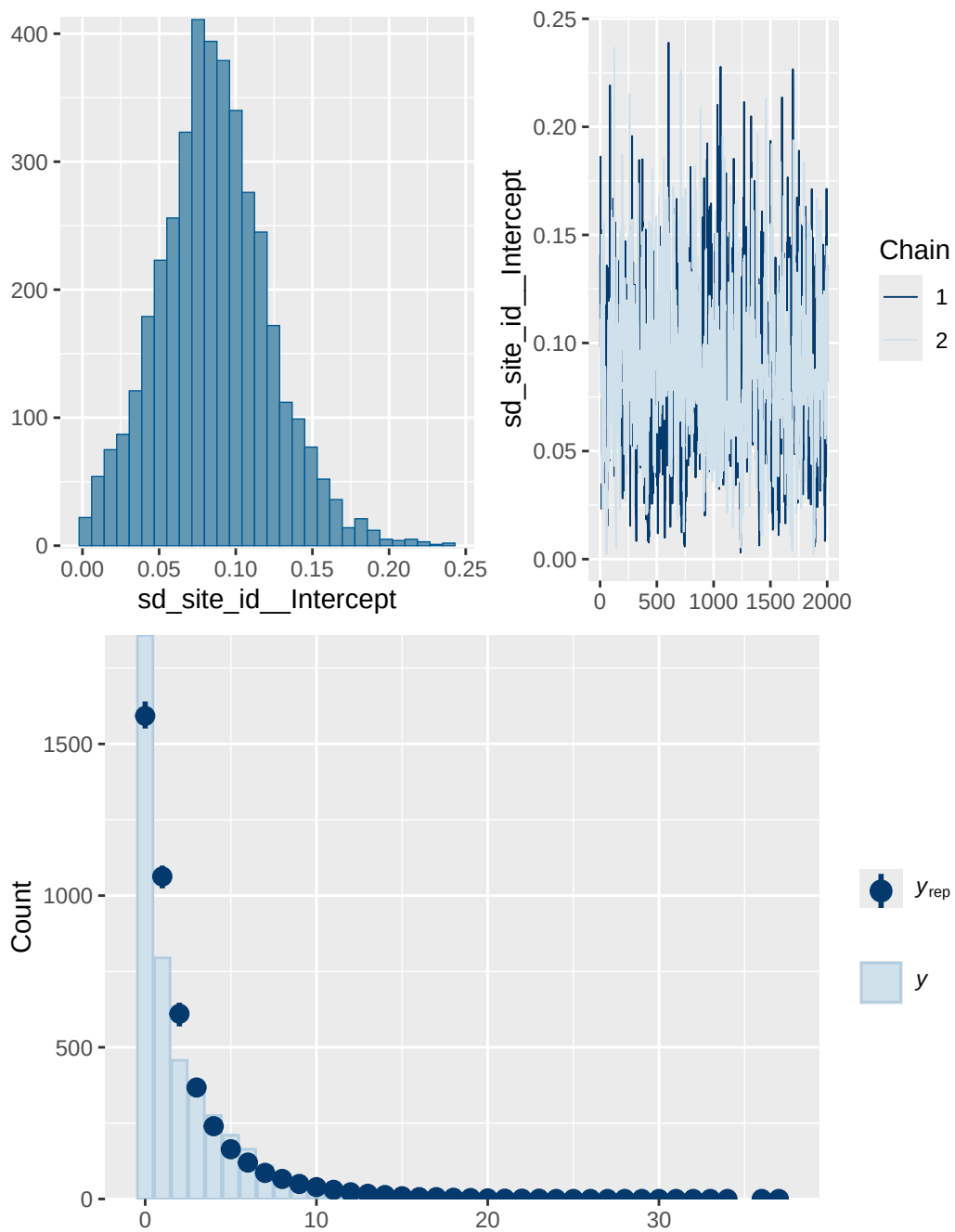
	Bulk_ESS	Tail_ESS
Intercept	1655	2118
breast_development_p_wave_2_sc	1639	2517
age_wave_2_sc	1140	2018
ethnraceblack_nh	1398	2258
ethnracehispanic	1093	2113
ethnraceother	1478	2384
inr_wave_0_sc	1366	2232
adhd_traits_wave_0_sc	1023	1712
stimulant_status_wave_3Y	969	1728

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

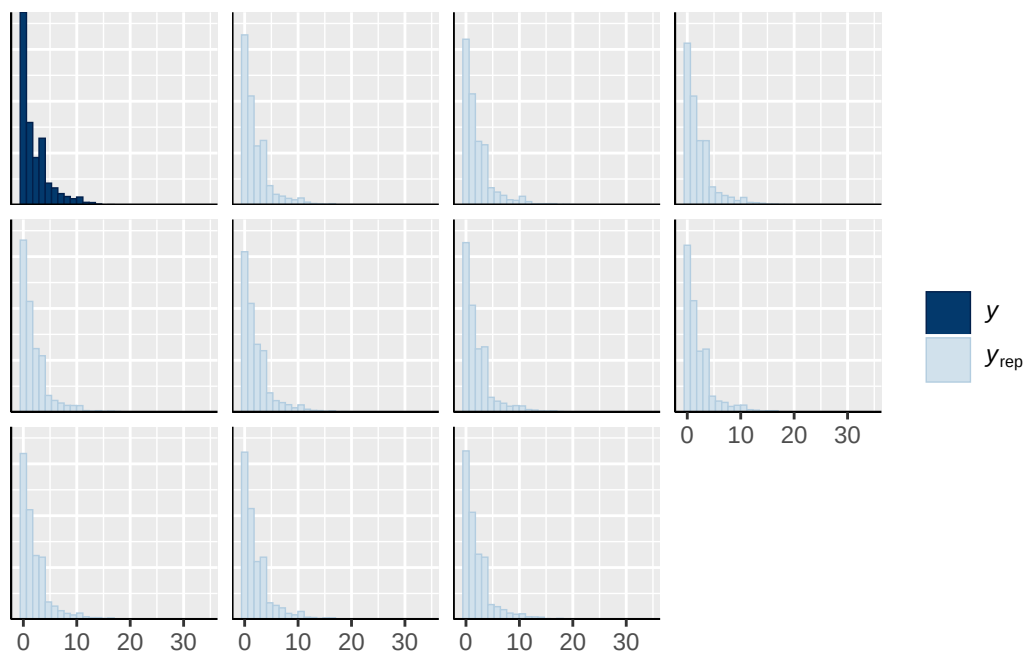
Model 1B longitudinal stimulant diagnostics

	prior	class	coef	group	resp
	normal(0, 1)	b			
	normal(0, 1)	b	adhd_traits_wave_0_sc		
	normal(0, 1)	b	age_wave_2_sc		
	normal(0, 1)	b	breast_development_p_wave_2_sc		
	normal(0, 1)	b	ethnraceblack_nh		
	normal(0, 1)	b	ethnracehispanic		
	normal(0, 1)	b	ethnraceother		
	normal(0, 1)	b	inr_wave_0_sc		
	normal(0, 1)	b	stimulant_status_wave_3Y		
student_t(3, 0, 2.7)	Intercept				
student_t(3, 0, 2.7)	sd				
student_t(3, 0, 2.7)	sd			family_id	
student_t(3, 0, 2.7)	sd		Intercept	family_id	
student_t(3, 0, 2.7)	sd			site_id	
student_t(3, 0, 2.7)	sd		Intercept	site_id	
dpar nlpar lb ub tag		source			
		user			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		default			
0		default			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			

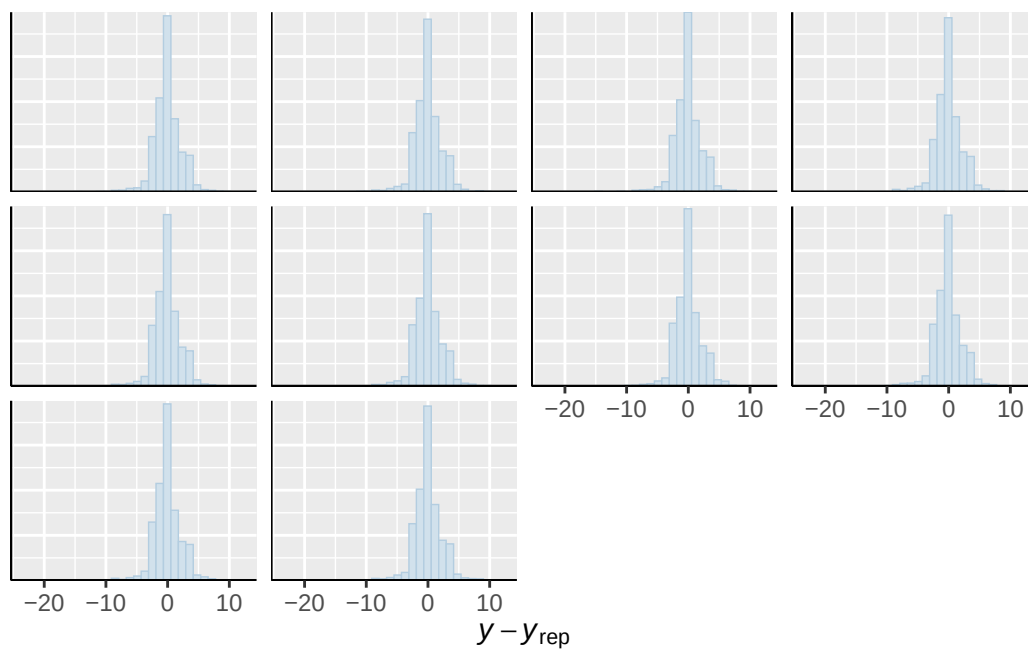




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

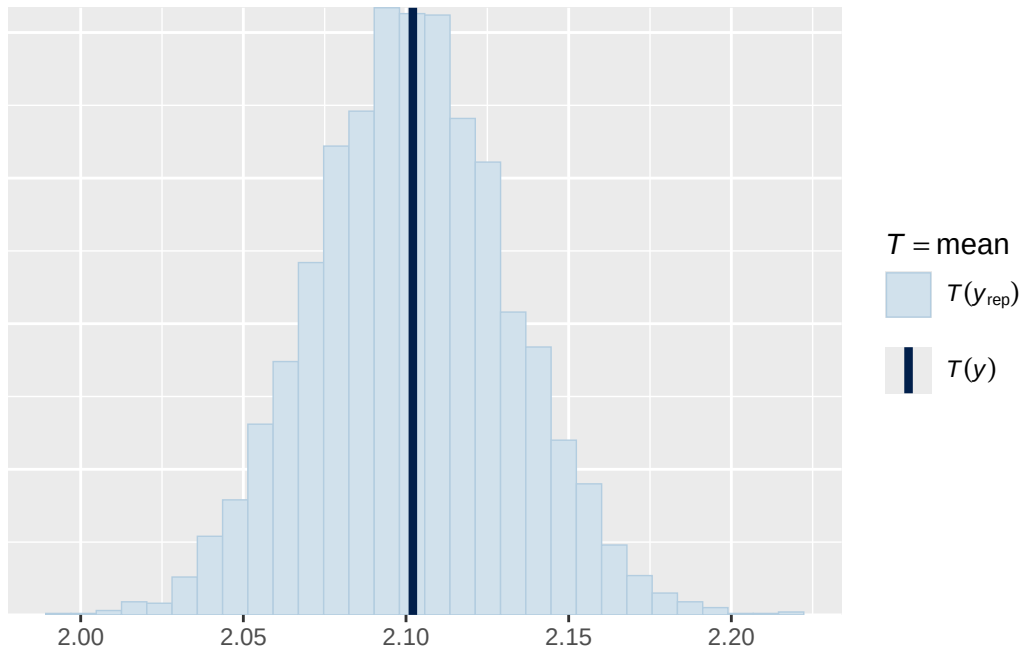


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

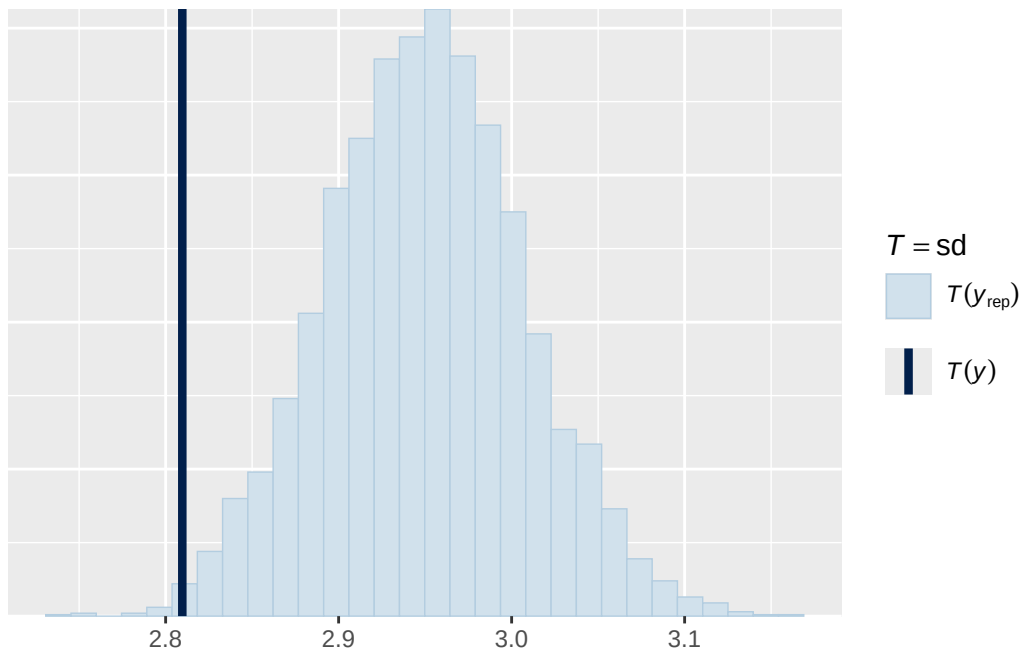


Using all posterior draws for ppc type 'stat' by default.

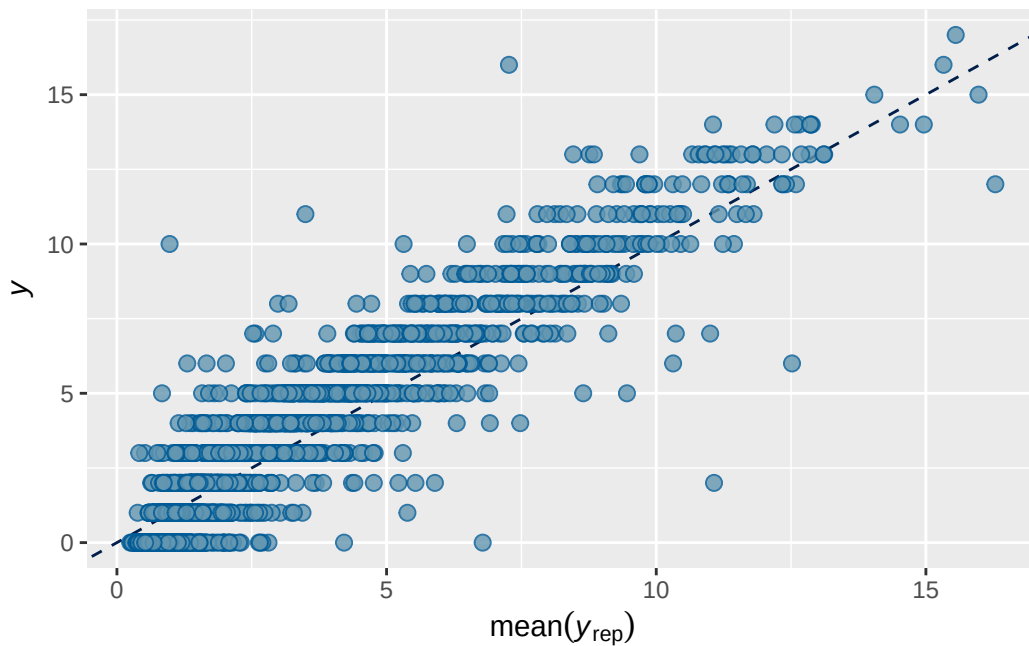
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 5B longitudinal stimulant output

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.8239053	0.005205048	0.8133844	0.8337296

Family: poisson
 Links: mu = log
 Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age.
 Data: analytic_df_wide (Number of observations: 4515)
 Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
 total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.986	0.020	0.946	1.026	1.001	2381	4669

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.065	0.037	0.005	0.145	1.009	611	1517

Regression Coefficients:

	Estimate	Est.Error		
Intercept	0.503	0.035		
breast_development_p_wave_2_sc	0.188	0.038		
adhd_traits_wave_2_sc	0.570	0.036		
age_wave_2_sc	-0.119	0.038		
ethnraceblack_nh	-0.129	0.066		
ethnracehispanic	-0.096	0.057		
ethnraceother	-0.082	0.064		
inr_wave_0_sc	-0.004	0.042		
externalising_wave_0_sc	0.914	0.035		
stimulant_status_wave_3Y	0.111	0.074		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.175	0.065		
	1-95% CI	u-95% CI	Rhat	
Intercept	0.435	0.571	1.001	
breast_development_p_wave_2_sc	0.114	0.264	1.001	
adhd_traits_wave_2_sc	0.498	0.641	1.001	
age_wave_2_sc	-0.193	-0.045	1.000	
ethnraceblack_nh	-0.260	0.000	1.003	
ethnracehispanic	-0.206	0.015	1.000	
ethnraceother	-0.209	0.044	1.001	
inr_wave_0_sc	-0.086	0.079	1.001	
externalising_wave_0_sc	0.846	0.982	1.001	
stimulant_status_wave_3Y	-0.033	0.257	1.001	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.044	0.304	1.000	
	Bulk_ESS	Tail_ESS		
Intercept	4138	4789		
breast_development_p_wave_2_sc	4101	5676		
adhd_traits_wave_2_sc	2777	4361		
age_wave_2_sc	3692	5187		
ethnraceblack_nh	2311	4174		
ethnracehispanic	3236	4672		
ethnraceother	2893	4509		
inr_wave_0_sc	3251	5037		
externalising_wave_0_sc	2506	3980		
stimulant_status_wave_3Y	2319	3887		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	2940	4482		

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Model 5B longitudinal stimulant diagnostics

```

prior      class
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
normal(0, 1)      b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd
student_t(3, 0, 2.9)      sd

coef      group resp dpar nlpar

adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
ethnraceblack_nh
ethnracehispanic
ethnraceother
externalising_wave_0_sc
inr_wave_0_sc
stimulant_status_wave_3Y

family_id
Intercept family_id
site_id
Intercept site_id

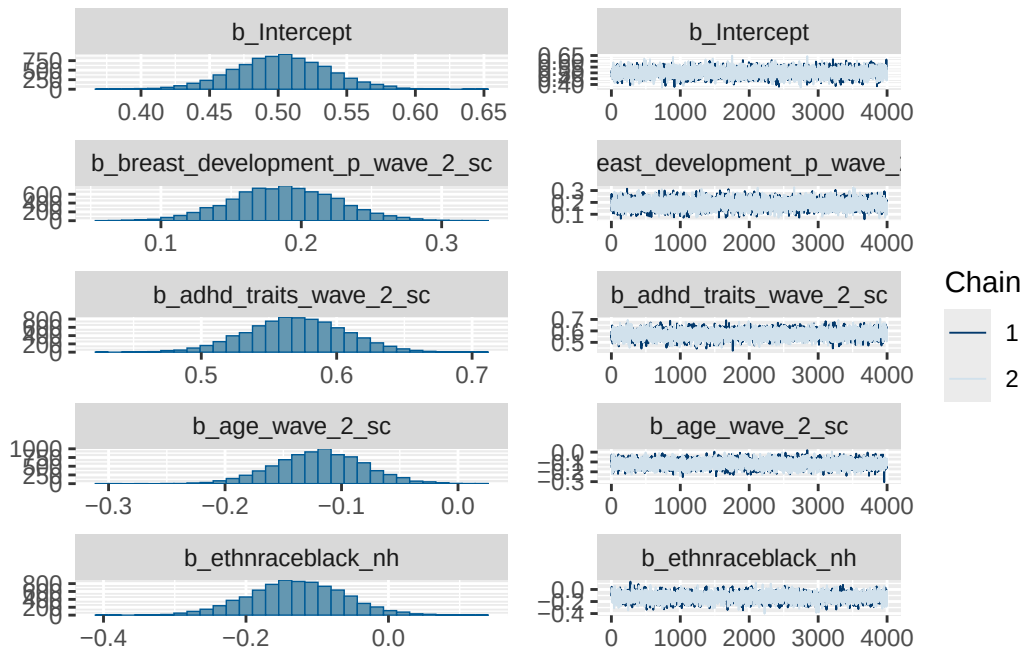
lb ub tag      source
user
(vectorized)
(vectorized)

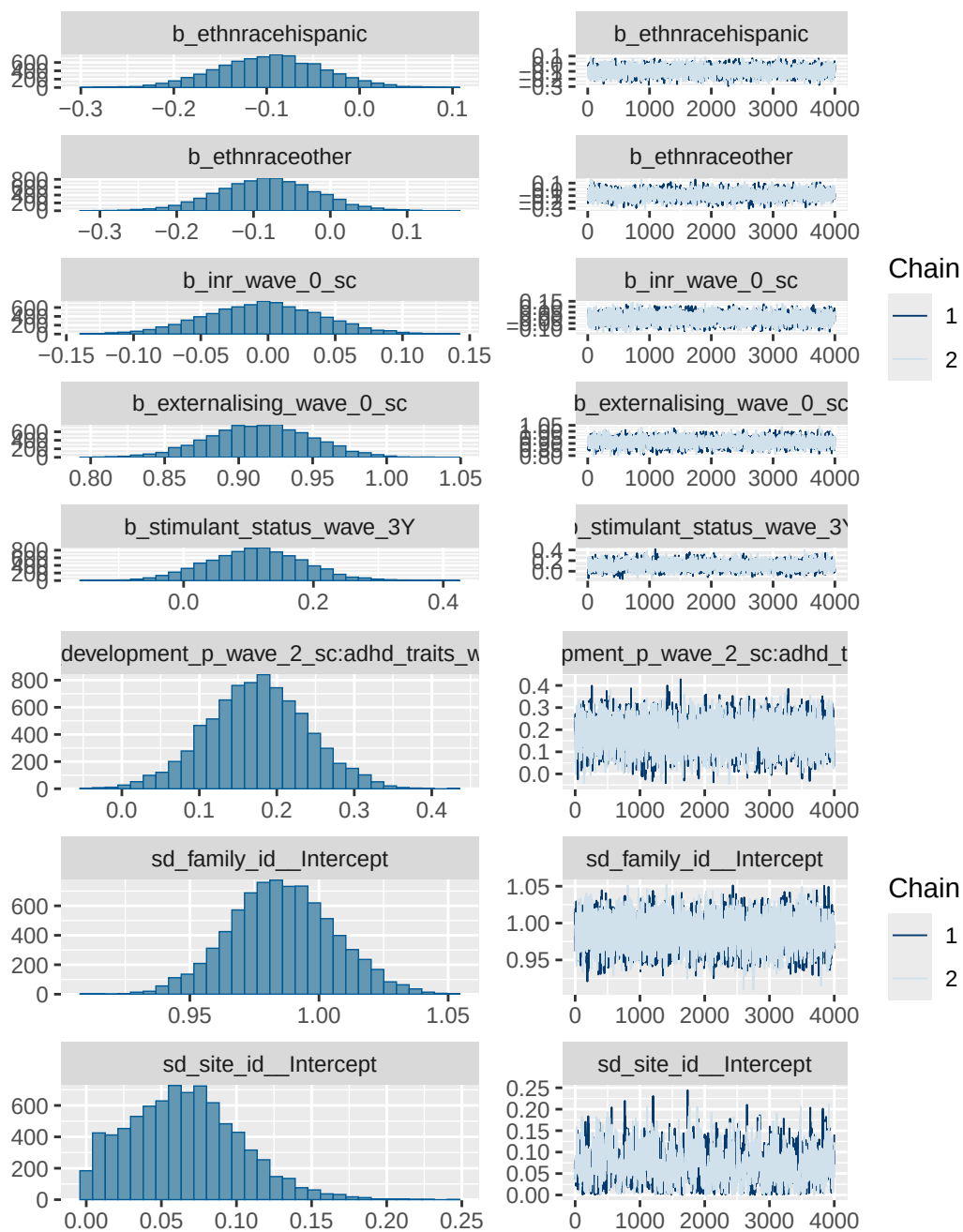
```

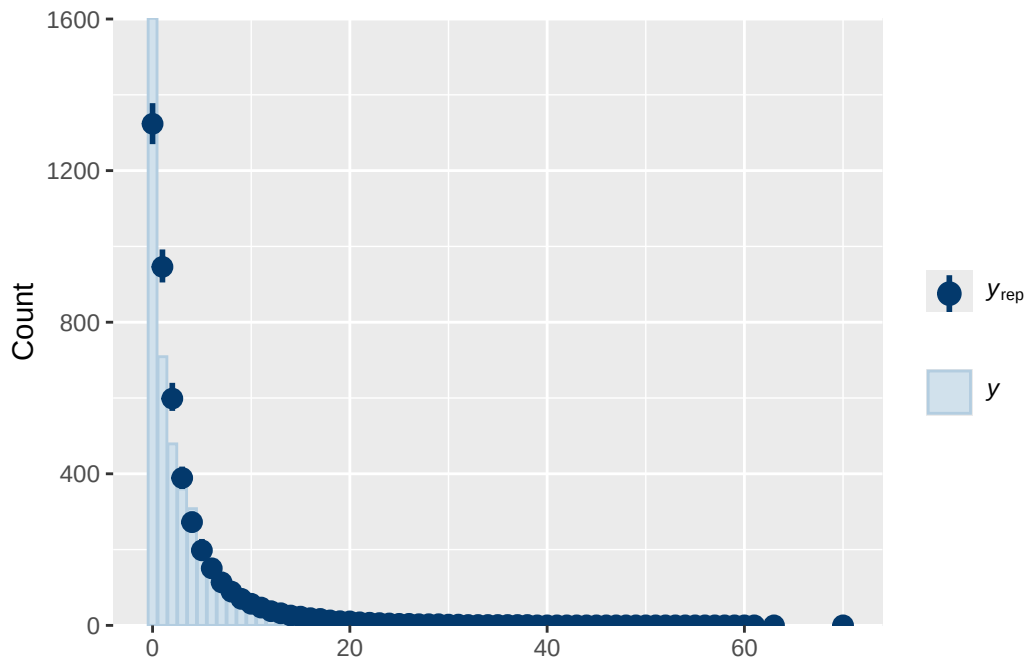
```

(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

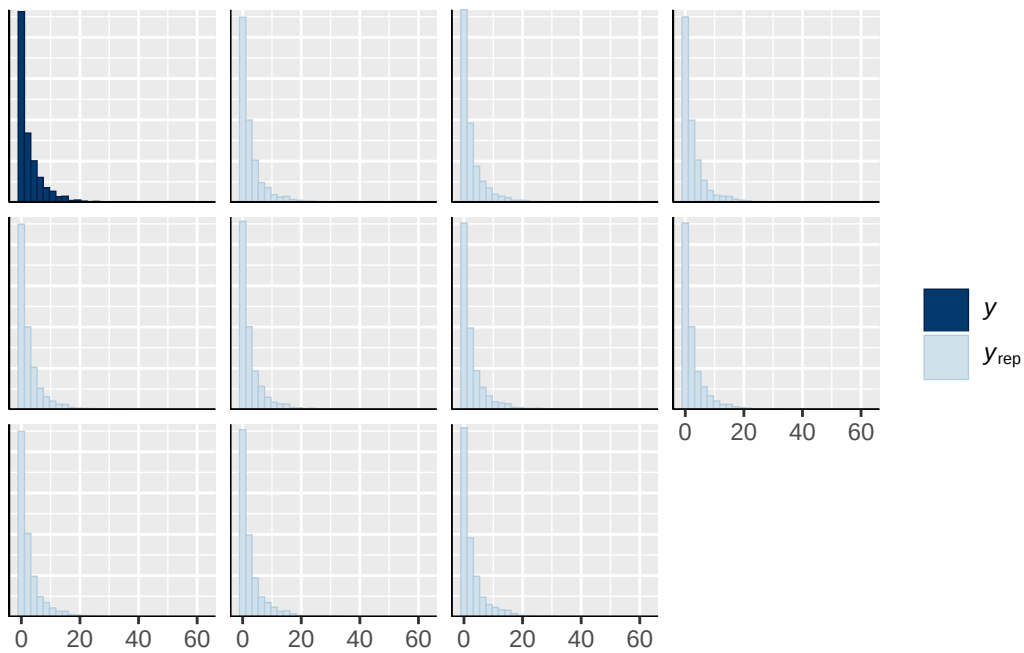
```



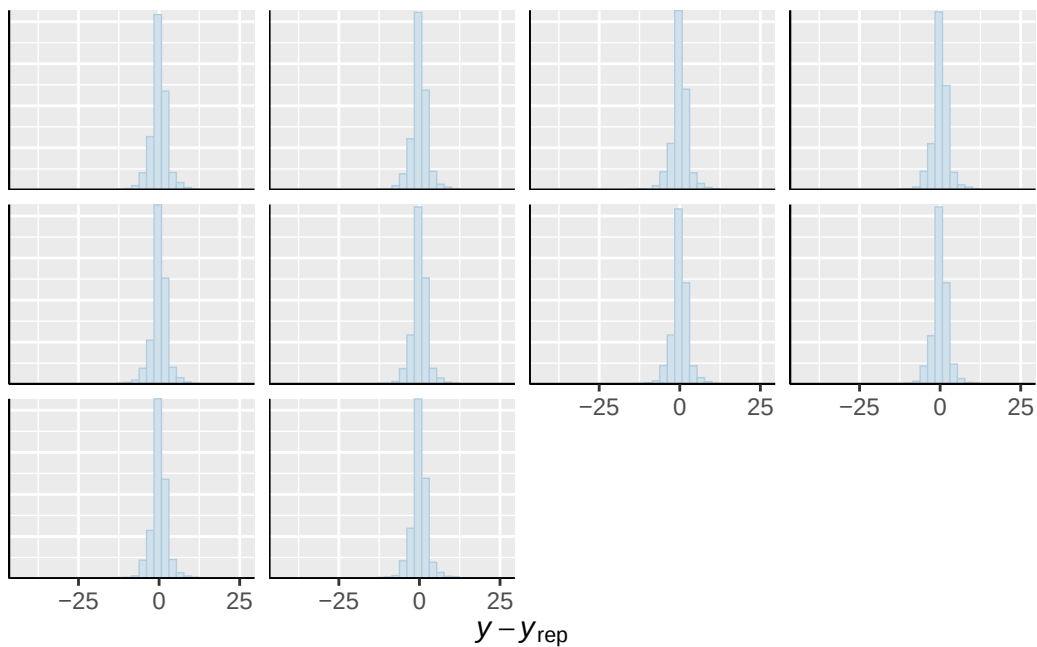




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

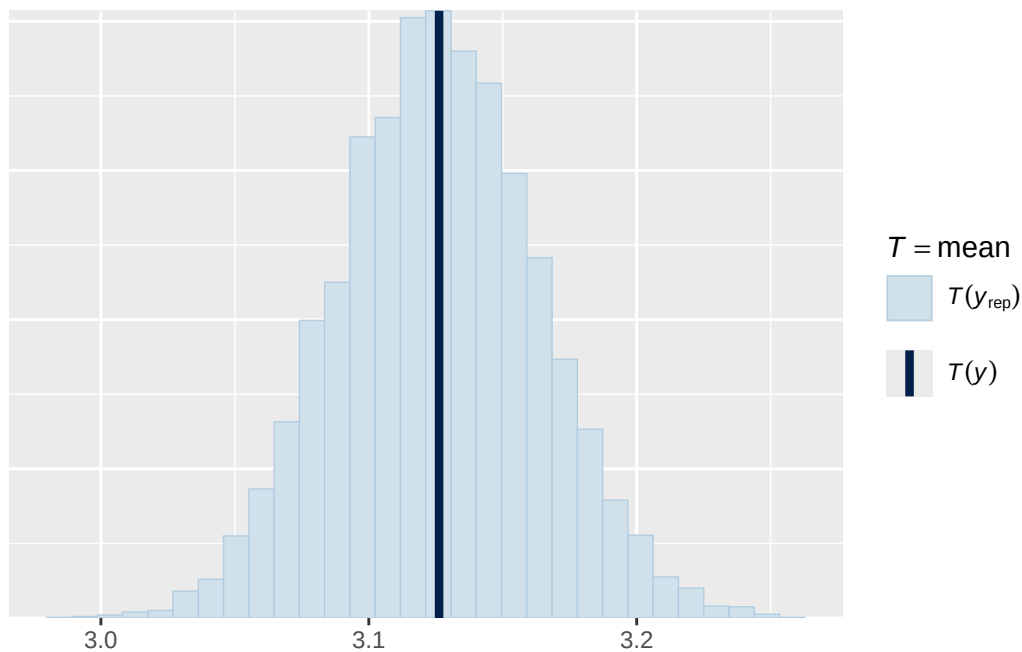


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

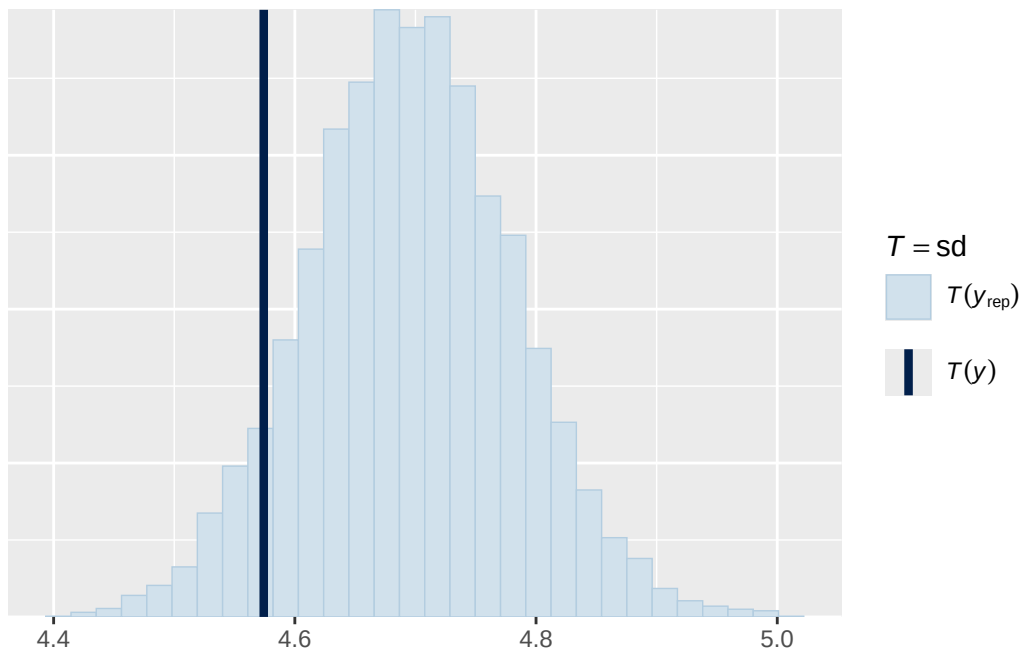


Using all posterior draws for ppc type 'stat' by default.

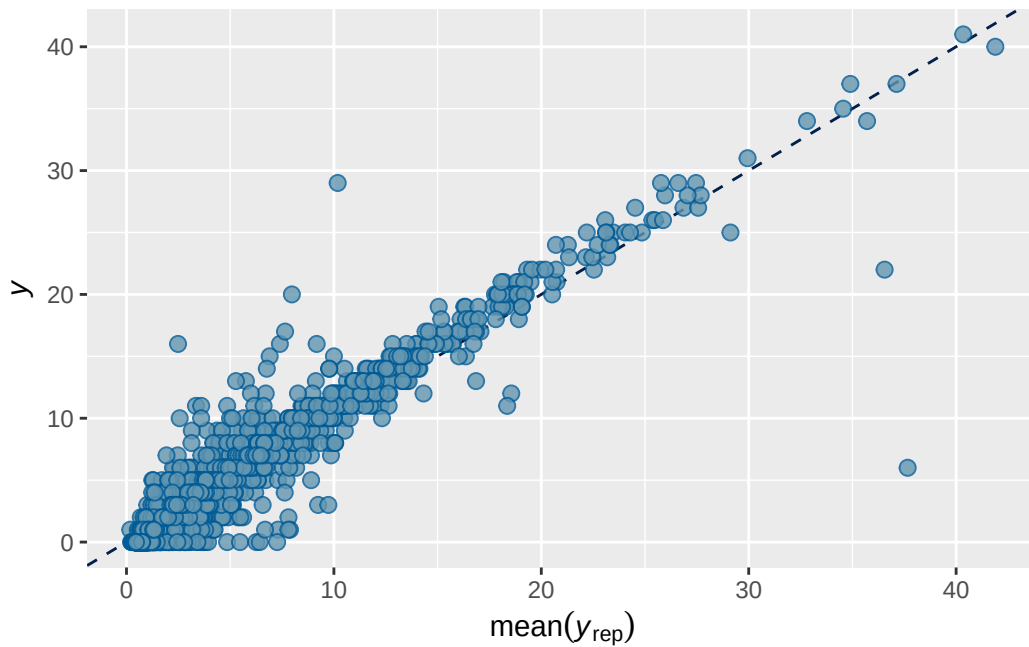
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use `stat_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1 longitudinal prior 1 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7348938 0.007531416 0.7199932 0.7493823

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4496)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3915)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.881    0.020   0.841   0.922 1.007     937    1457

```

```

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.092    0.036   0.020   0.165 1.007     504     518

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.150    0.039   0.072   0.224 1.003    1400
menarche_status_p_wave_2Y  0.099    0.040   0.022   0.178 1.001    1665
age_wave_2_sc       -0.113    0.040  -0.191  -0.038 1.001    1723
ethnraceblack_nh    -0.077    0.063  -0.197   0.049 1.001    1514
ethnracehispanic    -0.042    0.054  -0.146   0.066 1.001    1381
ethnraceother        0.003    0.060  -0.116   0.121 1.001    1324
inr_wave_0_sc        0.100    0.041   0.018   0.182 1.000    1489
adhd_traits_wave_0_sc    1.307    0.033   1.242   1.374 1.001    1073
      Tail_ESS
Intercept          2033
menarche_status_p_wave_2Y  2492
age_wave_2_sc       2637
ethnraceblack_nh    2412
ethnracehispanic    2077
ethnraceother       1677
inr_wave_0_sc       2112
adhd_traits_wave_0_sc  1886

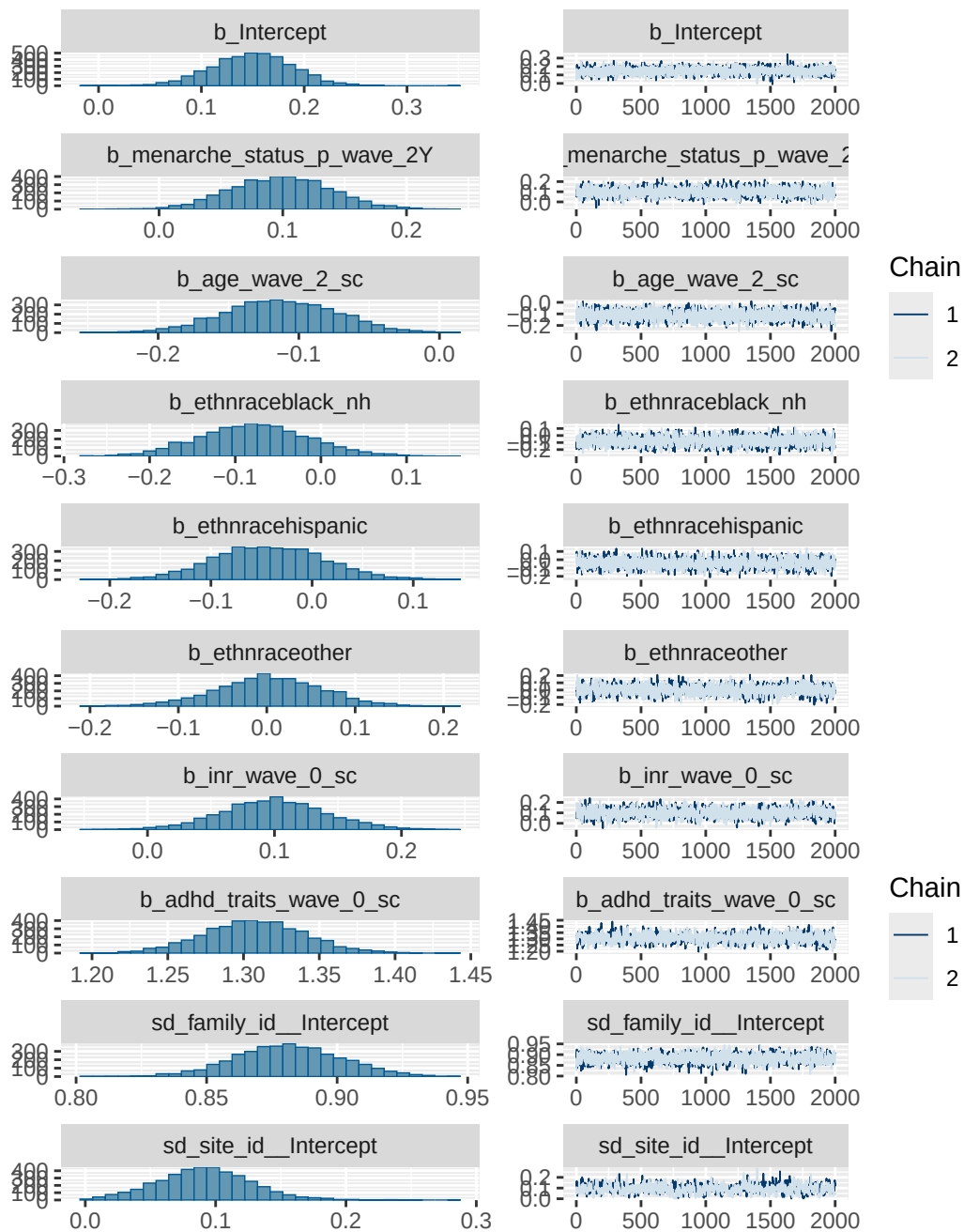
```

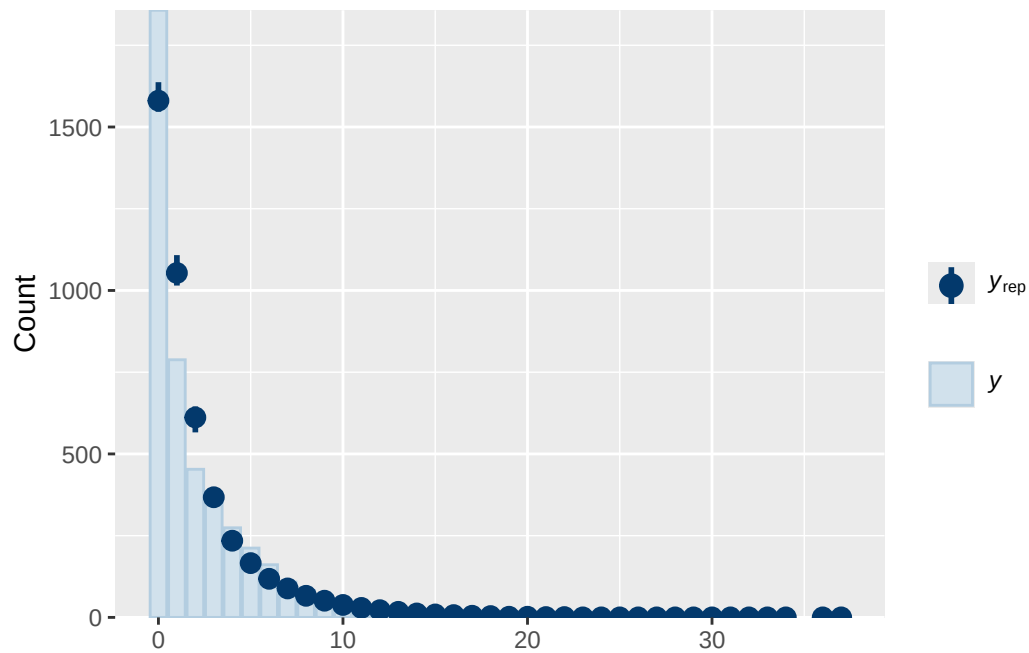
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

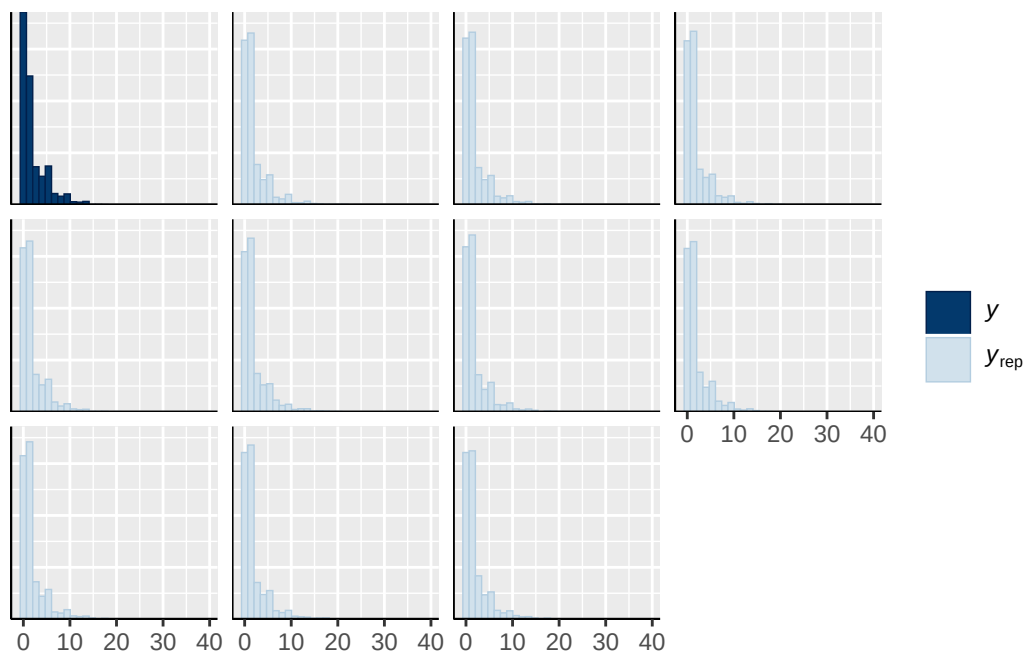
Model 1 longitudinal prior 1 diagnostics

	prior	class	coef	group	resp	dpar
	normal(0, 0.25)	b				
	normal(0, 0.25)	b	adhd_traits_wave_0_sc			
	normal(0, 0.25)	b	age_wave_2_sc			
	normal(0, 0.25)	b	ethnraceblack_nh			
	normal(0, 0.25)	b	ethnracehispanic			
	normal(0, 0.25)	b	ethnraceother			
	normal(0, 0.25)	b	inr_wave_0_sc			
	normal(0, 0.25)	b	menarche_status_p_wave_2Y			
student_t(3, 0, 2.7)	Intercept					
student_t(3, 0, 2.7)	sd					
student_t(3, 0, 2.7)	sd			family_id		
student_t(3, 0, 2.7)	sd		Intercept	family_id		
student_t(3, 0, 2.7)	sd			site_id		
student_t(3, 0, 2.7)	sd		Intercept	site_id		
nlpar	lb	ub	tag	source		
				user		
				(vectorized)		
				(vectorized)		
				(vectorized)		
				(vectorized)		
				(vectorized)		
				(vectorized)		
				(vectorized)		
				default		
0				default		
0				(vectorized)		
0				(vectorized)		
0				(vectorized)		
0				(vectorized)		

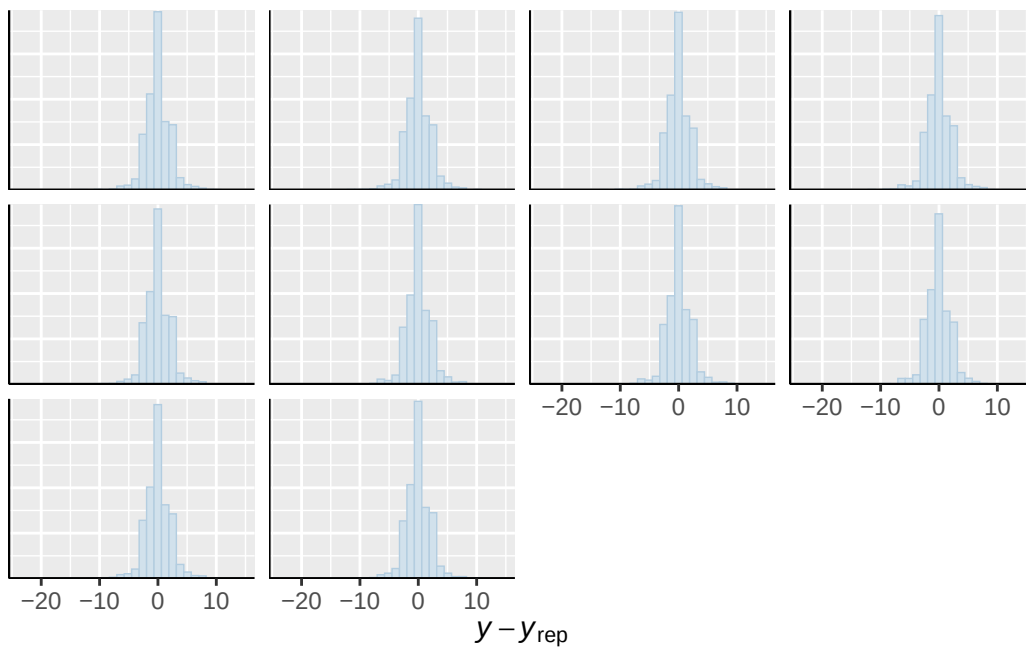




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

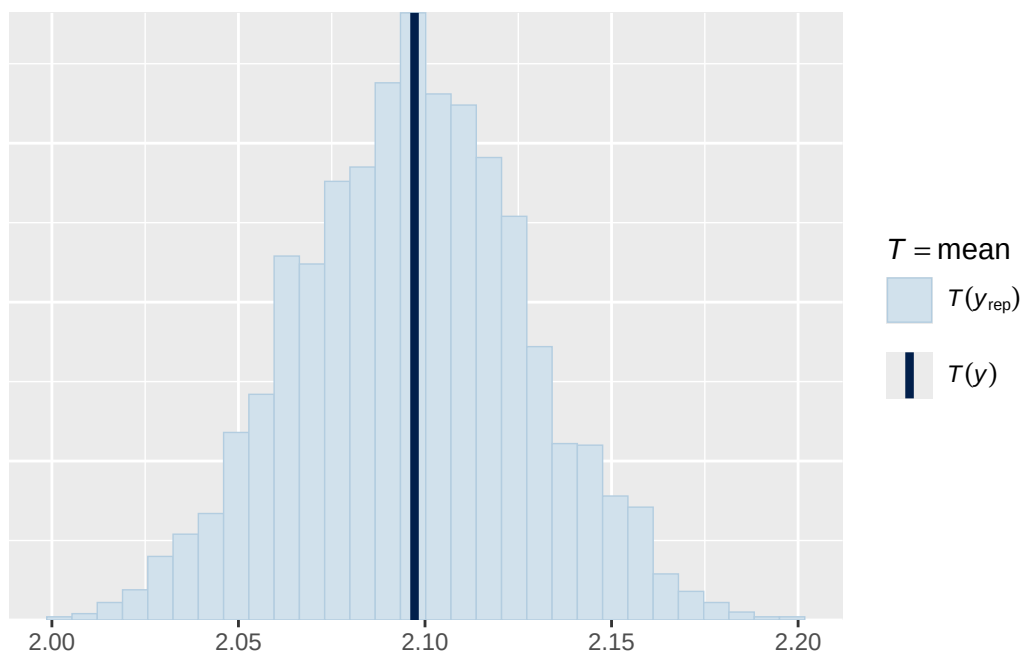


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

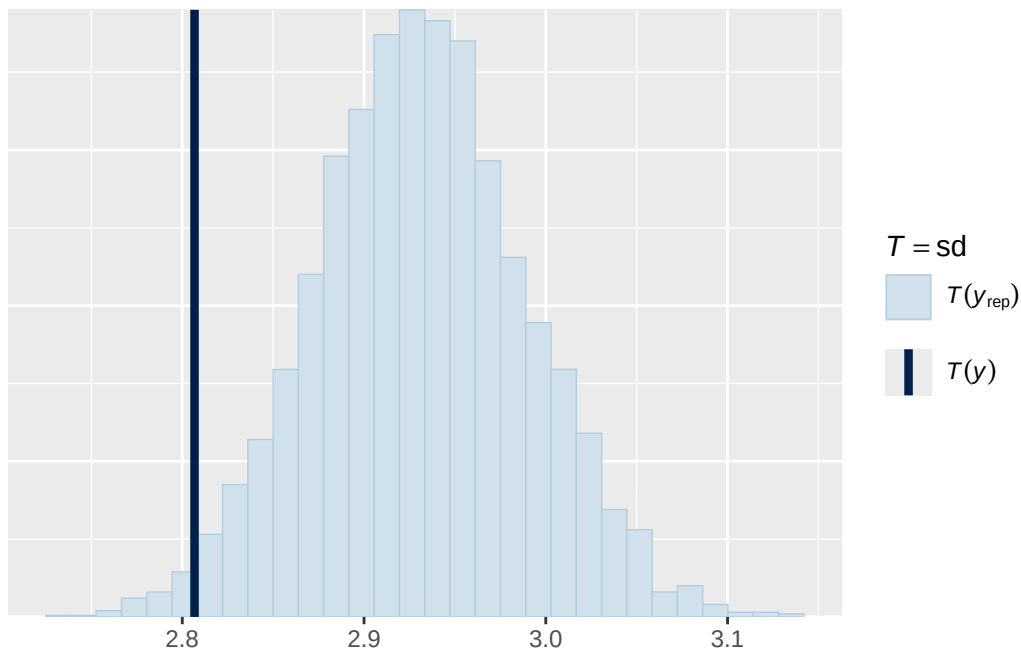


Using all posterior draws for ppc type 'stat' by default.

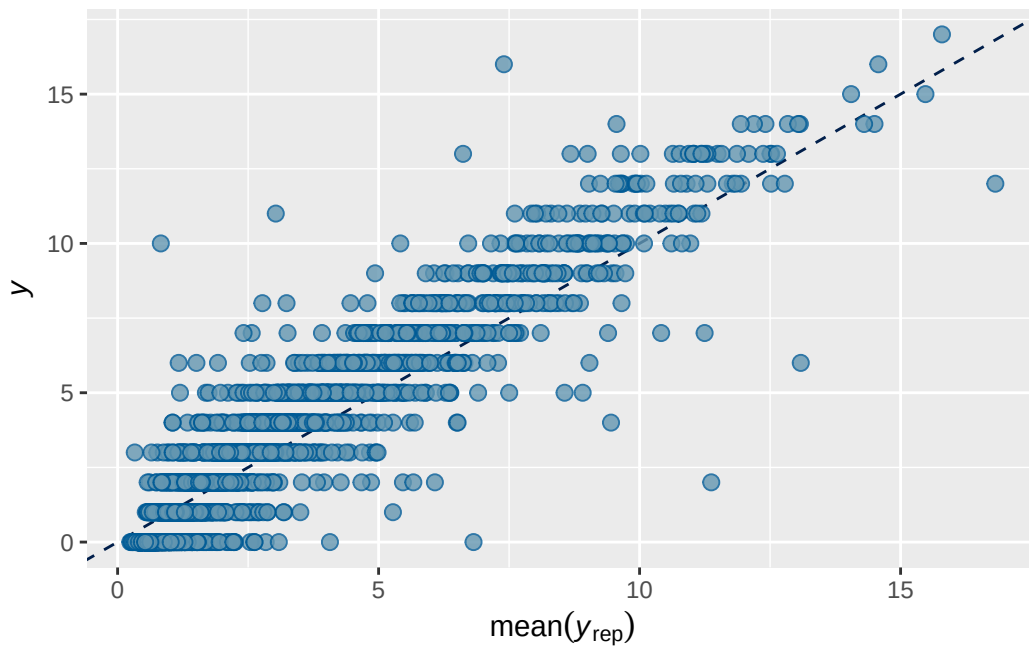
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1B longitudinal prior 1 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.7345825 0.007551707 0.7193221 0.7491023

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.883    0.021   0.844   0.924 1.001    1351    2068

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093    0.035   0.024   0.166 1.004     427     447

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat
Intercept          0.193    0.036   0.118   0.264 1.001
breast_development_p_wave_2_sc 0.118    0.040   0.039   0.194 1.002
age_wave_2_sc      -0.101    0.039  -0.176  -0.024 1.002
ethnraceblack_nh   -0.082    0.062  -0.205   0.039 1.000
ethnracehispanic   -0.031    0.055  -0.135   0.081 1.001
ethnraceother      -0.007    0.059  -0.125   0.107 1.001
inr_wave_0_sc       0.084    0.041   0.001   0.162 1.001
adhd_traits_wave_0_sc 1.300    0.033   1.233   1.366 1.001

      Bulk_ESS Tail_ESS
Intercept          1961    2153
breast_development_p_wave_2_sc 1659    2617
age_wave_2_sc      1453    2377
ethnraceblack_nh   1491    2487
ethnracehispanic   1159    1633
ethnraceother      1565    2433
inr_wave_0_sc       1462    2222
adhd_traits_wave_0_sc 1081    2297

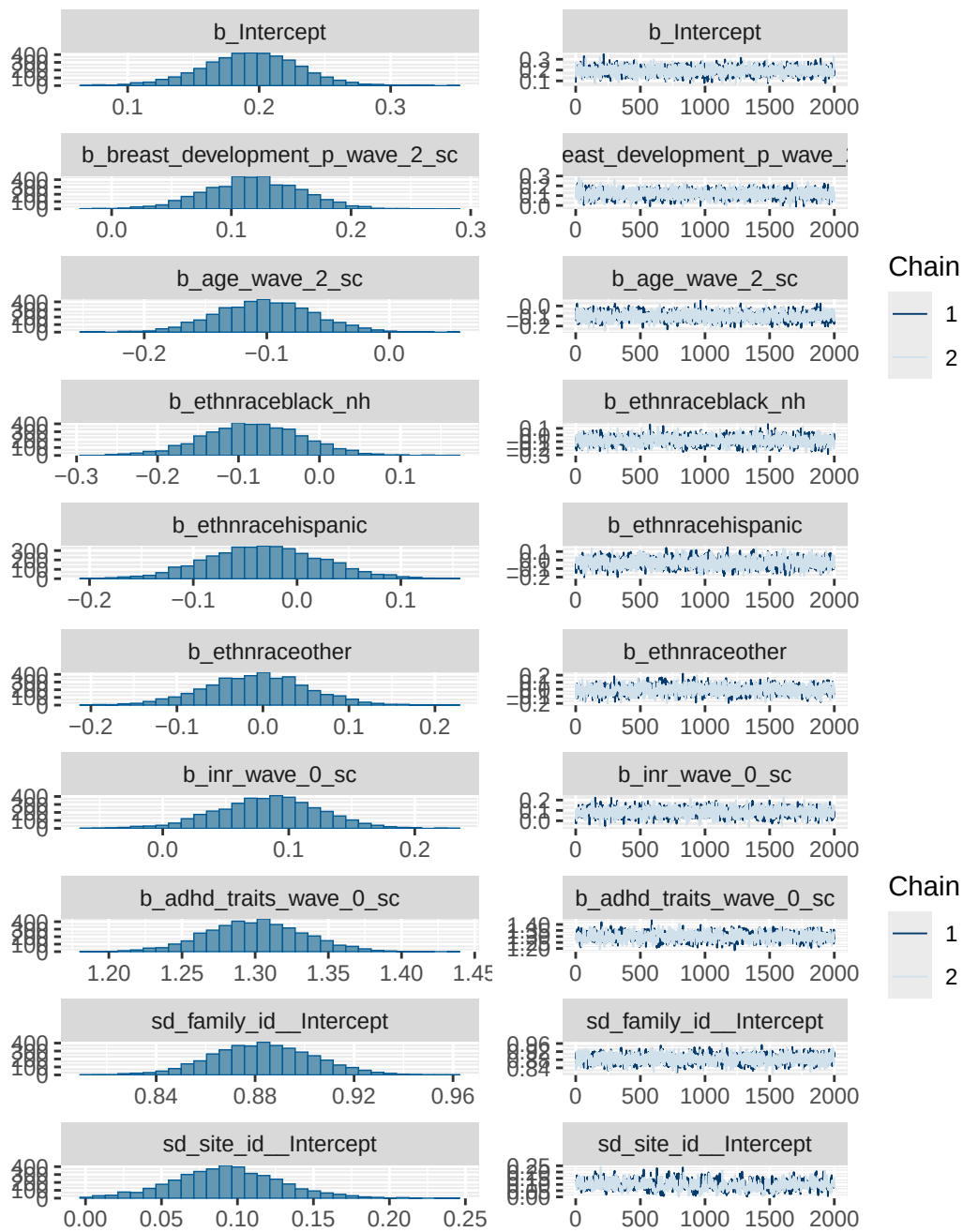
```

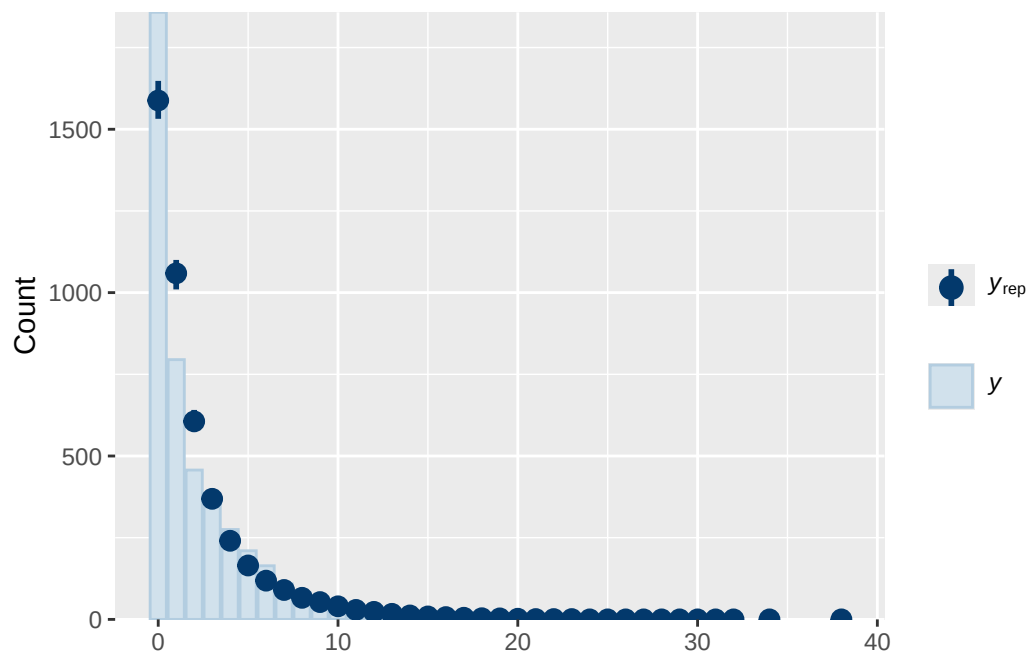
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

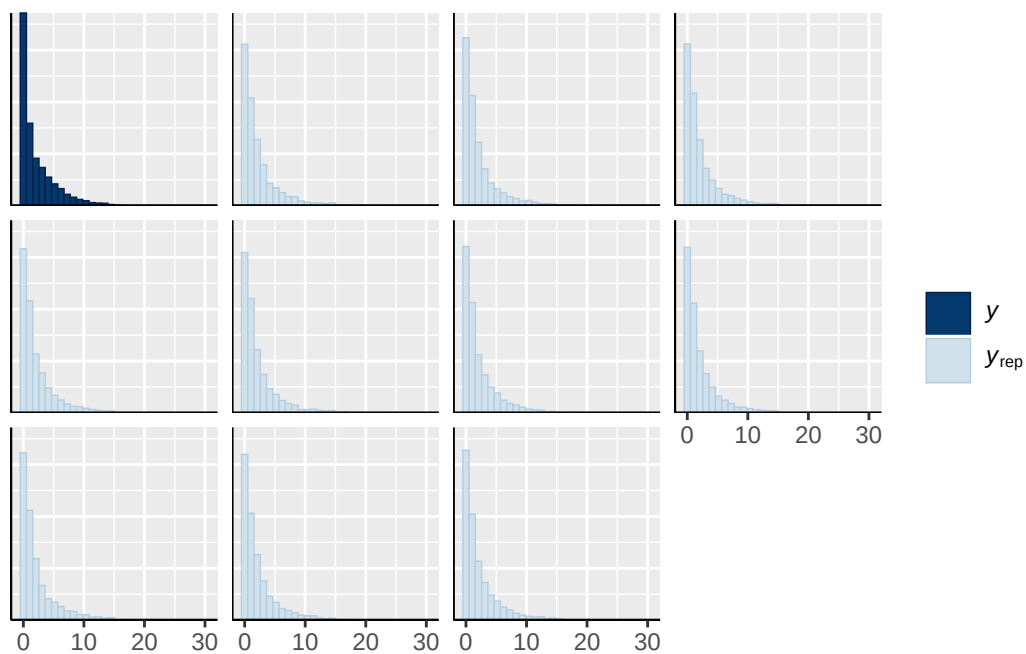
Model 1B longitudinal prior 1 diagnostics

	prior	class	coef	group	resp
	normal(0, 0.25)	b			
	normal(0, 0.25)	b	adhd_traits_wave_0_sc		
	normal(0, 0.25)	b	age_wave_2_sc		
	normal(0, 0.25)	b	breast_development_p_wave_2_sc		
	normal(0, 0.25)	b	ethnraceblack_nh		
	normal(0, 0.25)	b	ethnracehispanic		
	normal(0, 0.25)	b	ethnraceother		
	normal(0, 0.25)	b	inr_wave_0_sc		
student_t(3, 0, 2.7)	Intercept				
student_t(3, 0, 2.7)	sd				
student_t(3, 0, 2.7)	sd			family_id	
student_t(3, 0, 2.7)	sd		Intercept	family_id	
student_t(3, 0, 2.7)	sd			site_id	
student_t(3, 0, 2.7)	sd		Intercept	site_id	
dpar nlpar lb ub tag		source			
		user			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		default			
0		default			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			





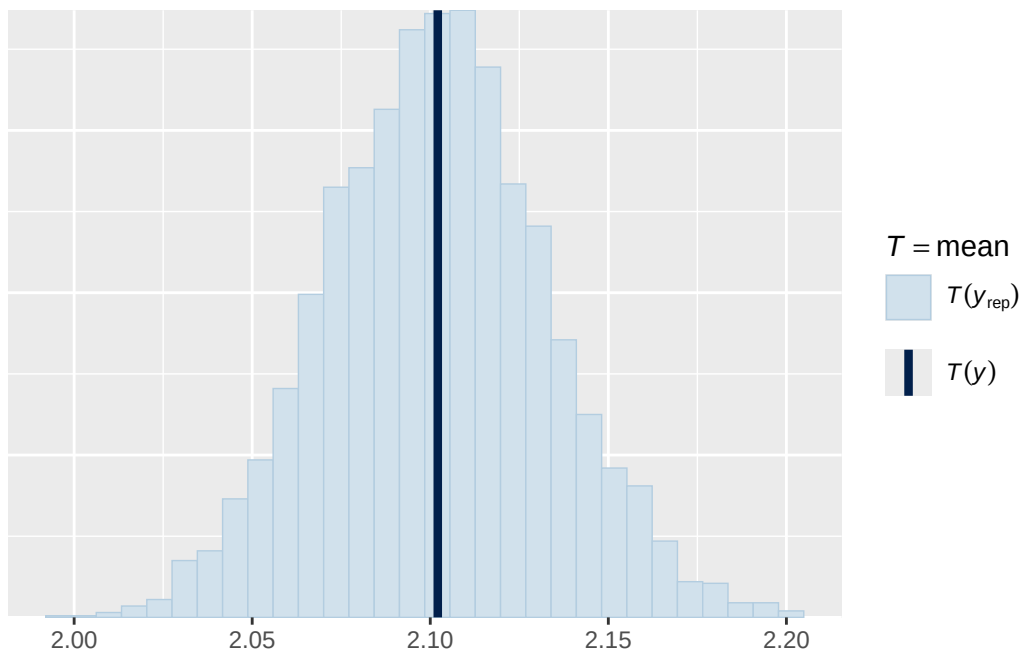
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



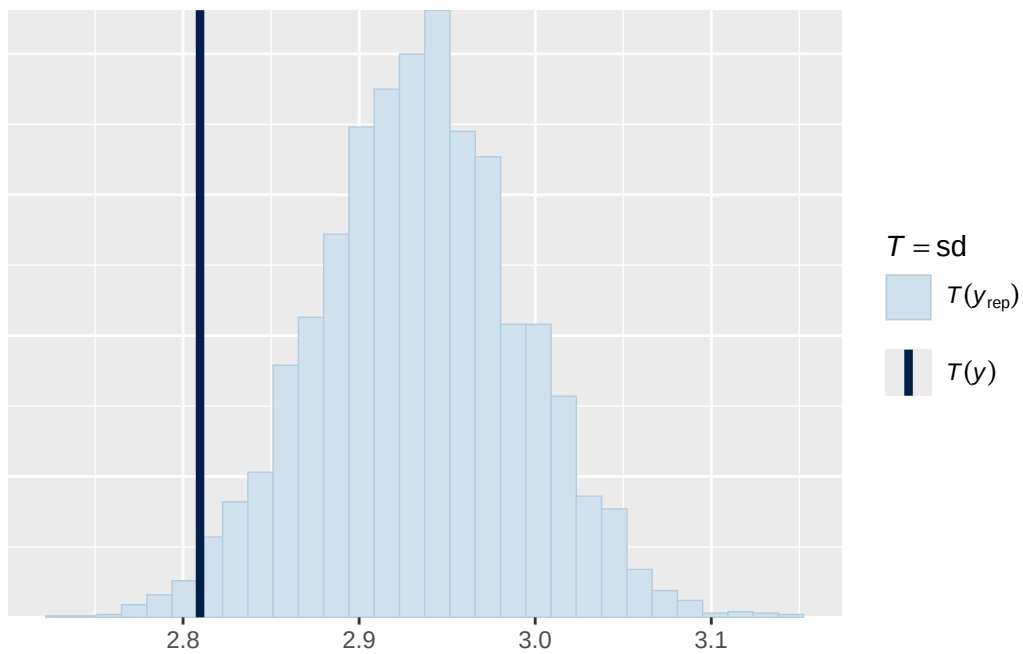
Using all posterior draws for ppc type 'stat' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.

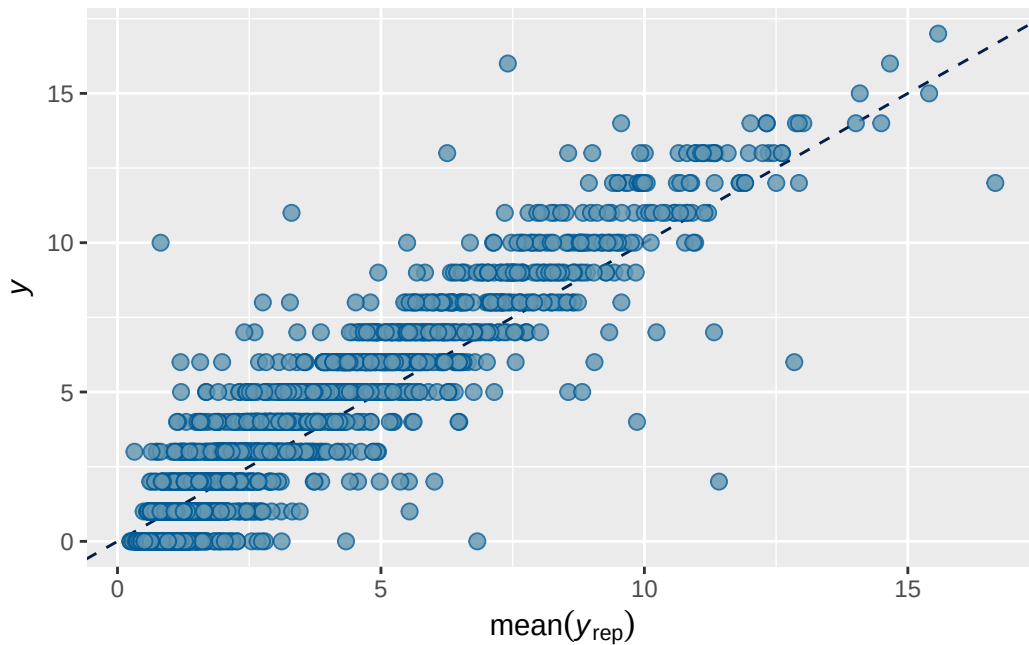
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 5B longitudinal prior 1 output

```

      Estimate   Est.Error    Q2.5    Q97.5
R2 0.8229693 0.005223904 0.8125308 0.8329675

```

Family: poisson

Links: mu = log

Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age

Data: analytic_df_wide (Number of observations: 4515)

Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
total post-warmup draws = 8000

Multilevel Hyperparameters:

~family_id (Number of levels: 3928)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.983	0.020	0.946	1.023	1.002	1810	3364

~site_id (Number of levels: 22)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.067	0.037	0.005	0.147	1.002	549	1105

Regression Coefficients:

	Estimate	Est.Error		
Intercept	0.508	0.035		
breast_development_p_wave_2_sc	0.185	0.038		
adhd_traits_wave_2_sc	0.583	0.034		
age_wave_2_sc	-0.114	0.036		
ethnraceblack_nh	-0.121	0.065		
ethnracehispanic	-0.090	0.056		
ethnraceother	-0.072	0.062		
inr_wave_0_sc	0.000	0.042		
externalising_wave_0_sc	0.903	0.034		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.159	0.063		
	1-95% CI	u-95% CI	Rhat	
Intercept	0.437	0.577	1.001	
breast_development_p_wave_2_sc	0.111	0.260	1.001	
adhd_traits_wave_2_sc	0.515	0.651	1.002	
age_wave_2_sc	-0.186	-0.043	1.001	
ethnraceblack_nh	-0.251	0.008	1.000	
ethnracehispanic	-0.199	0.020	1.001	
ethnraceother	-0.192	0.049	1.002	
inr_wave_0_sc	-0.081	0.085	1.001	
externalising_wave_0_sc	0.838	0.968	1.000	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.033	0.282	1.001	
	Bulk_ESS	Tail_ESS		
Intercept	1540	2546		
breast_development_p_wave_2_sc	2252	3825		
adhd_traits_wave_2_sc	1485	2841		
age_wave_2_sc	2118	4282		
ethnraceblack_nh	1606	2644		
ethnracehispanic	1581	2946		
ethnraceother	1532	2744		
inr_wave_0_sc	1856	3055		
externalising_wave_0_sc	1264	2791		
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	1515	3545		

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

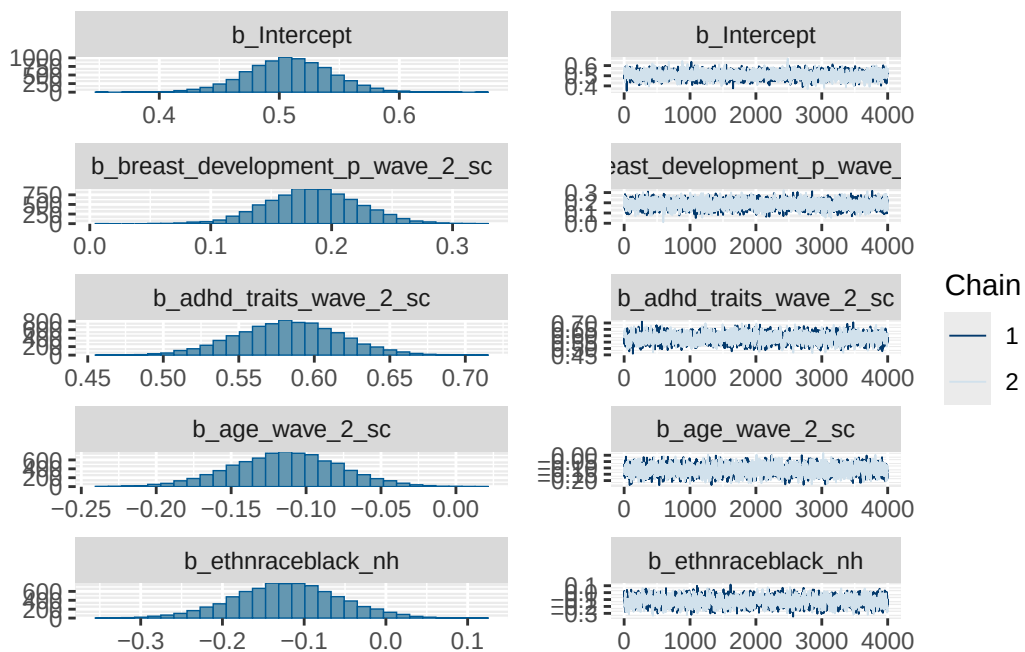
Model 5B longitudinal prior 1 diagnostics

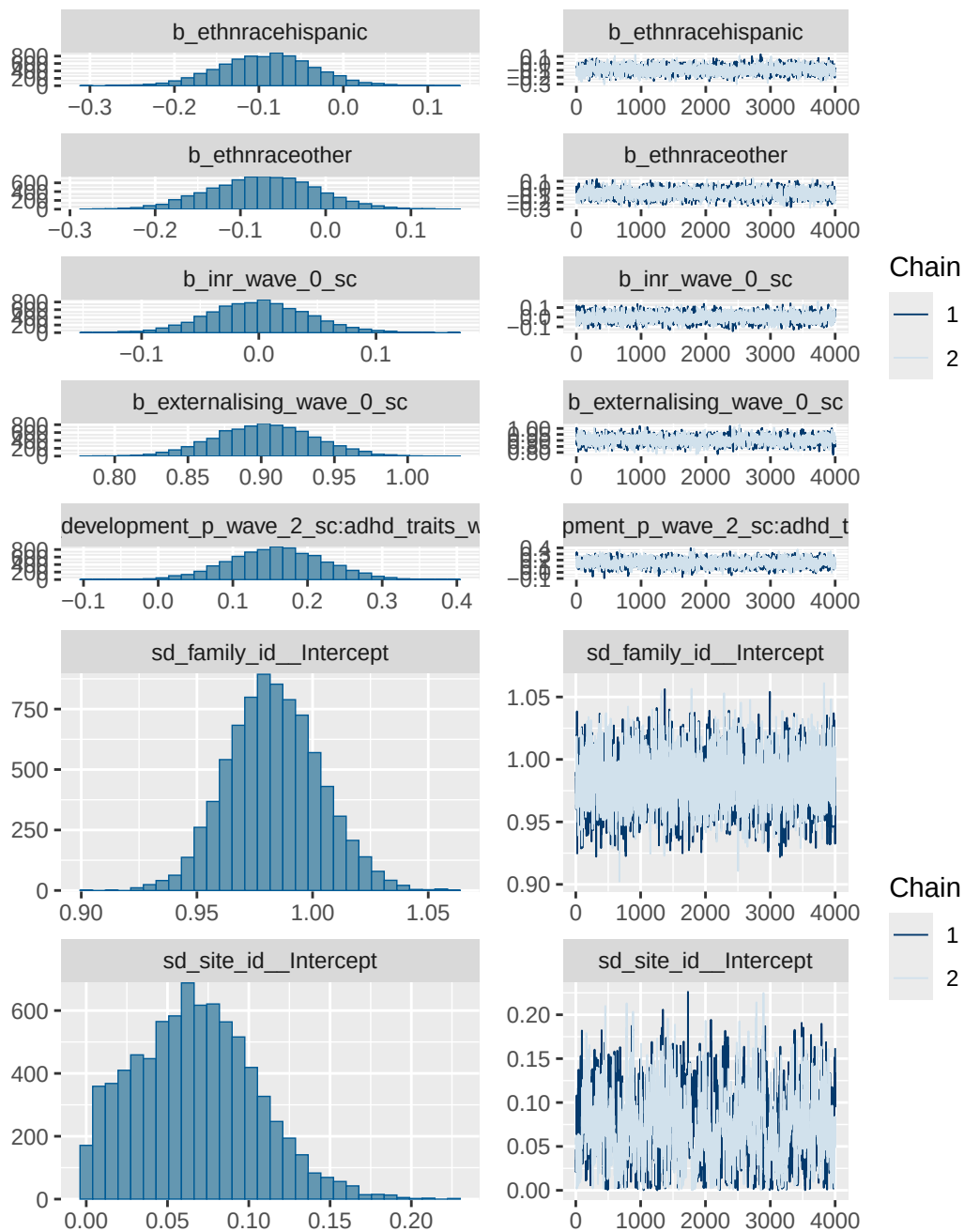
prior	class
normal(0, 0.25)	b

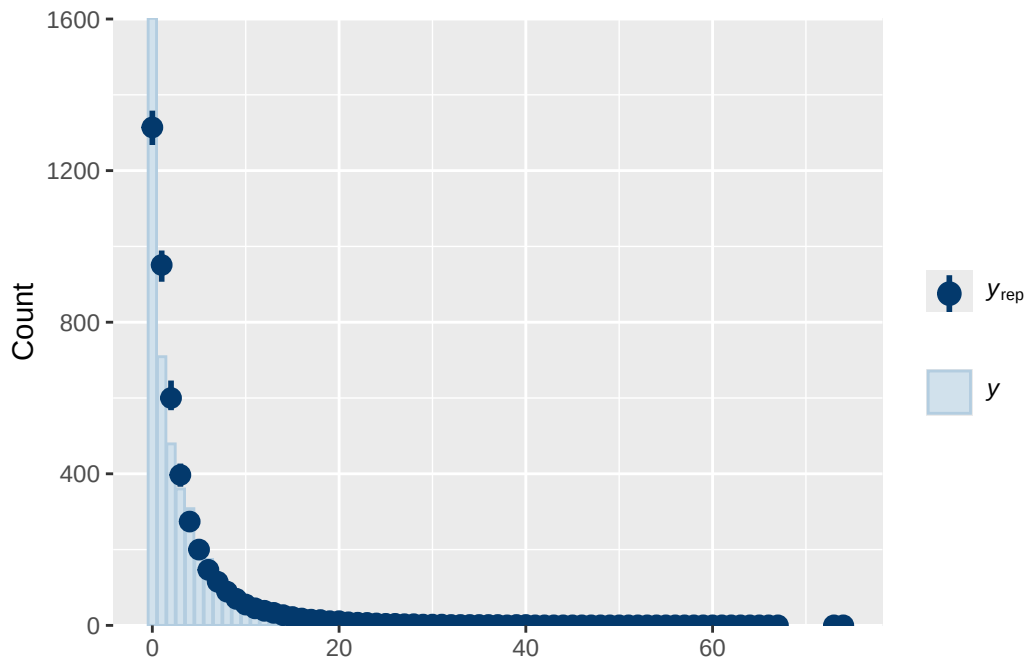

```

                default
0                default
0            (vectorized)
0            (vectorized)
0            (vectorized)
0            (vectorized)
0            (vectorized)

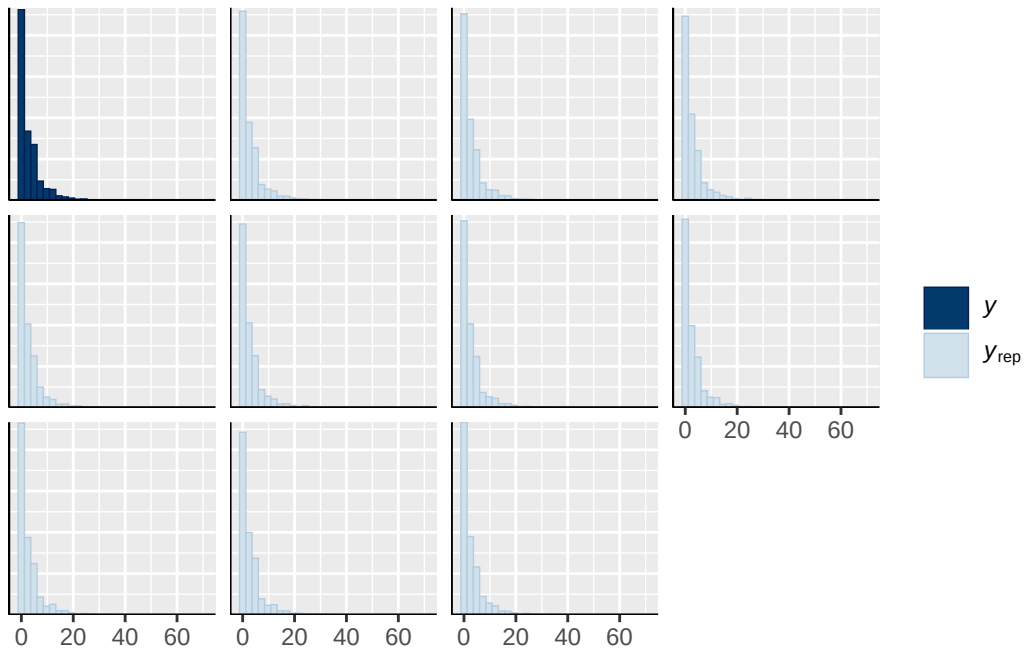
```



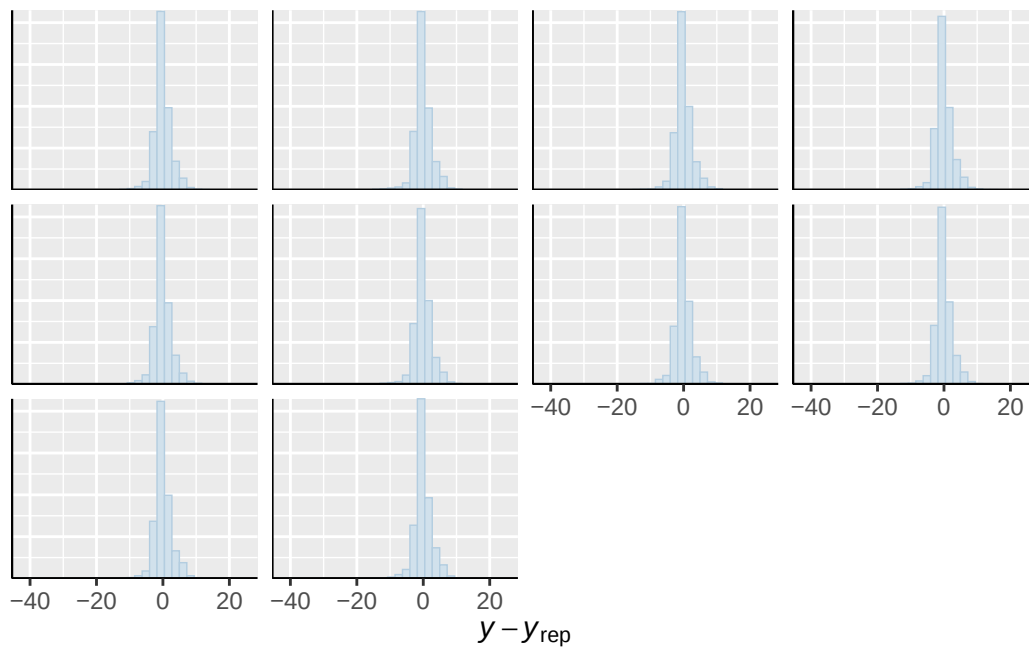




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

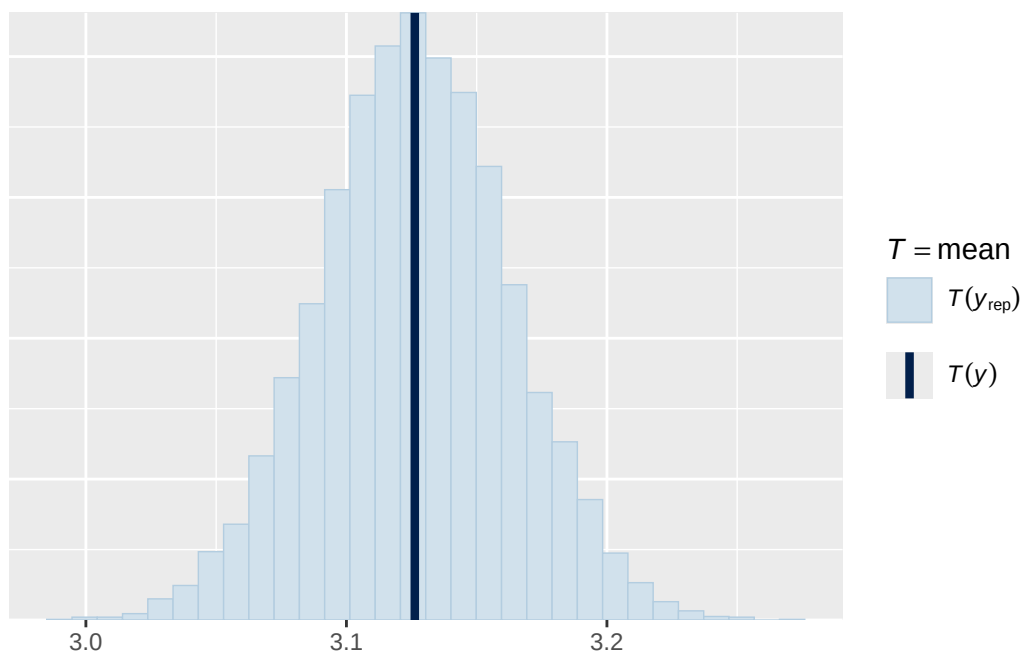


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

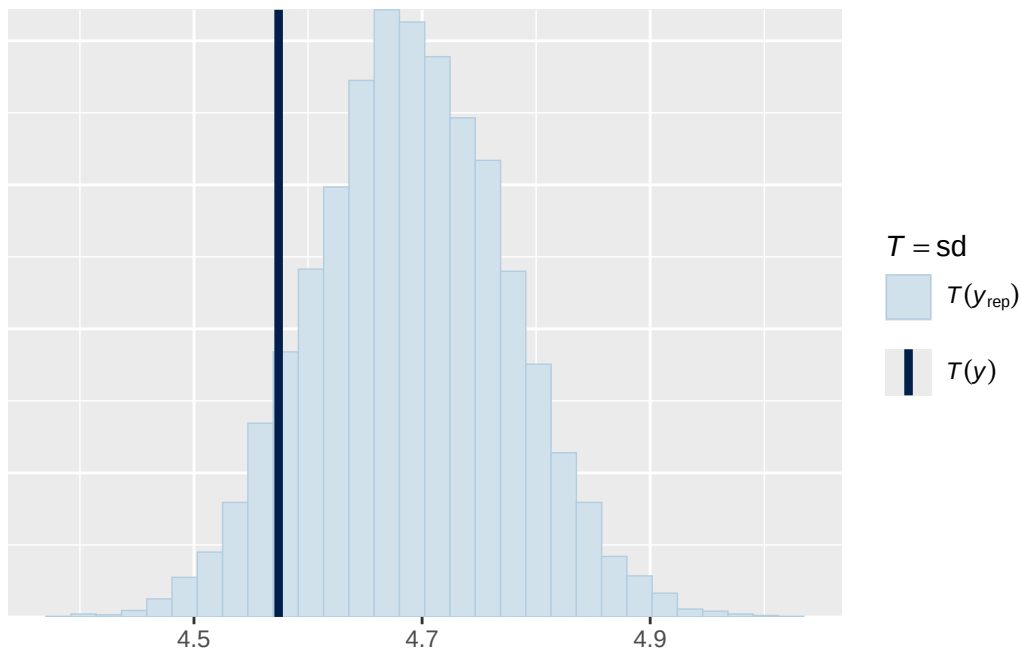


Using all posterior draws for ppc type 'stat' by default.

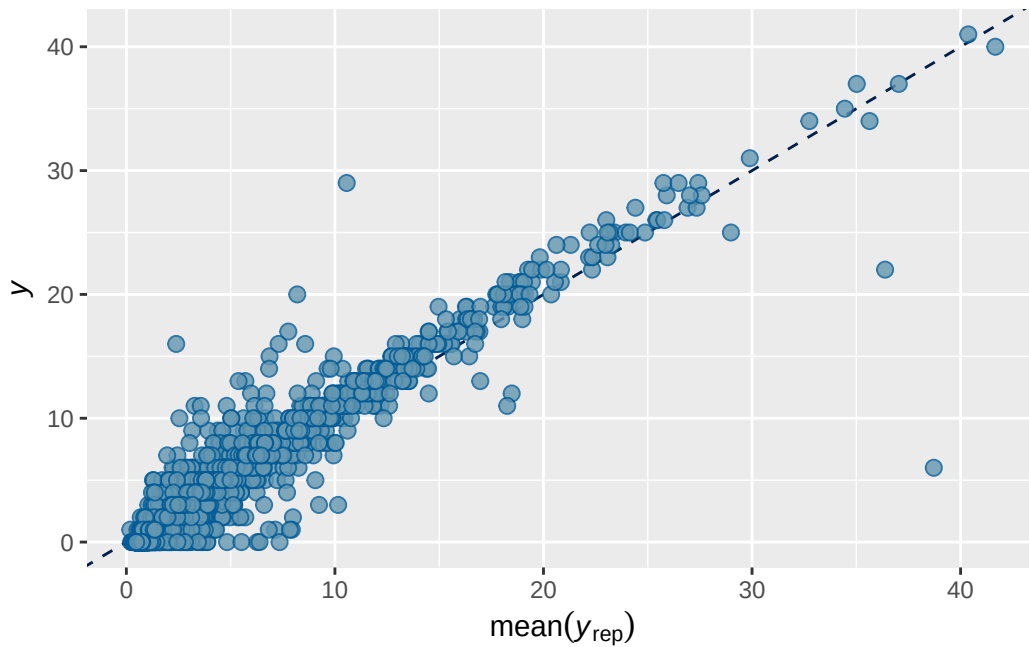
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1 longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5    Q97.5
R2 0.7393702 0.007767695 0.7238316 0.754201

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ menarche_status_p_wave_2 + age_wave_2_sc + ethnrace + inr_wave
Data: analytic_df_wide (Number of observations: 4494)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3913)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.899    0.021   0.857   0.940 1.000    1230    2009

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.090    0.036   0.021   0.163 1.006     556     535

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS
Intercept          0.148    0.040   0.070   0.227 1.001    1838
menarche_status_p_wave_2Y 0.101    0.042   0.019   0.182 1.001    1819
age_wave_2_sc       -0.111    0.041  -0.189  -0.032 1.003    1935
ethnraceblack_nh    -0.093    0.066  -0.224   0.039 1.000    1639
ethnracehispanic    -0.047    0.058  -0.158   0.068 1.000    1431
ethnraceother       -0.006    0.062  -0.128   0.118 1.000    1613
inr_wave_0_sc        0.062    0.043  -0.020   0.149 1.000    1648
adhd_traits_wave_0_sc  1.315    0.033   1.253   1.381 1.002    1270

      Tail_ESS
Intercept          2837
menarche_status_p_wave_2Y 2607
age_wave_2_sc       2664
ethnraceblack_nh    2775
ethnracehispanic    2236
ethnraceother       2739
inr_wave_0_sc        2050
adhd_traits_wave_0_sc 2183

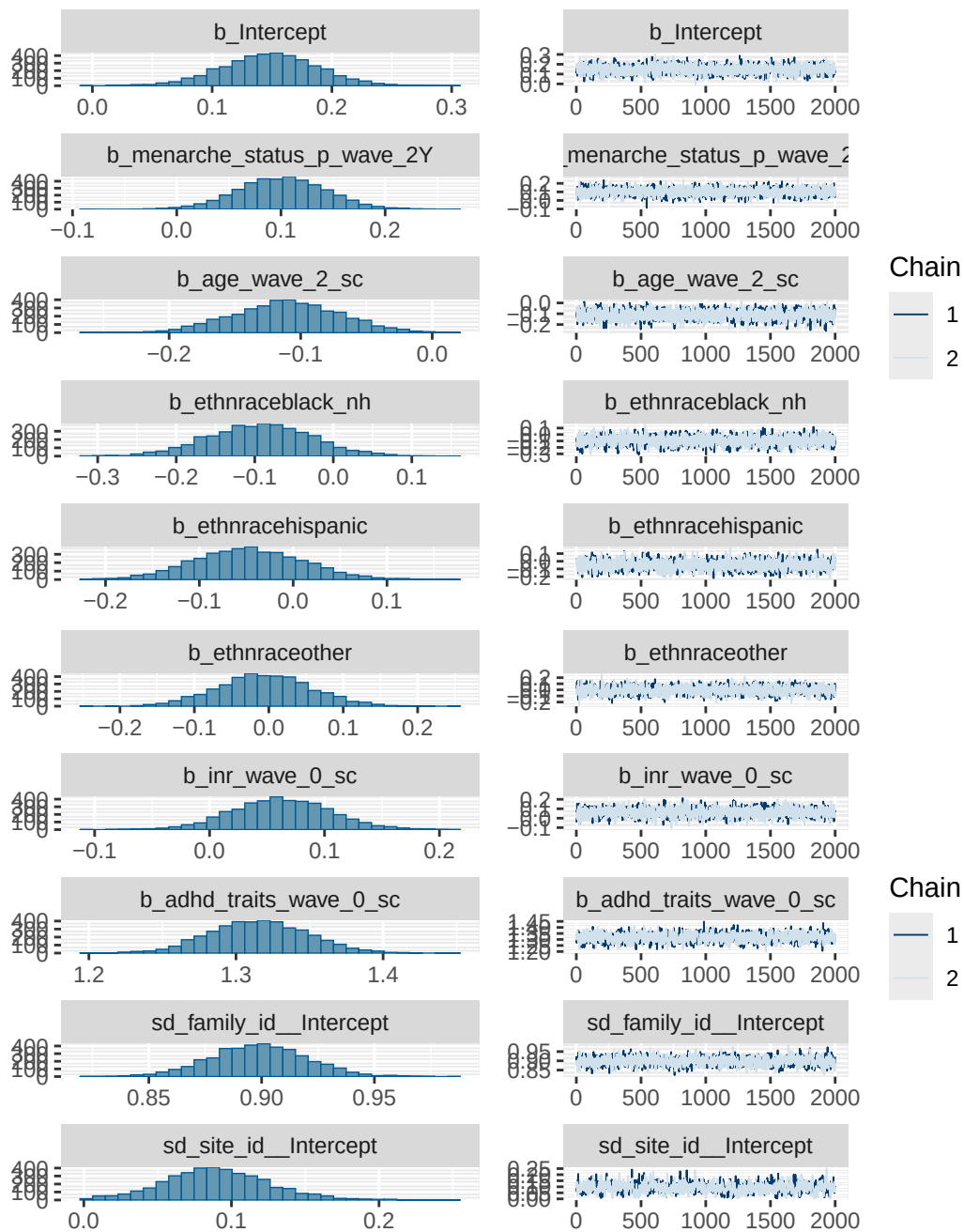
```

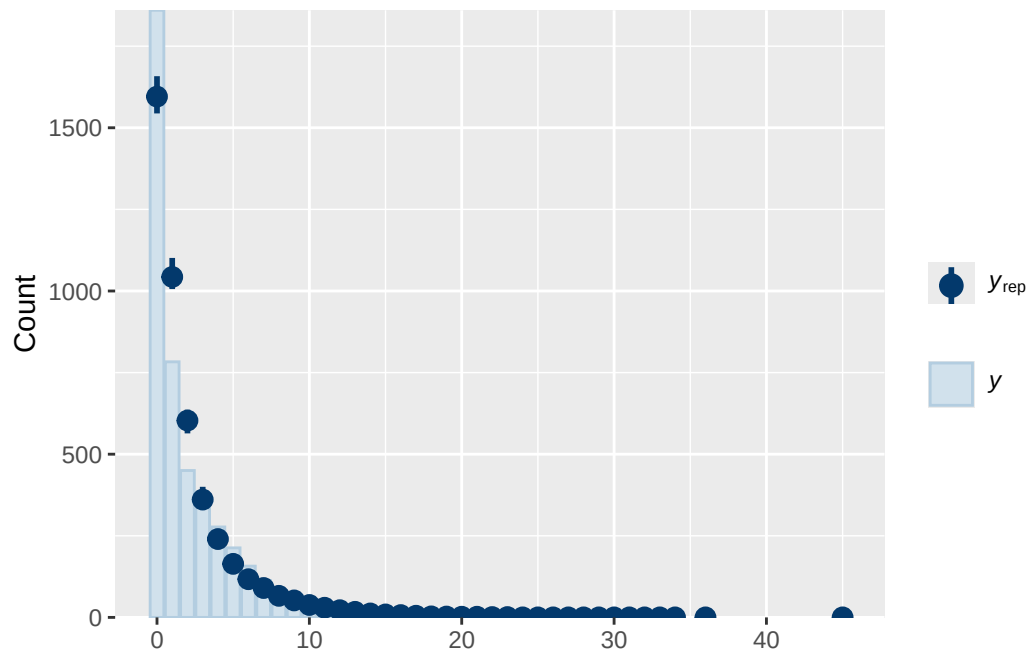
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

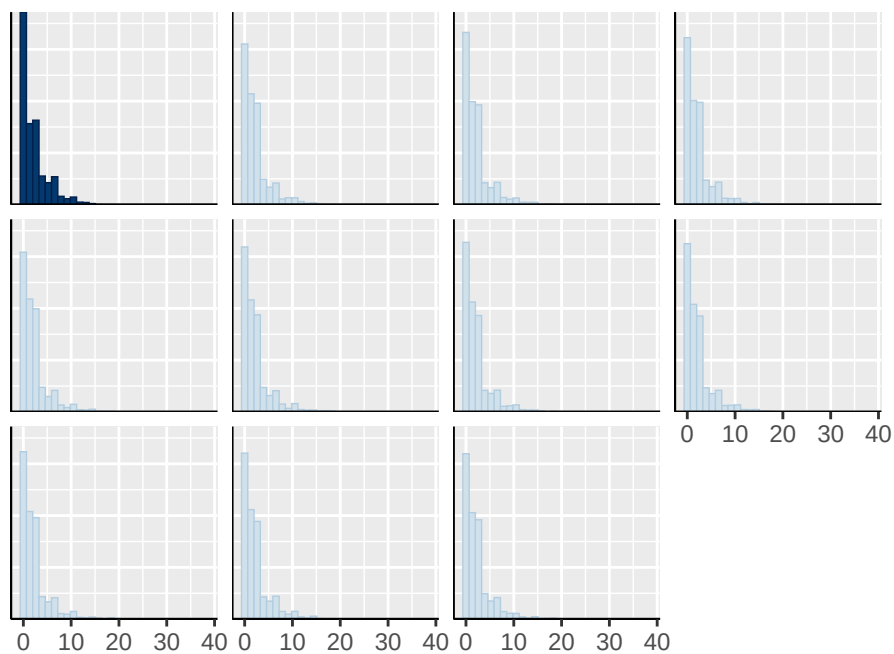
Model 1 longitudinal prior 2 diagnostics

	prior	class	coef	group	resp	dpar
	normal(0, 10)	b				
	normal(0, 10)	b	adhd_traits_wave_0_sc			
	normal(0, 10)	b	age_wave_2_sc			
	normal(0, 10)	b	ethnraceblack_nh			
	normal(0, 10)	b	ethnracehispanic			
	normal(0, 10)	b	ethnraceother			
	normal(0, 10)	b	inr_wave_0_sc			
	normal(0, 10)	b	menarche_status_p_wave_2Y			
student_t(3, 0, 2.9)	Intercept					
student_t(3, 0, 2.9)	sd					
student_t(3, 0, 2.9)	sd			family_id		
student_t(3, 0, 2.9)	sd		Intercept	family_id		
student_t(3, 0, 2.9)	sd			site_id		
student_t(3, 0, 2.9)	sd		Intercept	site_id		
nlpar lb ub tag		source				
		user				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		(vectorized)				
		default				
0		default				
0		(vectorized)				
0		(vectorized)				
0		(vectorized)				
0		(vectorized)				

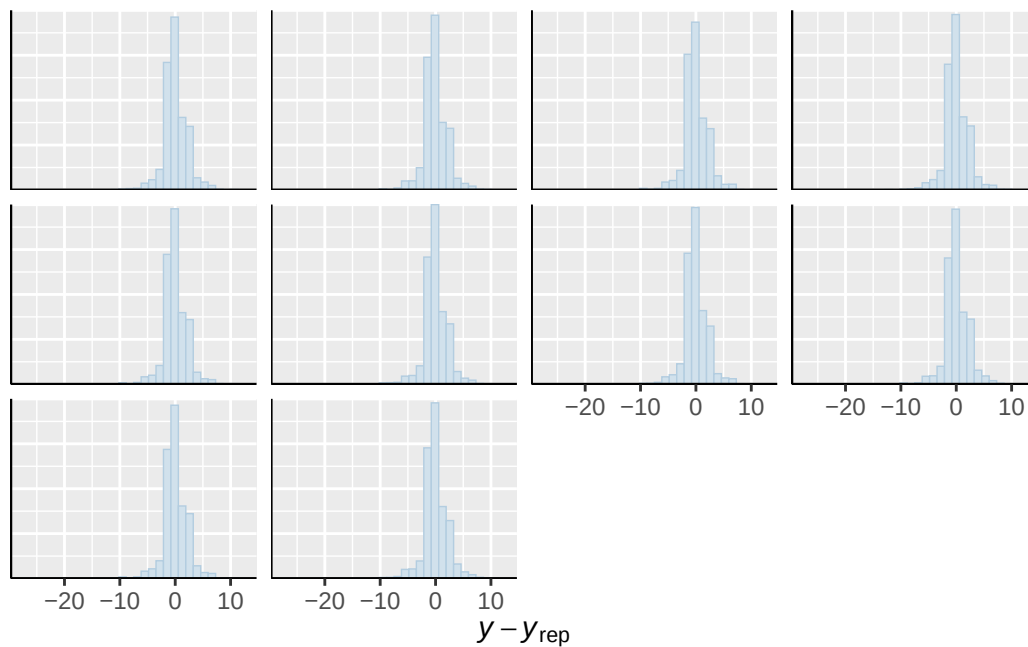




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

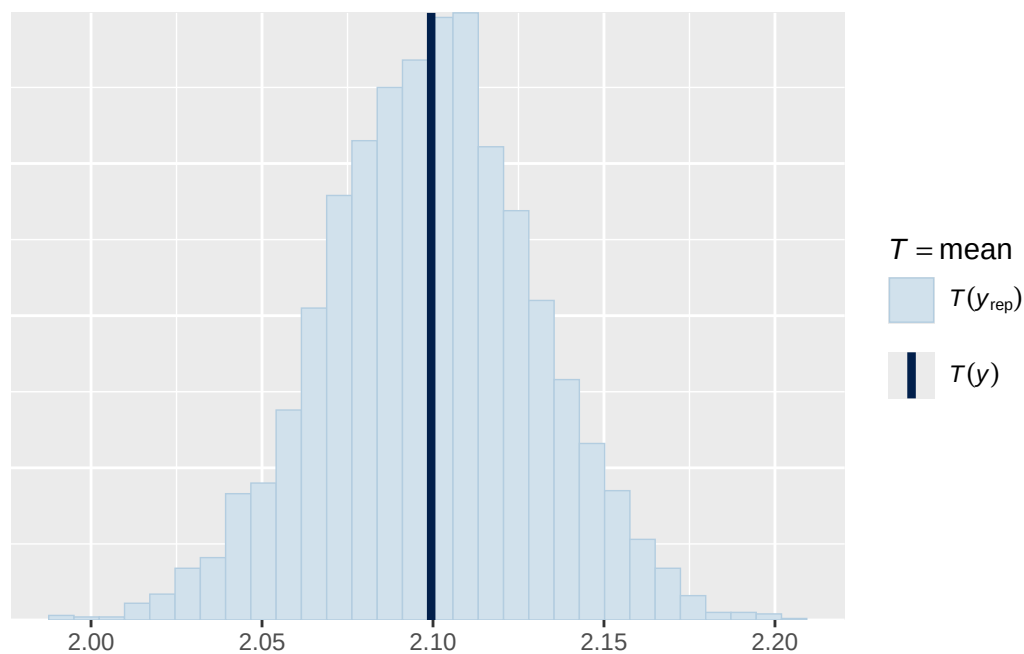


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

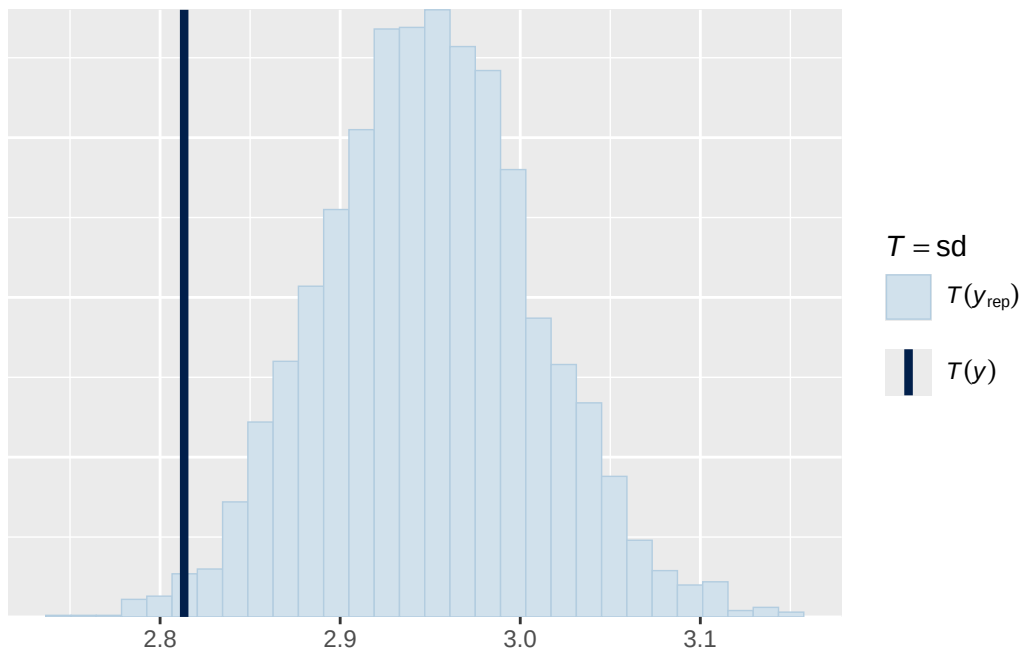


Using all posterior draws for ppc type 'stat' by default.

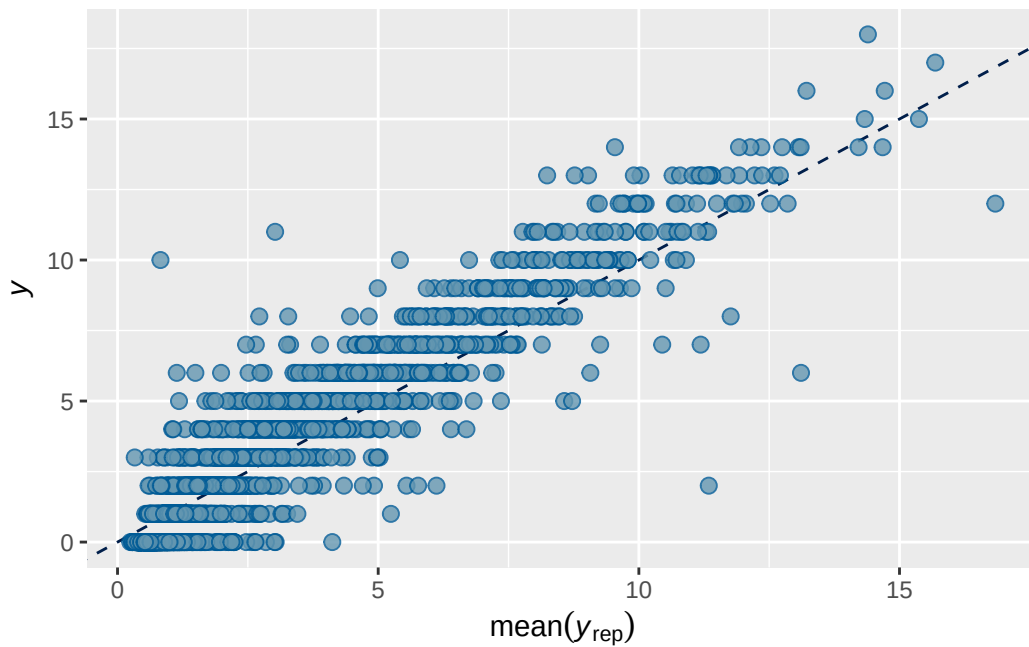
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use `stat_bin()` using `bins = 30`. Pick better value `binwidth`.



Using all posterior draws for ppc type 'stat' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 1B longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5    Q97.5
R2 0.7357083 0.007529665 0.7207626 0.750273

```

```

Family: poisson
Links: mu = log
Formula: adhd_traits_wave_3 ~ breast_development_p_wave_2_sc + age_wave_2_sc + ethnrace + inr_wave_0_sc
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.886    0.021   0.846   0.927 1.004    2360    4681

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.093    0.036   0.022   0.168 1.001    1068    1112

```

Regression Coefficients:

```

      Estimate Est.Error 1-95% CI u-95% CI  Rhat
Intercept      0.191    0.037   0.116   0.264 1.001
breast_development_p_wave_2_sc  0.125    0.040   0.047   0.204 1.000
age_wave_2_sc   -0.104    0.040  -0.182  -0.026 1.000
ethnraceblack_nh -0.091    0.065  -0.216   0.037 1.001
ethnracehispanic -0.035    0.057  -0.145   0.076 1.001
ethnraceother   -0.007    0.062  -0.129   0.118 1.001
inr_wave_0_sc    0.087    0.043   0.003   0.170 1.000
adhd_traits_wave_0_sc  1.322    0.034   1.256   1.388 1.001

      Bulk_ESS Tail_ESS
Intercept      3537    4081
breast_development_p_wave_2_sc  3892    4613
age_wave_2_sc   3724    5234
ethnraceblack_nh 2931    4747
ethnracehispanic 3005    4569
ethnraceother    3139    4952
inr_wave_0_sc    3087    4724
adhd_traits_wave_0_sc 2630    4155

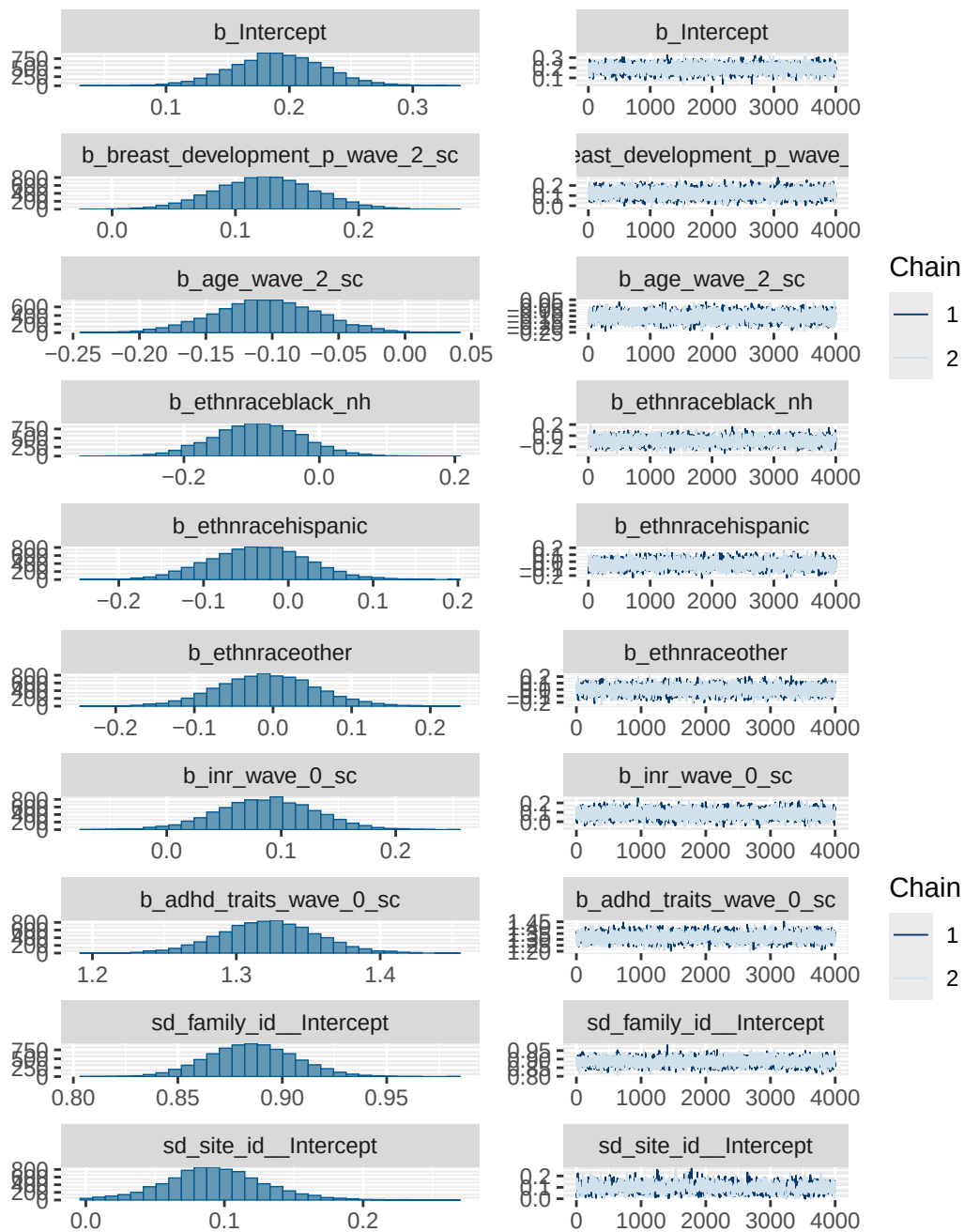
```

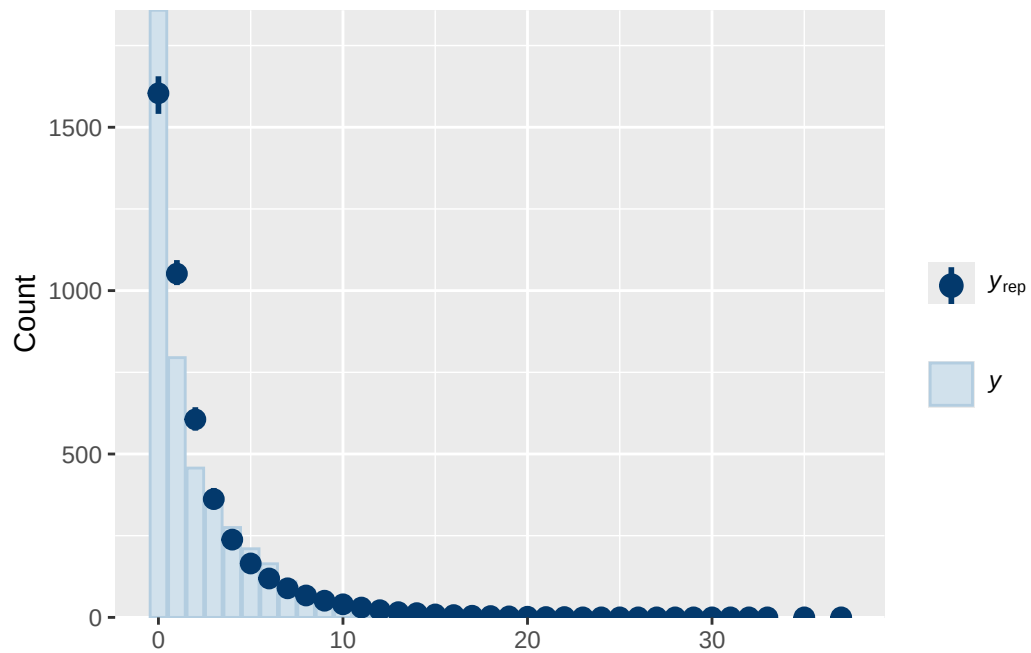
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS

and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

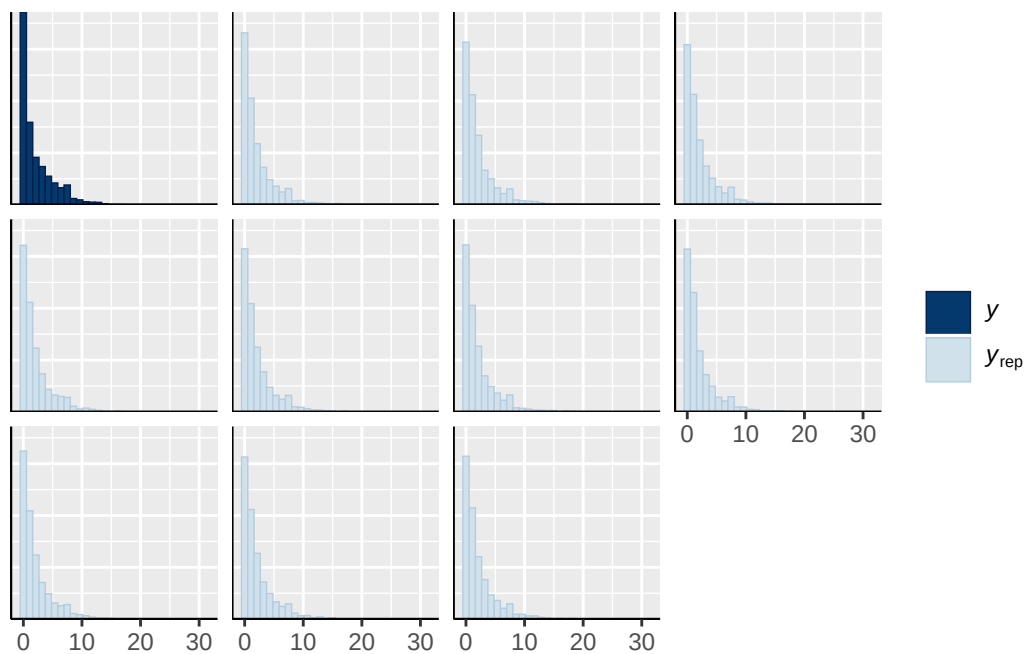
Model 1B longitudinal prior 2 diagnostics

	prior	class	coef	group	resp
	normal(0, 10)	b			
	normal(0, 10)	b	adhd_traits_wave_0_sc		
	normal(0, 10)	b	age_wave_2_sc		
	normal(0, 10)	b	breast_development_p_wave_2_sc		
	normal(0, 10)	b	ethnraceblack_nh		
	normal(0, 10)	b	ethnracehispanic		
	normal(0, 10)	b	ethnraceother		
	normal(0, 10)	b	inr_wave_0_sc		
student_t(3, 0, 2.7)	Intercept				
student_t(3, 0, 2.7)	sd				
student_t(3, 0, 2.7)	sd			family_id	
student_t(3, 0, 2.7)	sd		Intercept	family_id	
student_t(3, 0, 2.7)	sd			site_id	
student_t(3, 0, 2.7)	sd		Intercept	site_id	
dpar nlpar lb ub tag		source			
		user			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		(vectorized)			
		default			
0		default			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			
0		(vectorized)			

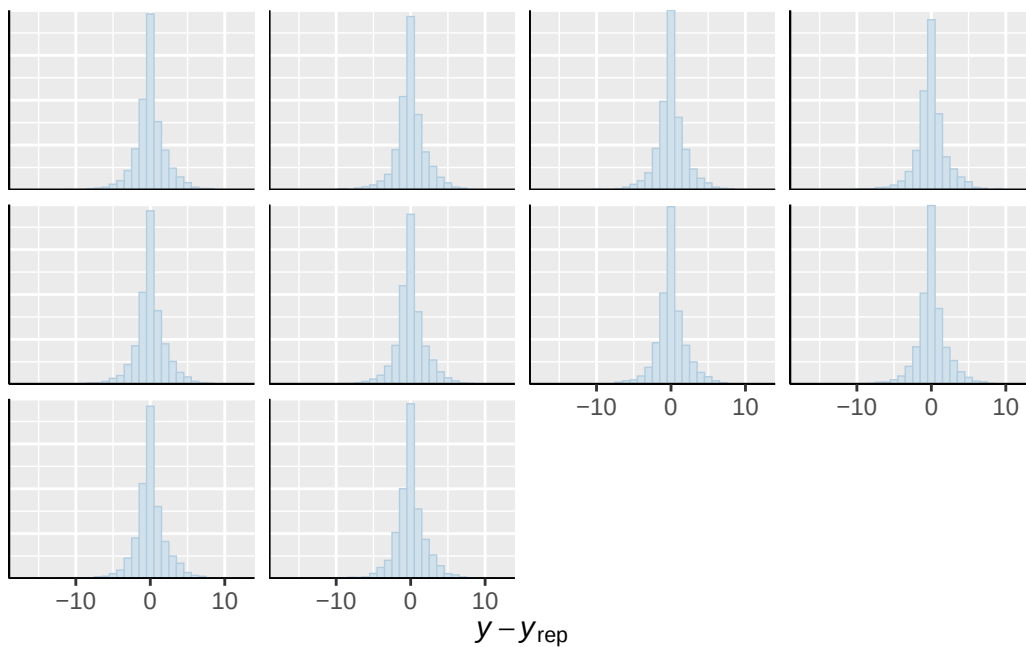




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

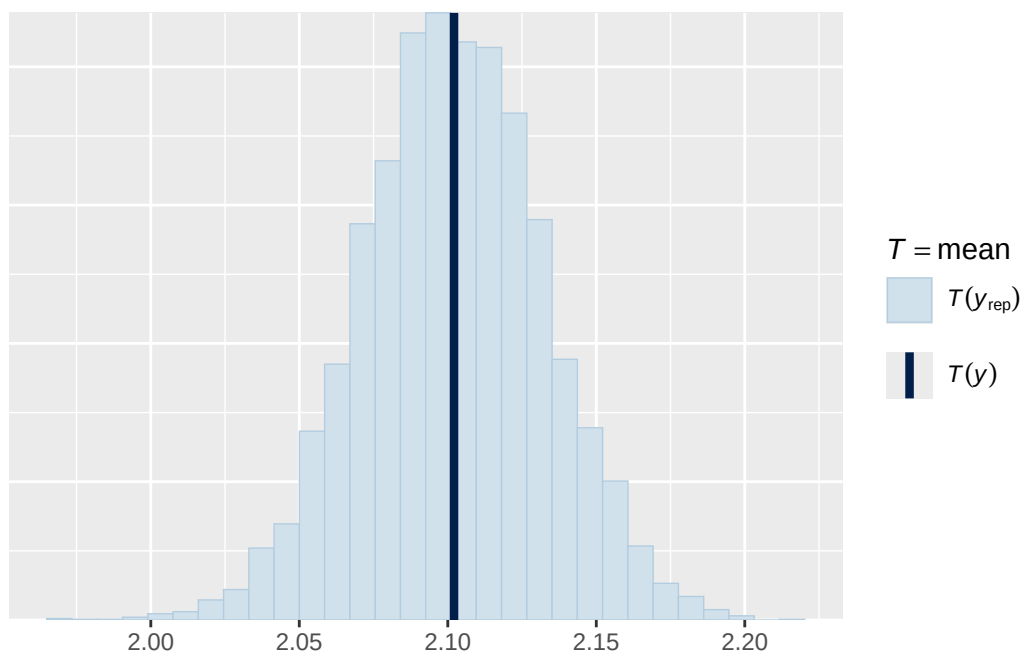


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

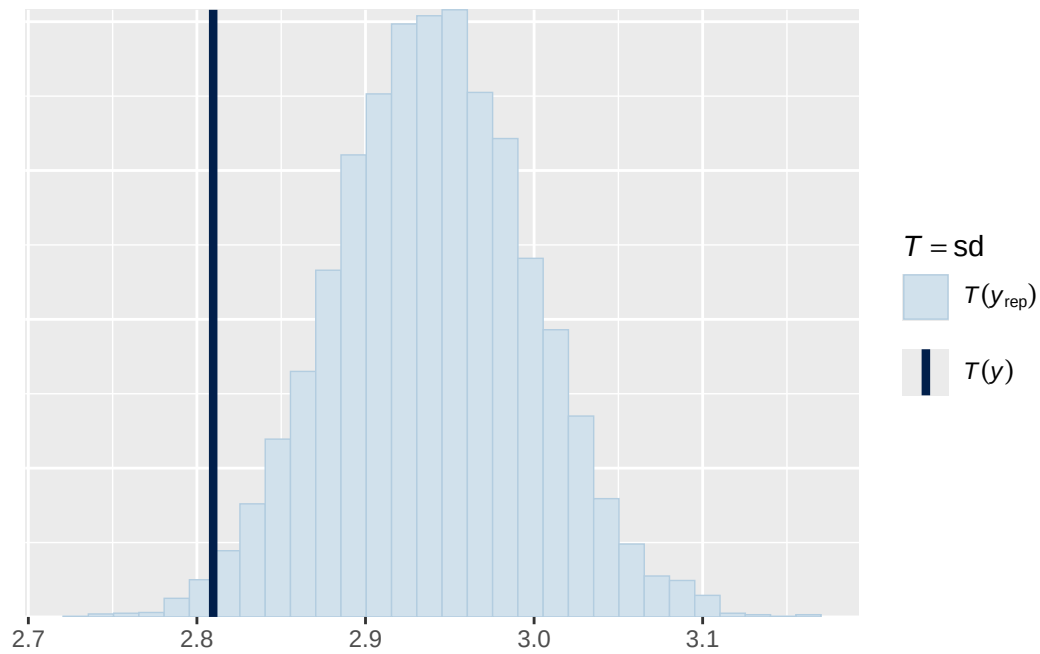


Using all posterior draws for ppc type 'stat' by default.

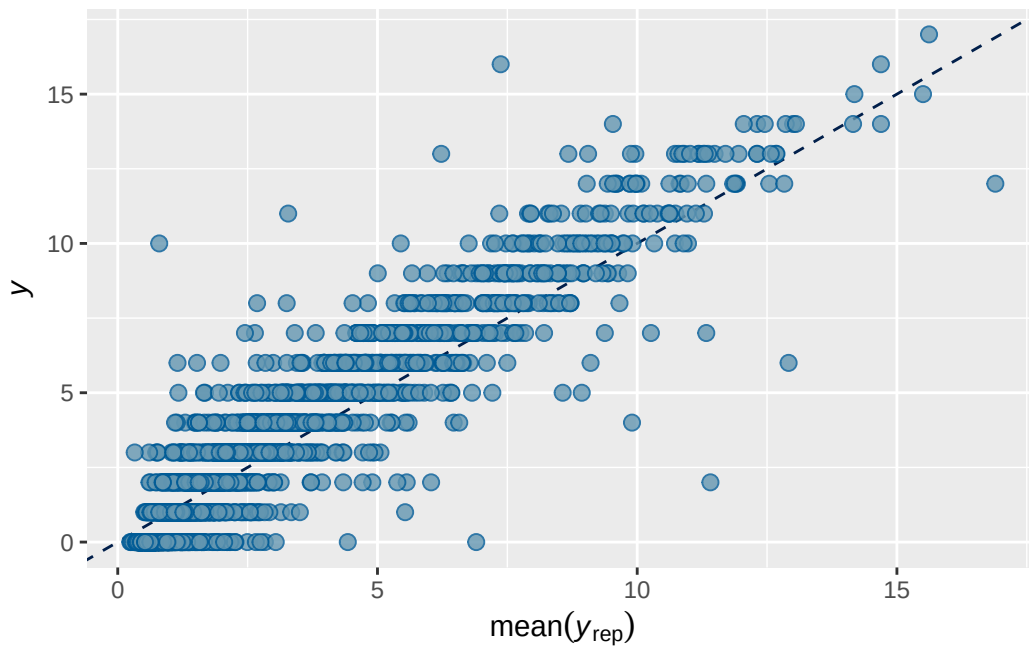
Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. Use ``stat_bin()'`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.



Model 5B longitudinal prior 2 output

```

      Estimate   Est.Error      Q2.5      Q97.5
R2 0.8232708 0.005211247 0.8127758 0.8330009

```

```

Family: poisson
Links: mu = log
Formula: externalising_wave_3 ~ breast_development_p_wave_2_sc * adhd_traits_wave_2_sc + age
Data: analytic_df_wide (Number of observations: 4515)
Draws: 2 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 8000

```

Multilevel Hyperparameters:

```

~family_id (Number of levels: 3928)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.985    0.019   0.947   1.023 1.000    2396    4408

~site_id (Number of levels: 22)
      Estimate Est.Error 1-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.066    0.038   0.004   0.150 1.003     747    1960

```

Regression Coefficients:

	Estimate	Est.Error			
Intercept	0.510	0.035			
breast_development_p_wave_2_sc	0.189	0.038			
adhd_traits_wave_2_sc	0.588	0.034			
age_wave_2_sc	-0.119	0.037			
ethnraceblack_nh	-0.134	0.066			
ethnracehispanic	-0.099	0.058			
ethnraceother	-0.082	0.064			
inr_wave_0_sc	-0.002	0.043			
externalising_wave_0_sc	0.912	0.035			
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.173	0.064			
	1-95% CI	u-95% CI	Rhat		
Intercept	0.440	0.577	1.000		
breast_development_p_wave_2_sc	0.115	0.264	1.001		
adhd_traits_wave_2_sc	0.520	0.653	1.001		
age_wave_2_sc	-0.193	-0.046	1.000		
ethnraceblack_nh	-0.261	-0.006	1.000		
ethnracehispanic	-0.212	0.015	1.000		
ethnraceother	-0.209	0.044	1.000		
inr_wave_0_sc	-0.087	0.082	1.001		

externalising_wave_0_sc	0.845	0.980	1.000
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	0.046	0.298	1.001
	Bulk_ESS	Tail_ESS	
Intercept	3957	5113	
breast_development_p_wave_2_sc	4506	5827	
adhd_traits_wave_2_sc	3055	4147	
age_wave_2_sc	4175	5517	
ethnraceblack_nh	2842	4278	
ethnracehispanic	3424	5003	
ethnraceother	3134	4414	
inr_wave_0_sc	3475	5000	
externalising_wave_0_sc	2660	4781	
breast_development_p_wave_2_sc:adhd_traits_wave_2_sc	3070	4827	

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

Model 5B longitudinal prior 2 diagnostics

```
prior      class
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
normal(0, 10)    b
student_t(3, 0, 2.9) Intercept
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
student_t(3, 0, 2.9)          sd
coef      group resp dpar nlpar
adhd_traits_wave_2_sc
age_wave_2_sc
breast_development_p_wave_2_sc
```

```

breast_development_p_wave_2_sc:adhd_traits_wave_2_sc
      ethnraceblack_nh
      ethnracehispanic
      ethnraceother
externalising_wave_0_sc
      inr_wave_0_sc

```

```

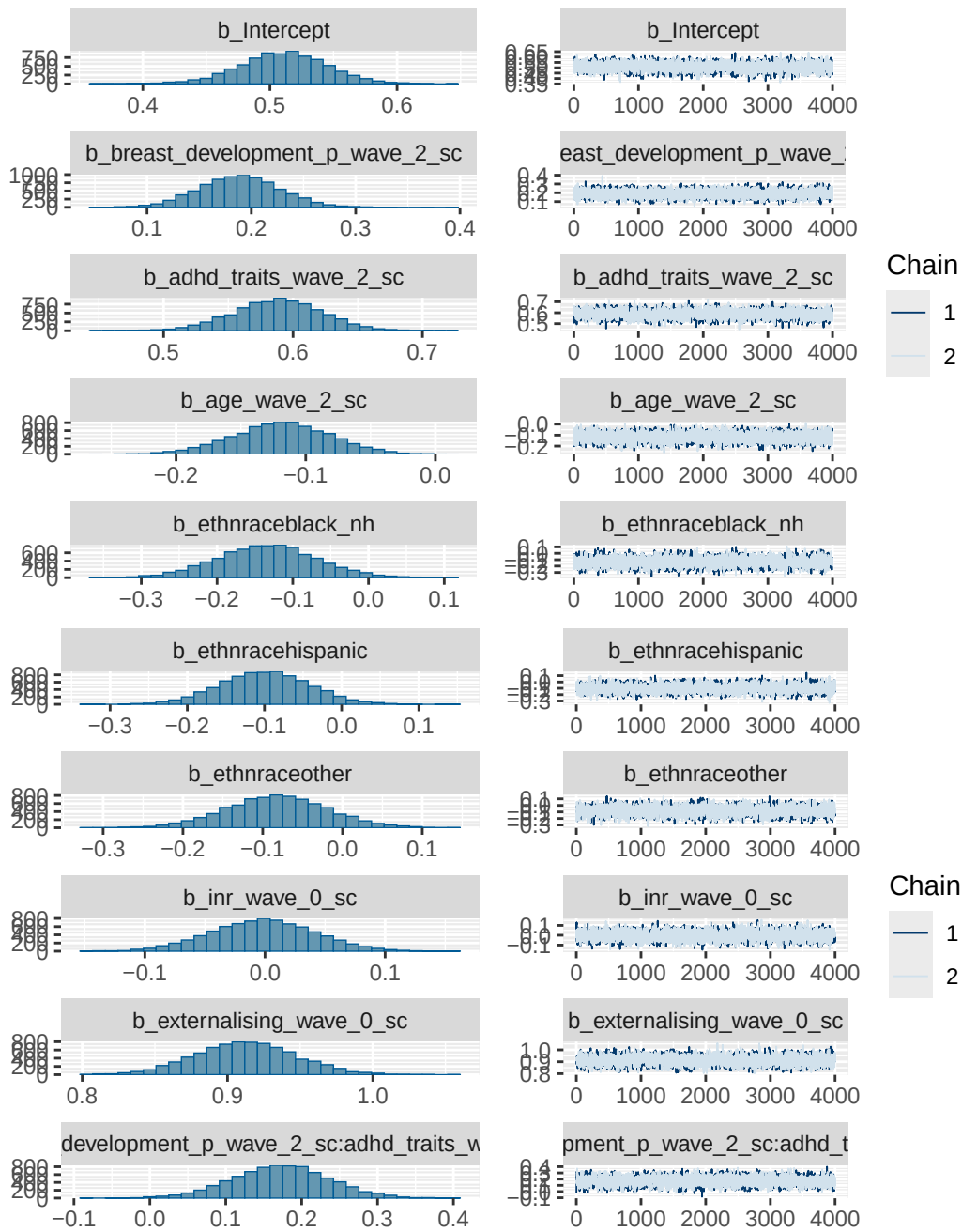
family_id
Intercept family_id
      site_id
Intercept site_id

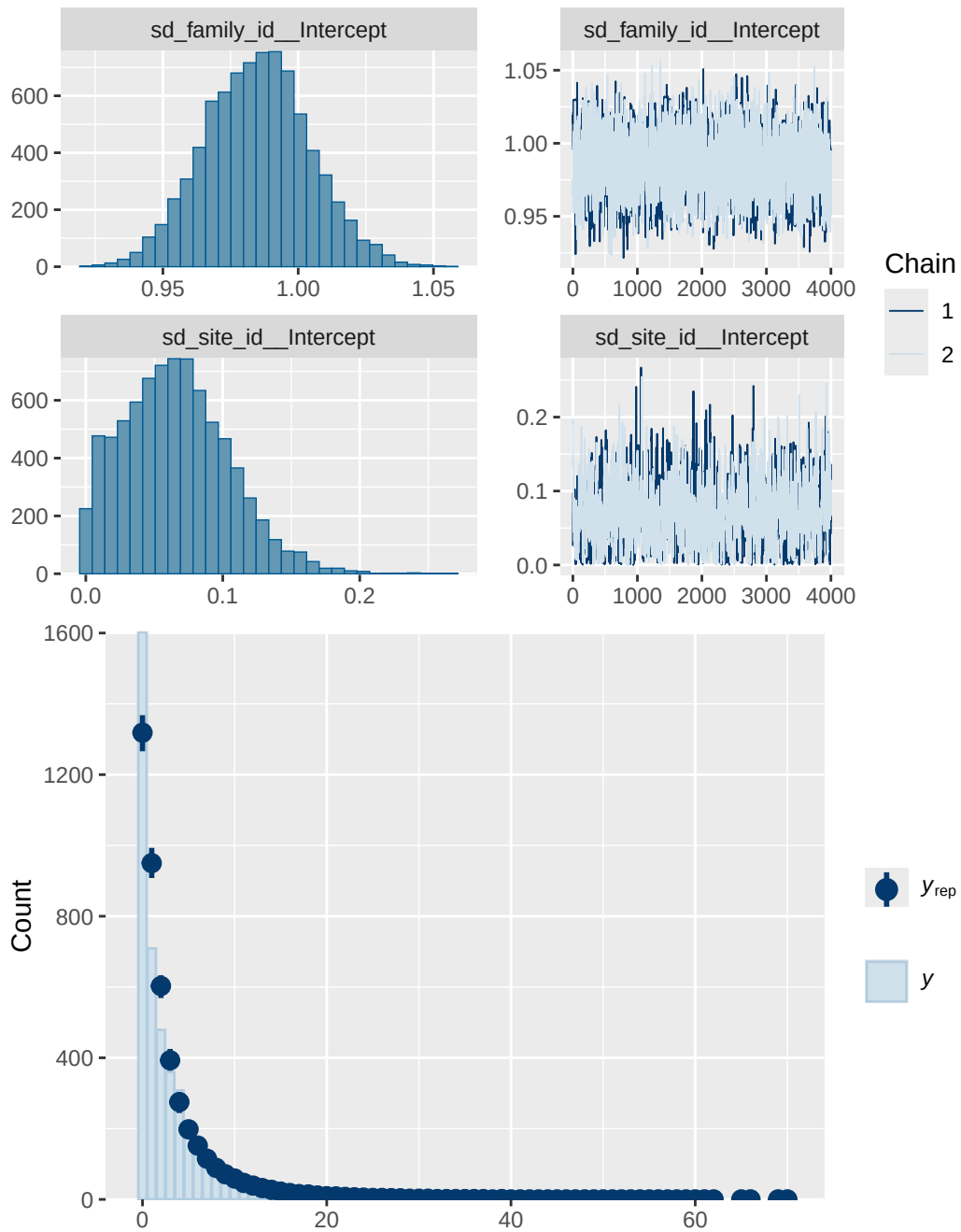
```

```

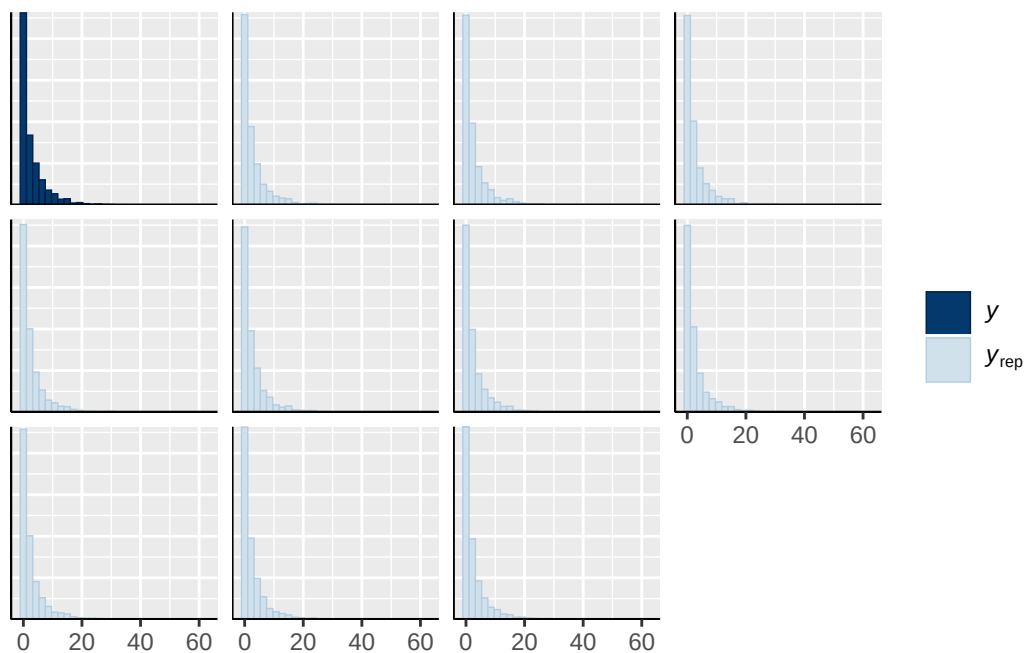
lb ub tag      source
      user
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      (vectorized)
      default
0      default
0      (vectorized)
0      (vectorized)
0      (vectorized)
0      (vectorized)

```

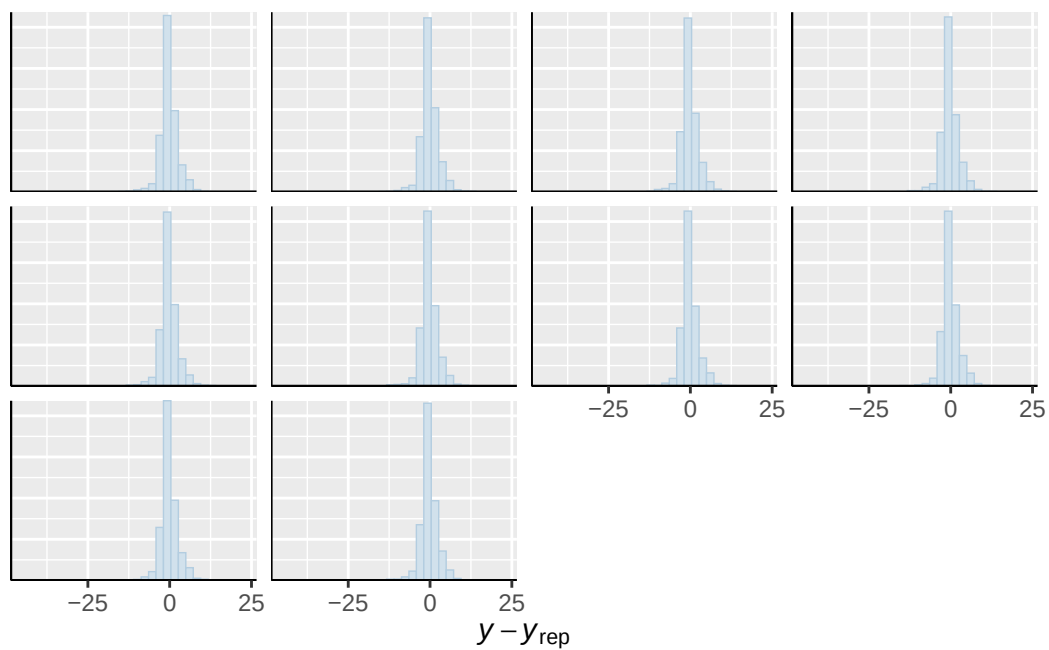




``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

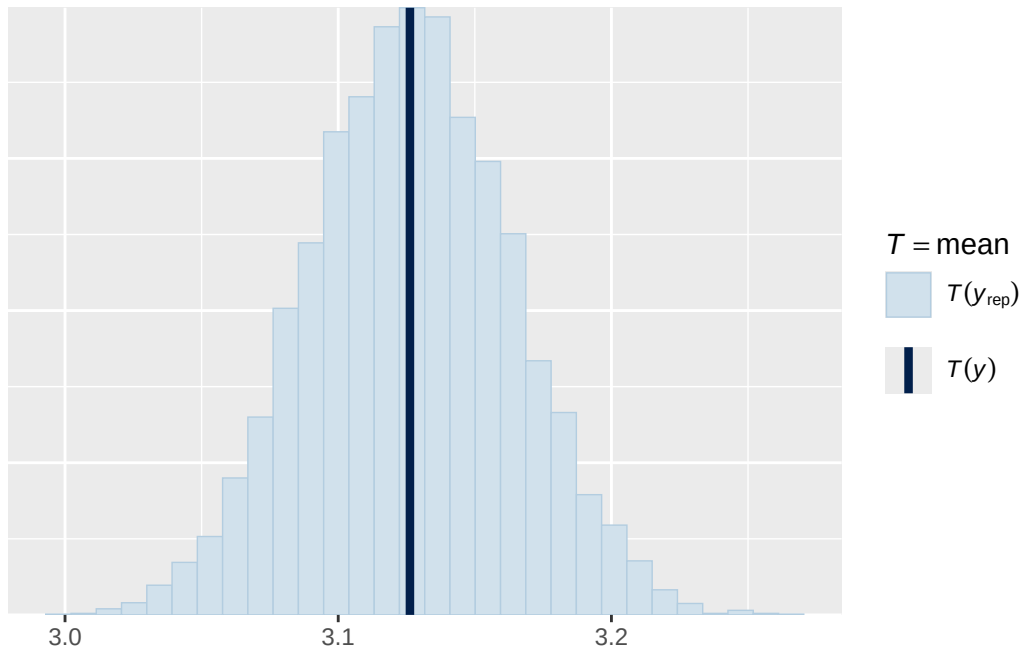


Using 10 posterior draws for ppc type 'error_hist' by default.
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

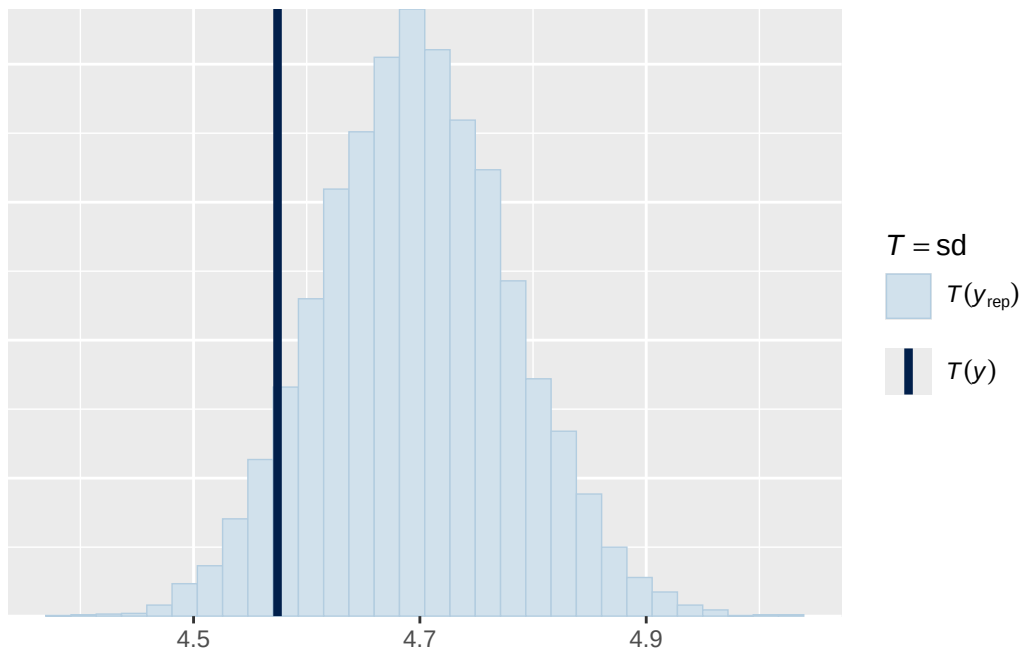


Using all posterior draws for ppc type 'stat' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'stat' by default. ``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Using all posterior draws for ppc type 'scatter_avg' by default.

